

3

Higher-Order Differential Equations

EXERCISES 3.1

Preliminary Theory: Linear Equations

1. From $y = c_1 e^x + c_2 e^{-x}$ we find $y' = c_1 e^x - c_2 e^{-x}$. Then $y(0) = c_1 + c_2 = 0$, $y'(0) = c_1 - c_2 = 1$ so that $c_1 = \frac{1}{2}$ and $c_2 = -\frac{1}{2}$. The solution is $y = \frac{1}{2}e^x - \frac{1}{2}e^{-x}$.
2. From $y = c_1 e^{4x} + c_2 e^{-x}$ we find $y' = 4c_1 e^{4x} - c_2 e^{-x}$. Then $y(0) = c_1 + c_2 = 1$, $y'(0) = 4c_1 - c_2 = 2$ so that $c_1 = \frac{3}{5}$ and $c_2 = \frac{2}{5}$. The solution is $y = \frac{3}{5}e^{4x} + \frac{2}{5}e^{-x}$.
3. From $y = c_1 x + c_2 x \ln x$ we find $y' = c_1 + c_2(1 + \ln x)$. Then $y(1) = c_1 = 3$, $y'(1) = c_1 + c_2 = -1$ so that $c_1 = 3$ and $c_2 = -4$. The solution is $y = 3x - 4x \ln x$.
4. From $y = c_1 + c_2 \cos x + c_3 \sin x$ we find $y' = -c_2 \sin x + c_3 \cos x$ and $y'' = -c_2 \cos x - c_3 \sin x$. Then $y(\pi) = c_1 - c_2 = 0$, $y'(\pi) = -c_3 = 2$, $y''(\pi) = c_2 = -1$ so that $c_1 = -1$, $c_2 = -1$, and $c_3 = -2$. The solution is $y = -1 - \cos x - 2 \sin x$.
5. From $y = c_1 + c_2 x^2$ we find $y' = 2c_2 x$. Then $y(0) = c_1 = 0$, $y'(0) = 2c_2 \cdot 0 = 0$ and hence $y'(0) = 1$ is not possible. Since $a_2(x) = x$ is 0 at $x = 0$, Theorem 3.1 is not violated.
6. In this case we have $y(0) = c_1 = 0$, $y'(0) = 2c_2 \cdot 0 = 0$ so $c_1 = 0$ and c_2 is arbitrary. Two solutions are $y = x^2$ and $y = 2x^2$.
7. From $x(0) = x_0 = c_1$ we see that $x(t) = x_0 \cos \omega t + c_2 \sin \omega t$ and $x'(t) = -x_0 \sin \omega t + c_2 \omega \cos \omega t$. Then $x'(0) = x_1 = c_2 \omega$ implies $c_2 = x_1/\omega$. Thus

$$x(t) = x_0 \cos \omega t + \frac{x_1}{\omega} \sin \omega t.$$

8. Solving the system

$$x(t_0) = c_1 \cos \omega t_0 + c_2 \sin \omega t_0 = x_0$$

$$x'(t_0) = -c_1 \omega \sin \omega t_0 + c_2 \omega \cos \omega t_0 = x_1$$

for c_1 and c_2 gives

$$c_1 = \frac{\omega x_0 \cos \omega t_0 - x_1 \sin \omega t_0}{\omega} \quad \text{and} \quad c_2 = \frac{x_1 \cos \omega t_0 + \omega x_0 \sin \omega t_0}{\omega}.$$

Thus

$$\begin{aligned} x(t) &= \frac{\omega x_0 \cos \omega t_0 - x_1 \sin \omega t_0}{\omega} \cos \omega t + \frac{x_1 \cos \omega t_0 + \omega x_0 \sin \omega t_0}{\omega} \sin \omega t \\ &= x_0 (\cos \omega t \cos \omega t_0 + \sin \omega t \sin \omega t_0) + \frac{x_1}{\omega} (\sin \omega t \cos \omega t_0 - \cos \omega t \sin \omega t_0) \\ &= x_0 \cos \omega(t - t_0) + \frac{x_1}{\omega} \sin \omega(t - t_0). \end{aligned}$$

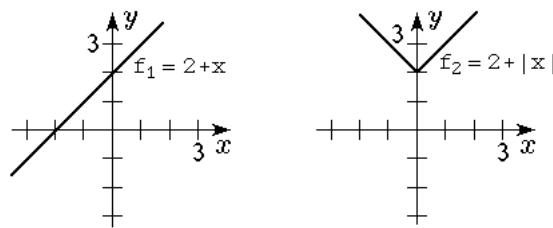
9. Since $a_2(x) = x - 2$ and $x_0 = 0$ the problem has a unique solution for $-\infty < x < 2$.

3.1 Preliminary Theory: Linear Equations

10. Since $a_0(x) = \tan x$ and $x_0 = 0$ the problem has a unique solution for $-\pi/2 < x < \pi/2$.
11. (a) We have $y(0) = c_1 + c_2 = 0$, $y''(1) = c_1 e + c_2 e^{-1} = 1$ so that $c_1 = e/(e^2 - 1)$ and $c_2 = -e/(e^2 - 1)$. The solution is $y = e(e^x - e^{-x})/(e^2 - 1)$.
- (b) We have $y(0) = c_3 \cosh 0 + c_4 \sinh 0 = c_3 = 0$ and $y(1) = c_3 \cosh 1 + c_4 \sinh 1 = c_4 \sinh 1 = 1$, so $c_3 = 0$ and $c_4 = 1/\sinh 1$. The solution is $y = (\sinh x)/(\sinh 1)$.
- (c) Starting with the solution in part (b) we have

$$y = \frac{1}{\sinh 1} \sinh x = \frac{2}{e^1 - e^{-1}} \frac{e^x - e^{-x}}{2} = \frac{e^x - e^{-x}}{e - 1/e} = \frac{e}{e^2 - 1}(e^x - e^{-x}).$$

12. In this case we have $y(0) = c_1 = 1$, $y'(1) = 2c_2 = 6$ so that $c_1 = 1$ and $c_2 = 3$. The solution is $y = 1 + 3x^2$.
13. From $y = c_1 e^x \cos x + c_2 e^x \sin x$ we find $y' = c_1 e^x(-\sin x + \cos x) + c_2 e^x(\cos x + \sin x)$.
- (a) We have $y(0) = c_1 = 1$, $y'(0) = c_1 + c_2 = 0$ so that $c_1 = 1$ and $c_2 = -1$. The solution is $y = e^x \cos x - e^x \sin x$.
- (b) We have $y(0) = c_1 = 1$, $y(\pi) = -e^\pi = -1$, which is not possible.
- (c) We have $y(0) = c_1 = 1$, $y(\pi/2) = c_2 e^{\pi/2} = 1$ so that $c_1 = 1$ and $c_2 = e^{-\pi/2}$. The solution is $y = e^x \cos x + e^{-\pi/2} e^x \sin x$.
- (d) We have $y(0) = c_1 = 0$, $y(\pi) = c_2 e^\pi \sin \pi = 0$ so that $c_1 = 0$ and c_2 is arbitrary. Solutions are $y = c_2 e^x \sin x$, for any real numbers c_2 .
14. (a) We have $y(-1) = c_1 + c_2 + 3 = 0$, $y(1) = c_1 + c_2 + 3 = 4$, which is not possible.
- (b) We have $y(0) = c_1 \cdot 0 + c_2 \cdot 0 + 3 = 1$, which is not possible.
- (c) We have $y(0) = c_1 \cdot 0 + c_2 \cdot 0 + 3 = 3$, $y(1) = c_1 + c_2 + 3 = 0$ so that c_1 is arbitrary and $c_2 = -3 - c_1$. Solutions are $y = c_1 x^2 - (c_1 + 3)x^4 + 3$.
- (d) We have $y(1) = c_1 + c_2 + 3 = 3$, $y(2) = 4c_1 + 16c_2 + 3 = 15$ so that $c_1 = -1$ and $c_2 = 1$. The solution is $y = -x^2 + x^4 + 3$.
15. Since $(-4)x + (3)x^2 + (1)(4x - 3x^2) = 0$ the set of functions is linearly dependent.
16. Since $(1)0 + (0)x + (0)e^x = 0$ the set of functions is linearly dependent. A similar argument shows that any set of functions containing $f(x) = 0$ will be linearly dependent.
17. Since $(-1/5)5 + (1)\cos^2 x + (1)\sin^2 x = 0$ the set of functions is linearly dependent.
18. Since $(1)\cos 2x + (1)1 + (-2)\cos^2 x = 0$ the set of functions is linearly dependent.
19. Since $(-4)x + (3)(x - 1) + (1)(x + 3) = 0$ the set of functions is linearly dependent.
20. From the graphs of $f_1(x) = 2 + x$ and $f_2(x) = 2 + |x|$ we see that the set of functions is linearly independent since they cannot be multiples of each other.



21. Suppose $c_1(1 + x) + c_2x + c_3x^2 = 0$. Then $c_1 + (c_1 + c_2)x + c_3x^2 = 0$ and so $c_1 = 0$, $c_1 + c_2 = 0$, and $c_3 = 0$. Since $c_1 = 0$ we also have $c_2 = 0$. Thus, the set of functions is linearly independent.
22. Since $(-1/2)e^x + (1/2)e^{-x} + (1)\sinh x = 0$ the set of functions is linearly dependent.

23. The functions satisfy the differential equation and are linearly independent since

$$W(e^{-3x}, e^{4x}) = 7e^x \neq 0$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 e^{-3x} + c_2 e^{4x}.$$

24. The functions satisfy the differential equation and are linearly independent since

$$W(\cosh 2x, \sinh 2x) = 2$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 \cosh 2x + c_2 \sinh 2x.$$

25. The functions satisfy the differential equation and are linearly independent since

$$W(e^x \cos 2x, e^x \sin 2x) = 2e^{2x} \neq 0$$

for $-\infty < x < \infty$. The general solution is $y = c_1 e^x \cos 2x + c_2 e^x \sin 2x$.

26. The functions satisfy the differential equation and are linearly independent since

$$W(e^{x/2}, xe^{x/2}) = e^x \neq 0$$

for $-\infty < x < \infty$. The general solution is

$$y = c_1 e^{x/2} + c_2 x e^{x/2}.$$

27. The functions satisfy the differential equation and are linearly independent since

$$W(x^3, x^4) = x^6 \neq 0$$

for $0 < x < \infty$. The general solution on this interval is

$$y = c_1 x^3 + c_2 x^4.$$

28. The functions satisfy the differential equation and are linearly independent since

$$W(\cos(\ln x), \sin(\ln x)) = 1/x \neq 0$$

for $0 < x < \infty$. The general solution on this interval is

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x).$$

29. The functions satisfy the differential equation and are linearly independent since

$$W(x, x^{-2}, x^{-2} \ln x) = 9x^{-6} \neq 0$$

for $0 < x < \infty$. The general solution on this interval is

$$y = c_1 x + c_2 x^{-2} + c_3 x^{-2} \ln x.$$

30. The functions satisfy the differential equation and are linearly independent since

$$W(1, x, \cos x, \sin x) = 1$$

for $-\infty < x < \infty$. The general solution on this interval is

$$y = c_1 + c_2 x + c_3 \cos x + c_4 \sin x.$$

3.1 Preliminary Theory: Linear Equations

31. The functions $y_1 = e^{2x}$ and $y_2 = e^{5x}$ form a fundamental set of solutions of the associated homogeneous equation, and $y_p = 6e^x$ is a particular solution of the nonhomogeneous equation.
32. The functions $y_1 = \cos x$ and $y_2 = \sin x$ form a fundamental set of solutions of the associated homogeneous equation, and $y_p = x \sin x + (\cos x) \ln(\cos x)$ is a particular solution of the nonhomogeneous equation.
33. The functions $y_1 = e^{2x}$ and $y_2 = xe^{2x}$ form a fundamental set of solutions of the associated homogeneous equation, and $y_p = x^2e^{2x} + x - 2$ is a particular solution of the nonhomogeneous equation.
34. The functions $y_1 = x^{-1/2}$ and $y_2 = x^{-1}$ form a fundamental set of solutions of the associated homogeneous equation, and $y_p = \frac{1}{15}x^2 - \frac{1}{6}x$ is a particular solution of the nonhomogeneous equation.
35. (a) We have $y'_{p_1} = 6e^{2x}$ and $y''_{p_1} = 12e^{2x}$, so

$$y''_{p_1} - 6y'_{p_1} + 5y_{p_1} = 12e^{2x} - 36e^{2x} + 15e^{2x} = -9e^{2x}.$$

Also, $y'_{p_2} = 2x + 3$ and $y''_{p_2} = 2$, so

$$y''_{p_2} - 6y'_{p_2} + 5y_{p_2} = 2 - 6(2x + 3) + 5(x^2 + 3x) = 5x^2 + 3x - 16.$$

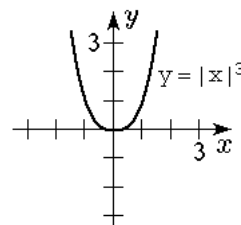
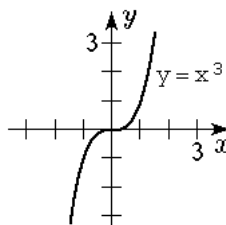
- (b) By the superposition principle for nonhomogeneous equations a particular solution of $y'' - 6y' + 5y = 5x^2 + 3x - 16 - 9e^{2x}$ is $y_p = x^2 + 3x + 3e^{2x}$. A particular solution of the second equation is

$$y_p = -2y_{p_2} - \frac{1}{9}y_{p_1} = -2x^2 - 6x - \frac{1}{3}e^{2x}.$$

36. (a) $y_{p_1} = 5$
(b) $y_{p_2} = -2x$
(c) $y_p = y_{p_1} + y_{p_2} = 5 - 2x$
(d) $y_p = \frac{1}{2}y_{p_1} - 2y_{p_2} = \frac{5}{2} + 4x$
37. (a) Since $D^2x = 0$, x and 1 are solutions of $y'' = 0$. Since they are linearly independent, the general solution is $y = c_1x + c_2$.
- (b) Since $D^3x^2 = 0$, x^2 , x , and 1 are solutions of $y''' = 0$. Since they are linearly independent, the general solution is $y = c_1x^2 + c_2x + c_3$.
- (c) Since $D^4x^3 = 0$, x^3 , x^2 , x , and 1 are solutions of $y^{(4)} = 0$. Since they are linearly independent, the general solution is $y = c_1x^3 + c_2x^2 + c_3x + c_4$.
- (d) By part (a), the general solution of $y'' = 0$ is $y_c = c_1x + c_2$. Since $D^2x^2 = 2! = 2$, $y_p = x^2$ is a particular solution of $y'' = 2$. Thus, the general solution is $y = c_1x + c_2 + x^2$.
- (e) By part (b), the general solution of $y''' = 0$ is $y_c = c_1x^2 + c_2x + c_3$. Since $D^3x^3 = 3! = 6$, $y_p = x^3$ is a particular solution of $y''' = 6$. Thus, the general solution is $y = c_1x^2 + c_2x + c_3 + x^3$.
- (f) By part (c), the general solution of $y^{(4)} = 0$ is $y_c = c_1x^3 + c_2x^2 + c_3x + c_4$. Since $D^4x^4 = 4! = 24$, $y_p = x^4$ is a particular solution of $y^{(4)} = 24$. Thus, the general solution is $y = c_1x^3 + c_2x^2 + c_3x + c_4 + x^4$.
38. By the superposition principle, if $y_1 = e^x$ and $y_2 = e^{-x}$ are both solutions of a homogeneous linear differential equation, then so are

$$\frac{1}{2}(y_1 + y_2) = \frac{e^x + e^{-x}}{2} = \cosh x \quad \text{and} \quad \frac{1}{2}(y_1 - y_2) = \frac{e^x - e^{-x}}{2} = \sinh x.$$

39. (a) From the graphs of $y_1 = x^3$ and $y_2 = |x|^3$ we see that the functions are linearly independent since they cannot be multiples of each other. It is easily shown that $y_1 = x^3$ is a solution of $x^2 y'' - 4xy' + 6y = 0$. To show that $y_2 = |x|^3$ is a solution let $y_2 = x^3$ for $x \geq 0$ and let $y_2 = -x^3$ for $x < 0$.



- (b) If $x \geq 0$ then $y_2 = x^3$ and

$$W(y_1, y_2) = \begin{vmatrix} x^3 & x^3 \\ 3x^2 & 3x^2 \end{vmatrix} = 0.$$

If $x < 0$ then $y_2 = -x^3$ and

$$W(y_1, y_2) = \begin{vmatrix} x^3 & -x^3 \\ 3x^2 & -3x^2 \end{vmatrix} = 0.$$

This does not violate Theorem 3.3 since $a_2(x) = x^2$ is zero at $x = 0$.

- (c) The functions $Y_1 = x^3$ and $Y_2 = x^2$ are solutions of $x^2 y'' - 4xy' + 6y = 0$. They are linearly independent since $W(x^3, x^2) = x^4 \neq 0$ for $-\infty < x < \infty$.
- (d) The function $y = x^3$ satisfies $y(0) = 0$ and $y'(0) = 0$.
- (e) Neither is the general solution on $(-\infty, \infty)$ since we form a general solution on an interval for which $a_2(x) \neq 0$ for every x in the interval.
40. Since $e^{x-3} = e^{-3}e^x = (e^{-5}e^2)e^x = e^{-5}e^{x+2}$, we see that e^{x-3} is a constant multiple of e^{x+2} and the set of functions is linearly dependent.
41. Since $0y_1 + 0y_2 + \cdots + 0y_k + 1y_{k+1} = 0$, the set of solutions is linearly dependent.
42. The set of solutions is linearly dependent. Suppose n of the solutions are linearly independent (if not, then the set of $n+1$ solutions is linearly dependent). Without loss of generality, let this set be $y_1, y_2, \dots, y_n$. Then $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$ is the general solution of the n th-order differential equation and for some choice, $c_1^*, c_2^*, \dots, c_n^*$, of the coefficients $y_{n+1} = c_1^* y_1 + c_2^* y_2 + \cdots + c_n^* y_n$. But then the set $y_1, y_2, \dots, y_n, y_{n+1}$ is linearly dependent.

EXERCISES 3.2

Reduction of Order

In Problems 1-8 we use reduction of order to find a second solution. In Problems 9-16 we use formula (5) from the text.

1. Define $y = u(x)e^{2x}$ so

$$y' = 2ue^{2x} + u'e^{2x}, \quad y'' = e^{2x}u'' + 4e^{2x}u' + 4e^{2x}u, \quad \text{and} \quad y'' - 4y' + 4y = e^{2x}u'' = 0.$$

Therefore $u'' = 0$ and $u = c_1 x + c_2$. Taking $c_1 = 1$ and $c_2 = 0$ we see that a second solution is $y_2 = xe^{2x}$.

3.2 Reduction of Order

2. Define $y = u(x)xe^{-x}$ so

$$y' = (1-x)e^{-x}u + xe^{-x}u', \quad y'' = xe^{-x}u'' + 2(1-x)e^{-x}u' - (2-x)e^{-x}u,$$

and

$$y'' + 2y' + y = e^{-x}(xu'' + 2u') = 0 \quad \text{or} \quad u'' + \frac{2}{x}u' = 0.$$

If $w = u'$ we obtain the linear first-order equation $w' + \frac{2}{x}w = 0$ which has the integrating factor $e^{\int \frac{2}{x} dx} = x^2$. Now

$$\frac{d}{dx}[x^2w] = 0 \quad \text{gives} \quad x^2w = c.$$

Therefore $w = u' = c/x^2$ and $u = c_1/x$. A second solution is $y_2 = \frac{1}{x}xe^{-x} = e^{-x}$.

3. Define $y = u(x)\cos 4x$ so

$$y' = -4u\sin 4x + u'\cos 4x, \quad y'' = u''\cos 4x - 8u'\sin 4x - 16u\cos 4x$$

and

$$y'' + 16y = (\cos 4x)u'' - 8(\sin 4x)u' = 0 \quad \text{or} \quad u'' - 8(\tan 4x)u' = 0.$$

If $w = u'$ we obtain the linear first-order equation $w' - 8(\tan 4x)w = 0$ which has the integrating factor $e^{-8\int \tan 4x dx} = \cos^2 4x$. Now

$$\frac{d}{dx}[(\cos^2 4x)w] = 0 \quad \text{gives} \quad (\cos^2 4x)w = c.$$

Therefore $w = u' = c\sec^2 4x$ and $u = c_1 \tan 4x$. A second solution is $y_2 = \tan 4x \cos 4x = \sin 4x$.

4. Define $y = u(x)\sin 3x$ so

$$y' = 3u\cos 3x + u'\sin 3x, \quad y'' = u''\sin 3x + 6u'\cos 3x - 9u\sin 3x,$$

and

$$y'' + 9y = (\sin 3x)u'' + 6(\cos 3x)u' = 0 \quad \text{or} \quad u'' + 6(\cot 3x)u' = 0.$$

If $w = u'$ we obtain the linear first-order equation $w' + 6(\cot 3x)w = 0$ which has the integrating factor $e^{\int 6\cot 3x dx} = \sin^2 3x$. Now

$$\frac{d}{dx}[(\sin^2 3x)w] = 0 \quad \text{gives} \quad (\sin^2 3x)w = c.$$

Therefore $w = u' = c\csc^2 3x$ and $u = c_1 \cot 3x$. A second solution is $y_2 = \cot 3x \sin 3x = \cos 3x$.

5. Define $y = u(x)\cosh x$ so

$$y' = u\sinh x + u'\cosh x, \quad y'' = u''\cosh x + 2u'\sinh x + u\cosh x$$

and

$$y'' - y = (\cosh x)u'' + 2(\sinh x)u' = 0 \quad \text{or} \quad u'' + 2(\tanh x)u' = 0.$$

If $w = u'$ we obtain the linear first-order equation $w' + 2(\tanh x)w = 0$ which has the integrating factor $e^{\int 2\tanh x dx} = \cosh^2 x$. Now

$$\frac{d}{dx}[(\cosh^2 x)w] = 0 \quad \text{gives} \quad (\cosh^2 x)w = c.$$

Therefore $w = u' = c\operatorname{sech}^2 x$ and $u = c \tanh x$. A second solution is $y_2 = \tanh x \cosh x = \sinh x$.

6. Define $y = u(x)e^{5x}$ so

$$y' = 5e^{5x}u + e^{5x}u', \quad y'' = e^{5x}u'' + 10e^{5x}u' + 25e^{5x}u$$

and

$$y'' - 25y = e^{5x}(u'' + 10u') = 0 \quad \text{or} \quad u'' + 10u' = 0.$$

If $w = u'$ we obtain the linear first-order equation $w' + 10w = 0$ which has the integrating factor $e^{10} \int dx = e^{10x}$. Now

$$\frac{d}{dx}[e^{10x}w] = 0 \quad \text{gives} \quad e^{10x}w = c.$$

Therefore $w = u' = ce^{-10x}$ and $u = c_1e^{-10x}$. A second solution is $y_2 = e^{-10x}e^{5x} = e^{-5x}$.

7. Define $y = u(x)e^{2x/3}$ so

$$y' = \frac{2}{3}e^{2x/3}u + e^{2x/3}u', \quad y'' = e^{2x/3}u'' + \frac{4}{3}e^{2x/3}u' + \frac{4}{9}e^{2x/3}u$$

and

$$9y'' - 12y' + 4y = 9e^{2x/3}u'' = 0.$$

Therefore $u'' = 0$ and $u = c_1x + c_2$. Taking $c_1 = 1$ and $c_2 = 0$ we see that a second solution is $y_2 = xe^{2x/3}$.

8. Define $y = u(x)e^{x/3}$ so

$$y' = \frac{1}{3}e^{x/3}u + e^{x/3}u', \quad y'' = e^{x/3}u'' + \frac{2}{3}e^{x/3}u' + \frac{1}{9}e^{x/3}u$$

and

$$6y'' + y' - y = e^{x/3}(6u'' + 5u') = 0 \quad \text{or} \quad u'' + \frac{5}{6}u' = 0.$$

If $w = u'$ we obtain the linear first-order equation $w' + \frac{5}{6}w = 0$ which has the integrating factor $e^{(5/6) \int dx} = e^{5x/6}$. Now

$$\frac{d}{dx}[e^{5x/6}w] = 0 \quad \text{gives} \quad e^{5x/6}w = c.$$

Therefore $w = u' = ce^{-5x/6}$ and $u = c_1e^{-5x/6}$. A second solution is $y_2 = e^{-5x/6}e^{x/3} = e^{-x/2}$.

9. Identifying $P(x) = -7/x$ we have

$$y_2 = x^4 \int \frac{e^{-\int (-7/x) dx}}{x^8} dx = x^4 \int \frac{1}{x} dx = x^4 \ln |x|.$$

A second solution is $y_2 = x^4 \ln |x|$.

10. Identifying $P(x) = 2/x$ we have

$$y_2 = x^2 \int \frac{e^{-\int (2/x) dx}}{x^4} dx = x^2 \int x^{-6} dx = -\frac{1}{5}x^{-3}.$$

A second solution is $y_2 = x^{-3}$.

11. Identifying $P(x) = 1/x$ we have

$$y_2 = \ln x \int \frac{e^{-\int dx/x}}{(\ln x)^2} dx = \ln x \int \frac{dx}{x(\ln x)^2} = \ln x \left(-\frac{1}{\ln x} \right) = -1.$$

A second solution is $y_2 = 1$.

12. Identifying $P(x) = 0$ we have

$$y_2 = x^{1/2} \ln x \int \frac{e^{-\int 0 dx}}{x(\ln x)^2} dx = x^{1/2} \ln x \left(-\frac{1}{\ln x} \right) = -x^{1/2}.$$

A second solution is $y_2 = x^{1/2}$.

3.2 Reduction of Order

13. Identifying $P(x) = -1/x$ we have

$$\begin{aligned} y_2 &= x \sin(\ln x) \int \frac{e^{-\int -dx/x}}{x^2 \sin^2(\ln x)} dx = x \sin(\ln x) \int \frac{x}{x^2 \sin^2(\ln x)} dx \\ &= x \sin(\ln x) \int \frac{\csc^2(\ln x)}{x} dx = [x \sin(\ln x)] [-\cot(\ln x)] = -x \cos(\ln x). \end{aligned}$$

A second solution is $y_2 = x \cos(\ln x)$.

14. Identifying $P(x) = -3/x$ we have

$$\begin{aligned} y_2 &= x^2 \cos(\ln x) \int \frac{e^{-\int -3 dx/x}}{x^4 \cos^2(\ln x)} dx = x^2 \cos(\ln x) \int \frac{x^3}{x^4 \cos^2(\ln x)} dx \\ &= x^2 \cos(\ln x) \int \frac{\sec^2(\ln x)}{x} dx = x^2 \cos(\ln x) \tan(\ln x) = x^2 \sin(\ln x). \end{aligned}$$

A second solution is $y_2 = x^2 \sin(\ln x)$.

15. Identifying $P(x) = 2(1+x)/(1-2x-x^2)$ we have

$$\begin{aligned} y_2 &= (x+1) \int \frac{e^{-\int 2(1+x)dx/(1-2x-x^2)}}{(x+1)^2} dx = (x+1) \int \frac{e^{\ln(1-2x-x^2)}}{(x+1)^2} dx \\ &= (x+1) \int \frac{1-2x-x^2}{(x+1)^2} dx = (x+1) \int \left[\frac{2}{(x+1)^2} - 1 \right] dx \\ &= (x+1) \left[-\frac{2}{x+1} - x \right] = -2 - x^2 - x. \end{aligned}$$

A second solution is $y_2 = x^2 + x + 2$.

16. Identifying $P(x) = -2x/(1-x^2)$ we have

$$y_2 = \int e^{-\int -2x dx/(1-x^2)} dx = \int e^{-\ln(1-x^2)} dx = \int \frac{1}{1-x^2} dx = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right|.$$

A second solution is $y_2 = \ln |(1+x)/(1-x)|$.

17. Define $y = u(x)e^{-2x}$ so

$$y' = -2ue^{-2x} + u'e^{-2x}, \quad y'' = u''e^{-2x} - 4u'e^{-2x} + 4ue^{-2x}$$

and

$$y'' - 4y = e^{-2x}u'' - 4e^{-2x}u' = 0 \quad \text{or} \quad u'' - 4u' = 0.$$

If $w = u'$ we obtain the linear first-order equation $w' - 4w = 0$ which has the integrating factor $e^{-4 \int dx} = e^{-4x}$. Now

$$\frac{d}{dx}[e^{-4x}w] = 0 \quad \text{gives} \quad e^{-4x}w = c.$$

Therefore $w = u' = ce^{4x}$ and $u = c_1e^{4x}$. A second solution is $y_2 = e^{-2x}e^{4x} = e^{2x}$. We see by observation that a particular solution is $y_p = -1/2$. The general solution is

$$y = c_1e^{-2x} + c_2e^{2x} - \frac{1}{2}.$$

18. Define $y = u(x) \cdot 1$ so

$$y' = u', \quad y'' = u'' \quad \text{and} \quad y'' + y' = u'' + u' = 1.$$

If $w = u'$ we obtain the linear first-order equation $w' + w = 1$ which has the integrating factor $e^{\int dx} = e^x$. Now

$$\frac{d}{dx}[e^x w] = e^x \quad \text{gives} \quad e^x w = e^x + c.$$

Therefore $w = u' = 1 + ce^{-x}$ and $u = x + c_1 e^{-x} + c_2$. The general solution is

$$y = u = x + c_1 e^{-x} + c_2.$$

19. Define $y = u(x)e^x$ so

$$y' = ue^x + u'e^x, \quad y'' = u''e^x + 2u'e^x + ue^x$$

and

$$y'' - 3y' + 2y = e^x u'' - e^x u' = 5e^{3x}.$$

If $w = u'$ we obtain the linear first-order equation $w' - w = 5e^{2x}$ which has the integrating factor $e^{-\int dx} = e^{-x}$. Now

$$\frac{d}{dx}[e^{-x} w] = 5e^x \quad \text{gives} \quad e^{-x} w = 5e^x + c_1.$$

Therefore $w = u' = 5e^{2x} + c_1 e^x$ and $u = \frac{5}{2}e^{2x} + c_1 e^x + c_2$. The general solution is

$$y = ue^x = \frac{5}{2}e^{3x} + c_1 e^{2x} + c_2 e^x.$$

20. Define $y = u(x)e^x$ so

$$y' = ue^x + u'e^x, \quad y'' = u''e^x + 2u'e^x + ue^x$$

and

$$y'' - 4y' + 3y = e^x u'' - e^x u' = x.$$

If $w = u'$ we obtain the linear first-order equation $w' - 2w = xe^{-x}$ which has the integrating factor $e^{-\int 2dx} = e^{-2x}$. Now

$$\frac{d}{dx}[e^{-2x} w] = xe^{-3x} \quad \text{gives} \quad e^{-2x} w = -\frac{1}{3}xe^{-3x} - \frac{1}{9}e^{-3x} + c_1.$$

Therefore $w = u' = -\frac{1}{3}xe^{-x} - \frac{1}{9}e^{-x} + c_1 e^{2x}$ and $u = \frac{1}{3}xe^{-x} + \frac{4}{9}e^{-x} + c_2 e^{2x} + c_3$. The general solution is

$$y = ue^x = \frac{1}{3}x + \frac{4}{9} + c_2 e^{3x} + c_3 e^x.$$

21. (a) For m_1 constant, let $y_1 = e^{m_1 x}$. Then $y_1' = m_1 e^{m_1 x}$ and $y_1'' = m_1^2 e^{m_1 x}$. Substituting into the differential equation we obtain

$$\begin{aligned} ay_1'' + by_1' + cy_1 &= am_1^2 e^{m_1 x} + bm_1 e^{m_1 x} + ce^{m_1 x} \\ &= e^{m_1 x}(am_1^2 + bm_1 + c) = 0. \end{aligned}$$

Thus, $y_1 = e^{m_1 x}$ will be a solution of the differential equation whenever $am_1^2 + bm_1 + c = 0$. Since a quadratic equation always has at least one real or complex root, the differential equation must have a solution of the form $y_1 = e^{m_1 x}$.

(b) Write the differential equation in the form

$$y'' + \frac{b}{a}y' + \frac{c}{a}y = 0,$$

3.2 Reduction of Order

and let $y_1 = e^{m_1 x}$ be a solution. Then a second solution is given by

$$\begin{aligned} y_2 &= e^{m_1 x} \int \frac{e^{-bx/a}}{e^{2m_1 x}} dx \\ &= e^{m_1 x} \int e^{-(b/a+2m_1)x} dx \\ &= -\frac{1}{b/a+2m_1} e^{m_1 x} e^{-(b/a+2m_1)x} \quad (m_1 \neq -b/2a) \\ &= -\frac{1}{b/a+2m_1} e^{-(b/a+m_1)x}. \end{aligned}$$

Thus, when $m_1 \neq -b/2a$, a second solution is given by $y_2 = e^{m_2 x}$ where $m_2 = -b/a - m_1$. When $m_1 = -b/2a$ a second solution is given by

$$y_2 = e^{m_1 x} \int dx = x e^{m_1 x}.$$

(c) The functions

$$\begin{aligned} \sin x &= \frac{1}{2i}(e^{ix} - e^{-ix}) & \cos x &= \frac{1}{2}(e^{ix} + e^{-ix}) \\ \sinh x &= \frac{1}{2}(e^x - e^{-x}) & \cosh x &= \frac{1}{2}(e^x + e^{-x}) \end{aligned}$$

are all expressible in terms of exponential functions.

- 22.** We have $y'_1 = 1$ and $y''_1 = 0$, so $xy''_1 - xy'_1 + y_1 = 0 - x + x = 0$ and $y_1(x) = x$ is a solution of the differential equation. Letting $y = u(x)y_1(x) = xu(x)$ we get

$$y' = xu'(x) + u(x) \quad \text{and} \quad y'' = xu''(x) + 2u'(x).$$

Then $xy'' - xy' + y = x^2u'' + 2xu' - x^2u' - xu + xu = x^2u'' - (x^2 - 2x)u' = 0$. If we make the substitution $w = u'$, the linear first-order differential equation becomes $x^2w' - (x^2 - x)w = 0$, which is separable:

$$\begin{aligned} \frac{dw}{dx} &= \left(1 - \frac{1}{x}\right)w \\ \frac{dw}{w} &= \left(1 - \frac{1}{x}\right)dx \\ \ln w &= x - \ln x + c \\ w &= c_1 \frac{e^x}{x}. \end{aligned}$$

Then $u' = c_1 e^x/x$ and $u = c_1 \int e^x dx/x$. To integrate e^x/x we use the series representation for e^x . Thus, a second solution is

$$\begin{aligned} y_2 &= xu(x) = c_1 x \int \frac{e^x}{x} dx \\ &= c_1 x \int \frac{1}{x} \left(1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots\right) dx \\ &= c_1 x \int \left(\frac{1}{x} + 1 + \frac{1}{2!}x + \frac{1}{3!}x^2 + \cdots\right) dx \\ &= c_1 x \left(\ln x + x + \frac{1}{2(2!)}x^2 + \frac{1}{3(3!)}x^3 + \cdots\right) \\ &= c_1 \left(x \ln x + x^2 + \frac{1}{2(2!)}x^3 + \frac{1}{3(3!)}x^4 + \cdots\right). \end{aligned}$$

An interval of definition is probably $(0, \infty)$ because of the $\ln x$ term.

3.3 Homogeneous Linear Equations with Constant Coefficients

23. (a) We have $y' = y'' = e^x$, so

$$xy'' - (x+10)y' + 10y = xe^x - (x+10)e^x + 10e^x = 0,$$

and $y = e^x$ is a solution of the differential equation.

(b) By (5) in the text a second solution is

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx = e^x \int \frac{e^{\int \frac{x+10}{x} dx}}{e^{2x}} dx = e^x \int \frac{e^{\int (1+10/x) dx}}{e^{2x}} dx \\ &= e^x \int \frac{e^{x+\ln x^{10}}}{e^{2x}} dx = e^x \int x^{10} e^{-x} dx \\ &= e^x (-3,628,800 - 3,628,800x - 1,814,400x^2 - 604,800x^3 - 151,200x^4 \\ &\quad - 30,240x^5 - 5,040x^6 - 720x^7 - 90x^8 - 10x^9 - x^{10}) e^{-x} \\ &= -3,628,800 - 3,628,800x - 1,814,400x^2 - 604,800x^3 - 151,200x^4 \\ &\quad - 30,240x^5 - 5,040x^6 - 720x^7 - 90x^8 - 10x^9 - x^{10}. \end{aligned}$$

(c) By Corollary (A) of Theorem 3.2, $-\frac{1}{10!} y_2 = \sum_{n=0}^{10} \frac{1}{n!} x^n$ is a solution.

EXERCISES 3.3

Homogeneous Linear Equations with Constant Coefficients

- From $4m^2 + m = 0$ we obtain $m = 0$ and $m = -1/4$ so that $y = c_1 + c_2 e^{-x/4}$.
- From $m^2 - 36 = 0$ we obtain $m = 6$ and $m = -6$ so that $y = c_1 e^{6x} + c_2 e^{-6x}$.
- From $m^2 - m - 6 = 0$ we obtain $m = 3$ and $m = -2$ so that $y = c_1 e^{3x} + c_2 e^{-2x}$.
- From $m^2 - 3m + 2 = 0$ we obtain $m = 1$ and $m = 2$ so that $y = c_1 e^x + c_2 e^{2x}$.
- From $m^2 + 8m + 16 = 0$ we obtain $m = -4$ and $m = -4$ so that $y = c_1 e^{-4x} + c_2 x e^{-4x}$.
- From $m^2 - 10m + 25 = 0$ we obtain $m = 5$ and $m = 5$ so that $y = c_1 e^{5x} + c_2 x e^{5x}$.
- From $12m^2 - 5m - 2 = 0$ we obtain $m = -1/4$ and $m = 2/3$ so that $y = c_1 e^{-x/4} + c_2 e^{2x/3}$.
- From $m^2 + 4m - 1 = 0$ we obtain $m = -2 \pm \sqrt{5}$ so that $y = c_1 e^{(-2+\sqrt{5})x} + c_2 e^{(-2-\sqrt{5})x}$.
- From $m^2 + 9 = 0$ we obtain $m = 3i$ and $m = -3i$ so that $y = c_1 \cos 3x + c_2 \sin 3x$.
- From $3m^2 + 1 = 0$ we obtain $m = i/\sqrt{3}$ and $m = -i/\sqrt{3}$ so that $y = c_1 \cos(x/\sqrt{3}) + c_2 (\sin x/\sqrt{3})$.
- From $m^2 - 4m + 5 = 0$ we obtain $m = 2 \pm i$ so that $y = e^{2x}(c_1 \cos x + c_2 \sin x)$.
- From $2m^2 + 2m + 1 = 0$ we obtain $m = -1/2 \pm i/2$ so that

$$y = e^{-x/2}[c_1 \cos(x/2) + c_2 \sin(x/2)].$$

- From $3m^2 + 2m + 1 = 0$ we obtain $m = -1/3 \pm \sqrt{2}i/3$ so that

$$y = e^{-x/3}[c_1 \cos(\sqrt{2}x/3) + c_2 \sin(\sqrt{2}x/3)].$$

3.3 Homogeneous Linear Equations with Constant Coefficients

14. From $2m^2 - 3m + 4 = 0$ we obtain $m = 3/4 \pm \sqrt{23}i/4$ so that

$$y = e^{3x/4}[c_1 \cos(\sqrt{23}x/4) + c_2 \sin(\sqrt{23}x/4)].$$

15. From $m^3 - 4m^2 - 5m = 0$ we obtain $m = 0$, $m = 5$, and $m = -1$ so that

$$y = c_1 + c_2 e^{5x} + c_3 e^{-x}.$$

16. From $m^3 - 1 = 0$ we obtain $m = 1$ and $m = -1/2 \pm \sqrt{3}i/2$ so that

$$y = c_1 e^x + e^{-x/2}[c_2 \cos(\sqrt{3}x/2) + c_3 \sin(\sqrt{3}x/2)].$$

17. From $m^3 - 5m^2 + 3m + 9 = 0$ we obtain $m = -1$, $m = 3$, and $m = 3$ so that

$$y = c_1 e^{-x} + c_2 e^{3x} + c_3 x e^{3x}.$$

18. From $m^3 + 3m^2 - 4m - 12 = 0$ we obtain $m = -2$, $m = 2$, and $m = -3$ so that

$$y = c_1 e^{-2x} + c_2 e^{2x} + c_3 e^{-3x}.$$

19. From $m^3 + m^2 - 2 = 0$ we obtain $m = 1$ and $m = -1 \pm i$ so that

$$u = c_1 e^t + e^{-t}(c_2 \cos t + c_3 \sin t).$$

20. From $m^3 - m^2 - 4 = 0$ we obtain $m = 2$ and $m = -1/2 \pm \sqrt{7}i/2$ so that

$$x = c_1 e^{2t} + e^{-t/2}[c_2 \cos(\sqrt{7}t/2) + c_3 \sin(\sqrt{7}t/2)].$$

21. From $m^3 + 3m^2 + 3m + 1 = 0$ we obtain $m = -1$, $m = -1$, and $m = -1$ so that

$$y = c_1 e^{-x} + c_2 x e^{-x} + c_3 x^2 e^{-x}.$$

22. From $m^3 - 6m^2 + 12m - 8 = 0$ we obtain $m = 2$, $m = 2$, and $m = 2$ so that

$$y = c_1 e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x}.$$

23. From $m^4 + m^3 + m^2 = 0$ we obtain $m = 0$, $m = 0$, and $m = -1/2 \pm \sqrt{3}i/2$ so that

$$y = c_1 + c_2 x + e^{-x/2}[c_3 \cos(\sqrt{3}x/2) + c_4 \sin(\sqrt{3}x/2)].$$

24. From $m^4 - 2m^2 + 1 = 0$ we obtain $m = 1$, $m = 1$, $m = -1$, and $m = -1$ so that

$$y = c_1 e^x + c_2 x e^x + c_3 e^{-x} + c_4 x e^{-x}.$$

25. From $16m^4 + 24m^2 + 9 = 0$ we obtain $m = \pm\sqrt{3}i/2$ and $m = \pm\sqrt{3}i/2$ so that

$$y = c_1 \cos(\sqrt{3}x/2) + c_2 \sin(\sqrt{3}x/2) + c_3 x \cos(\sqrt{3}x/2) + c_4 x \sin(\sqrt{3}x/2).$$

26. From $m^4 - 7m^2 - 18 = 0$ we obtain $m = 3$, $m = -3$, and $m = \pm\sqrt{2}i$ so that

$$y = c_1 e^{3x} + c_2 e^{-3x} + c_3 \cos \sqrt{2}x + c_4 \sin \sqrt{2}x.$$

27. From $m^5 + 5m^4 - 2m^3 - 10m^2 + m + 5 = 0$ we obtain $m = -1$, $m = -1$, $m = 1$, and $m = 1$, and $m = -5$ so that

$$u = c_1 e^{-r} + c_2 r e^{-r} + c_3 e^r + c_4 r e^r + c_5 e^{-5r}.$$

28. From $2m^5 - 7m^4 + 12m^3 + 8m^2 = 0$ we obtain $m = 0$, $m = 0$, $m = -1/2$, and $m = 2 \pm 2i$ so that

$$x = c_1 + c_2 s + c_3 e^{-s/2} + e^{2s}(c_4 \cos 2s + c_5 \sin 2s).$$

3.3 Homogeneous Linear Equations with Constant Coefficients

29. From $m^2 + 16 = 0$ we obtain $m = \pm 4i$ so that $y = c_1 \cos 4x + c_2 \sin 4x$. If $y(0) = 2$ and $y'(0) = -2$ then $c_1 = 2$, $c_2 = -1/2$, and $y = 2 \cos 4x - \frac{1}{2} \sin 4x$.

30. From $m^2 + 1 = 0$ we obtain $m = \pm i$ so that $y = c_1 \cos \theta + c_2 \sin \theta$. If $y(\pi/3) = 0$ and $y'(\pi/3) = 2$ then

$$\begin{aligned}\frac{1}{2}c_1 + \frac{\sqrt{3}}{2}c_2 &= 0 \\ -\frac{\sqrt{3}}{2}c_1 + \frac{1}{2}c_2 &= 2,\end{aligned}$$

so $c_1 = -\sqrt{3}$, $c_2 = 1$, and $y = -\sqrt{3} \cos \theta + \sin \theta$.

31. From $m^2 - 4m - 5 = 0$ we obtain $m = -1$ and $m = 5$, so that $y = c_1 e^{-x} + c_2 e^{5x}$. If $y(1) = 0$ and $y'(1) = 2$, then $c_1 e^{-1} + c_2 e^5 = 0$, $-c_1 e^{-1} + 5c_2 e^5 = 2$, so $c_1 = -e/3$, $c_2 = e^{-5}/3$, and $y = -\frac{1}{3}e^{1-x} + \frac{1}{3}e^{5x-5}$.

32. From $4m^2 - 4m - 3 = 0$ we obtain $m = -1/2$ and $m = 3/2$ so that $y = c_1 e^{-x/2} + c_2 e^{3x/2}$. If $y(0) = 1$ and $y'(0) = 5$ then $c_1 + c_2 = 1$, $-\frac{1}{2}c_1 + \frac{3}{2}c_2 = 5$, so $c_1 = -7/4$, $c_2 = 11/4$, and $y = -\frac{7}{4}e^{-x/2} + \frac{11}{4}e^{3x/2}$.

33. From $m^2 + m + 2 = 0$ we obtain $m = -1/2 \pm \sqrt{7}i/2$ so that $y = e^{-x/2}[c_1 \cos(\sqrt{7}x/2) + c_2 \sin(\sqrt{7}x/2)]$. If $y(0) = 0$ and $y'(0) = 0$ then $c_1 = 0$ and $c_2 = 0$ so that $y = 0$.

34. From $m^2 - 2m + 1 = 0$ we obtain $m = 1$ and $m = 1$ so that $y = c_1 e^x + c_2 x e^x$. If $y(0) = 5$ and $y'(0) = 10$ then $c_1 = 5$, $c_1 + c_2 = 10$ so $c_1 = 5$, $c_2 = 5$, and $y = 5e^x + 5xe^x$.

35. From $m^3 + 12m^2 + 36m = 0$ we obtain $m = 0$, $m = -6$, and $m = -6$ so that $y = c_1 + c_2 e^{-6x} + c_3 x e^{-6x}$. If $y(0) = 0$, $y'(0) = 1$, and $y''(0) = -7$ then

$$c_1 + c_2 = 0, \quad -6c_2 + c_3 = 1, \quad 36c_2 - 12c_3 = -7,$$

so $c_1 = 5/36$, $c_2 = -5/36$, $c_3 = 1/6$, and $y = \frac{5}{36} - \frac{5}{36}e^{-6x} + \frac{1}{6}xe^{-6x}$.

36. From $m^3 + 2m^2 - 5m - 6 = 0$ we obtain $m = -1$, $m = 2$, and $m = -3$ so that

$$y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{-3x}.$$

If $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 1$ then

$$c_1 + c_2 + c_3 = 0, \quad -c_1 + 2c_2 - 3c_3 = 0, \quad c_1 + 4c_2 + 9c_3 = 1,$$

so $c_1 = -1/6$, $c_2 = 1/15$, $c_3 = 1/10$, and

$$y = -\frac{1}{6}e^{-x} + \frac{1}{15}e^{2x} + \frac{1}{10}e^{-3x}.$$

37. From $m^2 - 10m + 25 = 0$ we obtain $m = 5$ and $m = 5$ so that $y = c_1 e^{5x} + c_2 x e^{5x}$. If $y(0) = 1$ and $y(1) = 0$ then $c_1 = 1$, $c_1 e^5 + c_2 e^5 = 0$, so $c_1 = 1$, $c_2 = -1$, and $y = e^{5x} - x e^{5x}$.

38. From $m^2 + 4 = 0$ we obtain $m = \pm 2i$ so that $y = c_1 \cos 2x + c_2 \sin 2x$. If $y(0) = 0$ and $y(\pi) = 0$ then $c_1 = 0$ and $y = c_2 \sin 2x$.

39. From $m^2 + 1 = 0$ we obtain $m = \pm i$ so that $y = c_1 \cos x + c_2 \sin x$ and $y' = -c_1 \sin x + c_2 \cos x$. From $y'(0) = c_1(0) + c_2(1) = c_2 = 0$ and $y'(\pi/2) = -c_1(1) = 0$ we find $c_1 = c_2 = 0$. A solution of the boundary-value problem is $y = 0$.

40. From $m^2 - 2m + 2 = 0$ we obtain $m = 1 \pm i$ so that $y = e^x(c_1 \cos x + c_2 \sin x)$. If $y(0) = 1$ and $y(\pi) = 1$ then $c_1 = 1$ and $y(\pi) = e^\pi \cos \pi = -e^\pi$. Since $-e^\pi \neq 1$, the boundary-value problem has no solution.

41. The auxiliary equation is $m^2 - 3 = 0$ which has roots $-\sqrt{3}$ and $\sqrt{3}$. By (10) the general solution is $y = c_1 e^{\sqrt{3}x} + c_2 e^{-\sqrt{3}x}$. By (11) the general solution is $y = c_1 \cosh \sqrt{3}x + c_2 \sinh \sqrt{3}x$. For $y = c_1 e^{\sqrt{3}x} + c_2 e^{-\sqrt{3}x}$ the

3.3 Homogeneous Linear Equations with Constant Coefficients

initial conditions imply $c_1 + c_2 = 1$, $\sqrt{3}c_1 - \sqrt{3}c_2 = 5$. Solving for c_1 and c_2 we find $c_1 = \frac{1}{2}(1 + 5\sqrt{3})$ and $c_2 = \frac{1}{2}(1 - 5\sqrt{3})$ so $y = \frac{1}{2}(1 + 5\sqrt{3})e^{\sqrt{3}x} + \frac{1}{2}(1 - 5\sqrt{3})e^{-\sqrt{3}x}$. For $y = c_1 \cosh \sqrt{3}x + c_2 \sinh \sqrt{3}x$ the initial conditions imply $c_1 = 1$, $\sqrt{3}c_2 = 5$. Solving for c_1 and c_2 we find $c_1 = 1$ and $c_2 = \frac{5}{3}\sqrt{3}$ so $y = \cosh \sqrt{3}x + \frac{5}{3}\sqrt{3} \sinh \sqrt{3}x$.

42. The auxiliary equation is $m^2 - 1 = 0$ which has roots -1 and 1 . By (10) the general solution is $y = c_1 e^x + c_2 e^{-x}$. By (11) the general solution is $y = c_1 \cosh x + c_2 \sinh x$. For $y = c_1 e^x + c_2 e^{-x}$ the boundary conditions imply $c_1 + c_2 = 1$, $c_1 e - c_2 e^{-1} = 0$. Solving for c_1 and c_2 we find $c_1 = 1/(1 + e^2)$ and $c_2 = e^2/(1 + e^2)$ so $y = e^x/(1 + e^2) + e^2 e^{-x}/(1 + e^2)$. For $y = c_1 \cosh x + c_2 \sinh x$ the boundary conditions imply $c_1 = 1$, $c_2 = -\tanh 1$, so $y = \cosh x - (\tanh 1) \sinh x$.
43. The auxiliary equation should have two positive roots, so that the solution has the form $y = c_1 e^{k_1 x} + c_2 e^{k_2 x}$. Thus, the differential equation is (f).
44. The auxiliary equation should have one positive and one negative root, so that the solution has the form $y = c_1 e^{k_1 x} + c_2 e^{-k_2 x}$. Thus, the differential equation is (a).
45. The auxiliary equation should have a pair of complex roots $\alpha \pm \beta i$ where $\alpha < 0$, so that the solution has the form $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$. Thus, the differential equation is (e).
46. The auxiliary equation should have a repeated negative root, so that the solution has the form $y = c_1 e^{-x} + c_2 x e^{-x}$. Thus, the differential equation is (c).
47. The differential equation should have the form $y'' + k^2 y = 0$ where $k = 1$ so that the period of the solution is 2π . Thus, the differential equation is (d).
48. The differential equation should have the form $y'' + k^2 y = 0$ where $k = 2$ so that the period of the solution is π . Thus, the differential equation is (b).
49. Since $(m - 4)(m + 5)^2 = m^3 + 6m^2 - 15m - 100$ the differential equation is $y''' + 6y'' - 15y' - 100y = 0$. The differential equation is not unique since any constant multiple of the left-hand side of the differential equation would lead to the auxiliary roots.
50. A third root must be $m_3 = 3 - i$ and the auxiliary equation is

$$\left(m + \frac{1}{2}\right)[m - (3 + i)][m - (3 - i)] = \left(m + \frac{1}{2}\right)(m^2 - 6m + 10) = m^3 - \frac{11}{2}m^2 + 7m + 5.$$

The differential equation is

$$y''' - \frac{11}{2}y'' + 7y' + 5y = 0.$$

51. From the solution $y_1 = e^{-4x} \cos x$ we conclude that $m_1 = -4 + i$ and $m_2 = -4 - i$ are roots of the auxiliary equation. Hence another solution must be $y_2 = e^{-4x} \sin x$. Now dividing the polynomial $m^3 + 6m^2 + m - 34$ by $[m - (-4 + i)][m - (-4 - i)] = m^2 + 8m + 17$ gives $m - 2$. Therefore $m_3 = 2$ is the third root of the auxiliary equation, and the general solution of the differential equation is

$$y = c_1 e^{-4x} \cos x + c_2 e^{-4x} \sin x + c_3 e^{2x}.$$

52. Factoring the difference of two squares we obtain

$$m^4 + 1 = (m^2 + 1)^2 - 2m^2 = (m^2 + 1 - \sqrt{2}m)(m^2 + 1 + \sqrt{2}m) = 0.$$

Using the quadratic formula on each factor we get $m = \pm\sqrt{2}/2 \pm \sqrt{2}i/2$. The solution of the differential equation is

$$y(x) = e^{\sqrt{2}x/2} \left(c_1 \cos \frac{\sqrt{2}}{2}x + c_2 \sin \frac{\sqrt{2}}{2}x \right) + e^{-\sqrt{2}x/2} \left(c_3 \cos \frac{\sqrt{2}}{2}x + c_4 \sin \frac{\sqrt{2}}{2}x \right).$$

3.3 Homogeneous Linear Equations with Constant Coefficients

53. Using the definition of $\sinh x$ and the formula for the cosine of the sum of two angles, we have

$$\begin{aligned} y &= \sinh x - 2 \cos(x + \pi/6) \\ &= \frac{1}{2}e^x - \frac{1}{2}e^{-x} - 2 \left[(\cos x) \left(\cos \frac{\pi}{6} \right) - (\sin x) \left(\sin \frac{\pi}{6} \right) \right] \\ &= \frac{1}{2}e^x - \frac{1}{2}e^{-x} - 2 \left(\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x \right) \\ &= \frac{1}{2}e^x - \frac{1}{2}e^{-x} - \sqrt{3} \cos x + \sin x. \end{aligned}$$

This form of the solution can be obtained from the general solution $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$ by choosing $c_1 = \frac{1}{2}$, $c_2 = -\frac{1}{2}$, $c_3 = -\sqrt{3}$, and $c_4 = 1$.

54. The auxiliary equation is $m^2 + \alpha = 0$ and we consider three cases where $\lambda = 0$, $\lambda = \alpha^2 > 0$, and $\lambda = -\alpha^2 < 0$:

Case I When $\alpha = 0$ the general solution of the differential equation is $y = c_1 + c_2 x$. The boundary conditions imply $0 = y(0) = c_1$ and $0 = y(\pi/2) = c_2 \pi/2$, so that $c_1 = c_2 = 0$ and the problem possesses only the trivial solution.

Case II When $\lambda = -\alpha^2 < 0$ the general solution of the differential equation is $y = c_1 e^{\alpha x} + c_2 e^{-\alpha x}$, or alternatively, $y = c_1 \cosh \alpha x + c_2 \sinh \alpha x$. Again, $y(0) = 0$ implies $c_1 = 0$ so $y = c_2 \sinh \alpha x$. The second boundary condition implies $0 = y(\pi/2) = c_2 \sinh \alpha \pi/2$ or $c_2 = 0$. In this case also, the problem possesses only the trivial solution.

Case III When $\lambda = \alpha^2 > 0$ the general solution of the differential equation is $y = c_1 \cos \alpha x + c_2 \sin \alpha x$. In this case also, $y(0) = 0$ yields $c_1 = 0$, so that $y = c_2 \sin \alpha x$. The second boundary condition implies $0 = c_2 \sin \alpha \pi/2$. When $\alpha \pi/2$ is an integer multiple of π , that is, when $\alpha = 2k$ for k a nonzero integer, the problem will have nontrivial solutions. Thus, for $\lambda = \alpha^2 = 4k^2$ the boundary-value problem will have nontrivial solutions $y = c_2 \sin 2kx$, where k is a nonzero integer. On the other hand, when α is not an even integer, the boundary-value problem will have only the trivial solution.

55. Applying integration by parts twice we have

$$\begin{aligned} \int e^{ax} f(x) dx &= \frac{1}{a} e^{ax} f(x) - \frac{1}{a} \int e^{ax} f'(x) dx \\ &= \frac{1}{a} e^{ax} f(x) - \frac{1}{a} \left[\frac{1}{a} e^{ax} f'(x) - \frac{1}{a} \int e^{ax} f''(x) dx \right] \\ &= \frac{1}{a} e^{ax} f(x) - \frac{1}{a^2} e^{ax} f'(x) + \frac{1}{a^2} \int e^{ax} f''(x) dx. \end{aligned}$$

Collecting the integrals we get

$$\int e^{ax} \left(f(x) - \frac{1}{a^2} f''(x) \right) dx = \frac{1}{a} e^{ax} f(x) - \frac{1}{a^2} e^{ax} f'(x).$$

In order for the technique to work we need to have

$$\int e^{ax} \left(f(x) - \frac{1}{a^2} f''(x) \right) dx = k \int e^{ax} f(x) dx$$

or

$$f(x) - \frac{1}{a^2} f''(x) = kf(x),$$

where $k \neq 0$. This is the second-order differential equation

$$f''(x) + a^2(k-1)f(x) = 0.$$

3.3 Homogeneous Linear Equations with Constant Coefficients

If $k < 1$, $k \neq 0$, the solution of the differential equation is a pair of exponential functions, in which case the original integrand is an exponential function and does not require integration by parts for its evaluation. Similarly, if $k = 1$, $f''(x) = 0$ and $f(x)$ has the form $f(x) = ax + b$. In this case a single application of integration by parts will suffice. Finally, if $k > 1$, the solution of the differential equation is

$$f(x) = c_1 \cos a\sqrt{k-1}x + c_2 \sin a\sqrt{k-1}x,$$

and we see that the technique will work for linear combinations of $\cos \alpha x$ and $\sin \alpha x$.

56. (a) The auxiliary equation is $m^2 - 64/L = 0$ which has roots $\pm 8/\sqrt{L}$. Thus, the general solution of the differential equation is $x = c_1 \cosh(8t/\sqrt{L}) + c_2 \sinh(8t/\sqrt{L})$.
- (b) Setting $x(0) = x_0$ and $x'(0) = 0$ we have $c_1 = x_0$, $8c_2/\sqrt{L} = 0$. Solving for c_1 and c_2 we get $c_1 = x_0$ and $c_2 = 0$, so $x(t) = x_0 \cosh(8t/\sqrt{L})$.
- (c) When $L = 20$ and $x_0 = 1$, $x(t) = \cosh(4t/\sqrt{5})$. The chain will last touch the peg when $x(t) = 10$. Solving $x(t) = 10$ for t we get $t_1 = \frac{1}{4}\sqrt{5} \cosh^{-1} 10 \approx 1.67326$. The velocity of the chain at this instant is $x'(t_1) = 12\sqrt{11/5} \approx 17.7989$ ft/s.
57. Using a CAS to solve the auxiliary equation $m^3 - 6m^2 + 2m + 1 = 0$ we find $m_1 = -0.270534$, $m_2 = 0.658675$, and $m_3 = 5.61186$. The general solution is

$$y = c_1 e^{-0.270534x} + c_2 e^{0.658675x} + c_3 e^{5.61186x}.$$

58. Using a CAS to solve the auxiliary equation $6.11m^3 + 8.59m^2 + 7.93m + 0.778 = 0$ we find $m_1 = -0.110241$, $m_2 = -0.647826 + 0.857532i$, and $m_3 = -0.647826 - 0.857532i$. The general solution is

$$y = c_1 e^{-0.110241x} + e^{-0.647826x} (c_2 \cos 0.857532x + c_3 \sin 0.857532x).$$

59. Using a CAS to solve the auxiliary equation $3.15m^4 - 5.34m^2 + 6.33m - 2.03 = 0$ we find $m_1 = -1.74806$, $m_2 = 0.501219$, $m_3 = 0.62342 + 0.588965i$, and $m_4 = 0.62342 - 0.588965i$. The general solution is

$$y = c_1 e^{-1.74806x} + c_2 e^{0.501219x} + e^{0.62342x} (c_3 \cos 0.588965x + c_4 \sin 0.588965x).$$

60. Using a CAS to solve the auxiliary equation $m^4 + 2m^2 - m + 2 = 0$ we find $m_1 = 1/2 + \sqrt{3}i/2$, $m_2 = 1/2 - \sqrt{3}i/2$, $m_3 = -1/2 + \sqrt{7}i/2$, and $m_4 = -1/2 - \sqrt{7}i/2$. The general solution is

$$y = e^{x/2} \left(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x \right) + e^{-x/2} \left(c_3 \cos \frac{\sqrt{7}}{2}x + c_4 \sin \frac{\sqrt{7}}{2}x \right).$$

61. From $2m^4 + 3m^3 - 16m^2 + 15m - 4 = 0$ we obtain $m = -4$, $m = \frac{1}{2}$, $m = 1$, and $m = 1$, so that $y = c_1 e^{-4x} + c_2 e^{x/2} + c_3 e^x + c_4 x e^x$. If $y(0) = -2$, $y'(0) = 6$, $y''(0) = 3$, and $y'''(0) = \frac{1}{2}$, then

$$c_1 + c_2 + c_3 = -2$$

$$-4c_1 + \frac{1}{2}c_2 + c_3 + c_4 = 6$$

$$16c_1 + \frac{1}{4}c_2 + c_3 + 2c_4 = 3$$

$$-64c_1 + \frac{1}{8}c_2 + c_3 + 3c_4 = \frac{1}{2},$$

so $c_1 = -\frac{4}{75}$, $c_2 = -\frac{116}{3}$, $c_3 = \frac{918}{25}$, $c_4 = -\frac{58}{5}$, and

$$y = -\frac{4}{75}e^{-4x} - \frac{116}{3}e^{x/2} + \frac{918}{25}e^x - \frac{58}{5}xe^x.$$

62. From $m^4 - 3m^3 + 3m^2 - m = 0$ we obtain $m = 0$, $m = 1$, $m = 1$, and $m = 1$ so that $y = c_1 + c_2e^x + c_3xe^x + c_4x^2e^x$. If $y(0) = 0$, $y'(0) = 0$, $y''(0) = 1$, and $y'''(0) = 1$ then

$$c_1 + c_2 = 0, \quad c_2 + c_3 = 0, \quad c_2 + 2c_3 + 2c_4 = 1, \quad c_2 + 3c_3 + 6c_4 = 1,$$

so $c_1 = 2$, $c_2 = -2$, $c_3 = 2$, $c_4 = -1/2$, and

$$y = 2 - 2e^x + 2xe^x - \frac{1}{2}x^2e^x.$$

EXERCISES 3.4

Undetermined Coefficients

1. From $m^2 + 3m + 2 = 0$ we find $m_1 = -1$ and $m_2 = -2$. Then $y_c = c_1e^{-x} + c_2e^{-2x}$ and we assume $y_p = A$. Substituting into the differential equation we obtain $2A = 6$. Then $A = 3$, $y_p = 3$ and

$$y = c_1e^{-x} + c_2e^{-2x} + 3.$$

2. From $4m^2 + 9 = 0$ we find $m_1 = -\frac{3}{2}i$ and $m_2 = \frac{3}{2}i$. Then $y_c = c_1 \cos \frac{3}{2}x + c_2 \sin \frac{3}{2}x$ and we assume $y_p = A$. Substituting into the differential equation we obtain $9A = 15$. Then $A = \frac{5}{3}$, $y_p = \frac{5}{3}$ and

$$y = c_1 \cos \frac{3}{2}x + c_2 \sin \frac{3}{2}x + \frac{5}{3}.$$

3. From $m^2 - 10m + 25 = 0$ we find $m_1 = m_2 = 5$. Then $y_c = c_1e^{5x} + c_2xe^{5x}$ and we assume $y_p = Ax + B$. Substituting into the differential equation we obtain $25A = 30$ and $-10A + 25B = 3$. Then $A = \frac{6}{5}$, $B = \frac{6}{5}$, $y_p = \frac{6}{5}x + \frac{6}{5}$, and

$$y = c_1e^{5x} + c_2xe^{5x} + \frac{6}{5}x + \frac{6}{5}.$$

4. From $m^2 + m - 6 = 0$ we find $m_1 = -3$ and $m_2 = 2$. Then $y_c = c_1e^{-3x} + c_2e^{2x}$ and we assume $y_p = Ax + B$. Substituting into the differential equation we obtain $-6A = 2$ and $A - 6B = 0$. Then $A = -\frac{1}{3}$, $B = -\frac{1}{18}$, $y_p = -\frac{1}{3}x - \frac{1}{18}$, and

$$y = c_1e^{-3x} + c_2e^{2x} - \frac{1}{3}x - \frac{1}{18}.$$

5. From $\frac{1}{4}m^2 + m + 1 = 0$ we find $m_1 = m_2 = -2$. Then $y_c = c_1e^{-2x} + c_2xe^{-2x}$ and we assume $y_p = Ax^2 + Bx + C$. Substituting into the differential equation we obtain $A = 1$, $2A + B = -2$, and $\frac{1}{2}A + B + C = 0$. Then $A = 1$, $B = -4$, $C = \frac{7}{2}$, $y_p = x^2 - 4x + \frac{7}{2}$, and

$$y = c_1e^{-2x} + c_2xe^{-2x} + x^2 - 4x + \frac{7}{2}.$$

6. From $m^2 - 8m + 20 = 0$ we find $m_1 = 4 + 2i$ and $m_2 = 4 - 2i$. Then $y_c = e^{4x}(c_1 \cos 2x + c_2 \sin 2x)$ and we assume $y_p = Ax^2 + Bx + C + (Dx + E)e^x$. Substituting into the differential equation we obtain

$$2A - 8B + 20C = 0$$

$$-6D + 13E = 0$$

$$-16A + 20B = 0$$

$$13D = -26$$

$$20A = 100.$$

3.4 Undetermined Coefficients

Then $A = 5$, $B = 4$, $C = \frac{11}{10}$, $D = -2$, $E = -\frac{12}{13}$, $y_p = 5x^2 + 4x + \frac{11}{10} + (-2x - \frac{12}{13})e^x$ and

$$y = e^{4x}(c_1 \cos 2x + c_2 \sin 2x) + 5x^2 + 4x + \frac{11}{10} + \left(-2x - \frac{12}{13}\right)e^x.$$

7. From $m^2 + 3 = 0$ we find $m_1 = \sqrt{3}i$ and $m_2 = -\sqrt{3}i$. Then $y_c = c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x$ and we assume $y_p = (Ax^2 + Bx + C)e^{3x}$. Substituting into the differential equation we obtain $2A + 6B + 12C = 0$, $12A + 12B = 0$, and $12A = -48$. Then $A = -4$, $B = 4$, $C = -\frac{4}{3}$, $y_p = (-4x^2 + 4x - \frac{4}{3})e^{3x}$ and

$$y = c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x + \left(-4x^2 + 4x - \frac{4}{3}\right)e^{3x}.$$

8. From $4m^2 - 4m - 3 = 0$ we find $m_1 = \frac{3}{2}$ and $m_2 = -\frac{1}{2}$. Then $y_c = c_1 e^{3x/2} + c_2 e^{-x/2}$ and we assume $y_p = A \cos 2x + B \sin 2x$. Substituting into the differential equation we obtain $-19 - 8B = 1$ and $8A - 19B = 0$. Then $A = -\frac{19}{425}$, $B = -\frac{8}{425}$, $y_p = -\frac{19}{425} \cos 2x - \frac{8}{425} \sin 2x$, and

$$y = c_1 e^{3x/2} + c_2 e^{-x/2} - \frac{19}{425} \cos 2x - \frac{8}{425} \sin 2x.$$

9. From $m^2 - m = 0$ we find $m_1 = 1$ and $m_2 = 0$. Then $y_c = c_1 e^x + c_2$ and we assume $y_p = Ax$. Substituting into the differential equation we obtain $-A = -3$. Then $A = 3$, $y_p = 3x$ and $y = c_1 e^x + c_2 + 3x$.

10. From $m^2 + 2m = 0$ we find $m_1 = -2$ and $m_2 = 0$. Then $y_c = c_1 e^{-2x} + c_2$ and we assume $y_p = Ax^2 + Bx + Cx e^{-2x}$. Substituting into the differential equation we obtain $2A + 2B = 5$, $4A = 2$, and $-2C = -1$. Then $A = \frac{1}{2}$, $B = 2$, $C = \frac{1}{2}$, $y_p = \frac{1}{2}x^2 + 2x + \frac{1}{2}x e^{-2x}$, and

$$y = c_1 e^{-2x} + c_2 + \frac{1}{2}x^2 + 2x + \frac{1}{2}x e^{-2x}.$$

11. From $m^2 - m + \frac{1}{4} = 0$ we find $m_1 = m_2 = \frac{1}{2}$. Then $y_c = c_1 e^{x/2} + c_2 x e^{x/2}$ and we assume $y_p = A + Bx^2 e^{x/2}$. Substituting into the differential equation we obtain $\frac{1}{4}A = 3$ and $2B = 1$. Then $A = 12$, $B = \frac{1}{2}$, $y_p = 12 + \frac{1}{2}x^2 e^{x/2}$, and

$$y = c_1 e^{x/2} + c_2 x e^{x/2} + 12 + \frac{1}{2}x^2 e^{x/2}.$$

12. From $m^2 - 16 = 0$ we find $m_1 = 4$ and $m_2 = -4$. Then $y_c = c_1 e^{4x} + c_2 e^{-4x}$ and we assume $y_p = A x e^{4x}$. Substituting into the differential equation we obtain $8A = 2$. Then $A = \frac{1}{4}$, $y_p = \frac{1}{4}x e^{4x}$ and

$$y = c_1 e^{4x} + c_2 e^{-4x} + \frac{1}{4}x e^{4x}.$$

13. From $m^2 + 4 = 0$ we find $m_1 = 2i$ and $m_2 = -2i$. Then $y_c = c_1 \cos 2x + c_2 \sin 2x$ and we assume $y_p = Ax \cos 2x + Bx \sin 2x$. Substituting into the differential equation we obtain $4B = 0$ and $-4A = 3$. Then $A = -\frac{3}{4}$, $B = 0$, $y_p = -\frac{3}{4}x \cos 2x$, and

$$y = c_1 \cos 2x + c_2 \sin 2x - \frac{3}{4}x \cos 2x.$$

14. From $m^2 - 4 = 0$ we find $m_1 = 2$ and $m_2 = -2$. Then $y_c = c_1 e^{2x} + c_2 e^{-2x}$ and we assume that $y_p = (Ax^2 + Bx + C) \cos 2x + (Dx^2 + Ex + F) \sin 2x$. Substituting into the differential equation we obtain

$$-8A = 0$$

$$-8B + 8D = 0$$

$$2A - 8C + 4E = 0$$

$$-8D = 1$$

$$-8A - 8E = 0$$

$$-4B + 2D - 8F = -3.$$

3.4 Undetermined Coefficients

Then $A = 0$, $B = -\frac{1}{8}$, $C = 0$, $D = -\frac{1}{8}$, $E = 0$, $F = \frac{13}{32}$, so $y_p = -\frac{1}{8}x \cos 2x + \left(-\frac{1}{8}x^2 + \frac{13}{32}\right) \sin 2x$, and

$$y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{8}x \cos 2x + \left(-\frac{1}{8}x^2 + \frac{13}{32}\right) \sin 2x.$$

15. From $m^2 + 1 = 0$ we find $m_1 = i$ and $m_2 = -i$. Then $y_c = c_1 \cos x + c_2 \sin x$ and we assume $y_p = (Ax^2 + Bx) \cos x + (Cx^2 + Dx) \sin x$. Substituting into the differential equation we obtain $4C = 0$, $2A + 2D = 0$, $-4A = 2$, and $-2B + 2C = 0$. Then $A = -\frac{1}{2}$, $B = 0$, $C = 0$, $D = \frac{1}{2}$, $y_p = -\frac{1}{2}x^2 \cos x + \frac{1}{2}x \sin x$, and

$$y = c_1 \cos x + c_2 \sin x - \frac{1}{2}x^2 \cos x + \frac{1}{2}x \sin x.$$

16. From $m^2 - 5m = 0$ we find $m_1 = 5$ and $m_2 = 0$. Then $y_c = c_1 e^{5x} + c_2$ and we assume $y_p = Ax^4 + Bx^3 + Cx^2 + Dx$. Substituting into the differential equation we obtain $-20A = 2$, $12A - 15B = -4$, $6B - 10C = -1$, and $2C - 5D = 6$. Then $A = -\frac{1}{10}$, $B = \frac{14}{75}$, $C = \frac{53}{250}$, $D = -\frac{697}{625}$, $y_p = -\frac{1}{10}x^4 + \frac{14}{75}x^3 + \frac{53}{250}x^2 - \frac{697}{625}x$, and

$$y = c_1 e^{5x} + c_2 - \frac{1}{10}x^4 + \frac{14}{75}x^3 + \frac{53}{250}x^2 - \frac{697}{625}x.$$

17. From $m^2 - 2m + 5 = 0$ we find $m_1 = 1 + 2i$ and $m_2 = 1 - 2i$. Then $y_c = e^x(c_1 \cos 2x + c_2 \sin 2x)$ and we assume $y_p = Axe^x \cos 2x + Bxe^x \sin 2x$. Substituting into the differential equation we obtain $4B = 1$ and $-4A = 0$. Then $A = 0$, $B = \frac{1}{4}$, $y_p = \frac{1}{4}xe^x \sin 2x$, and

$$y = e^x(c_1 \cos 2x + c_2 \sin 2x) + \frac{1}{4}xe^x \sin 2x.$$

18. From $m^2 - 2m + 2 = 0$ we find $m_1 = 1 + i$ and $m_2 = 1 - i$. Then $y_c = e^x(c_1 \cos x + c_2 \sin x)$ and we assume $y_p = Ae^{2x} \cos x + Be^{2x} \sin x$. Substituting into the differential equation we obtain $A + 2B = 1$ and $-2A + B = -3$. Then $A = \frac{7}{5}$, $B = -\frac{1}{5}$, $y_p = \frac{7}{5}e^{2x} \cos x - \frac{1}{5}e^{2x} \sin x$ and

$$y = e^x(c_1 \cos x + c_2 \sin x) + \frac{7}{5}e^{2x} \cos x - \frac{1}{5}e^{2x} \sin x.$$

19. From $m^2 + 2m + 1 = 0$ we find $m_1 = m_2 = -1$. Then $y_c = c_1 e^{-x} + c_2 x e^{-x}$ and we assume $y_p = A \cos x + B \sin x + C \cos 2x + D \sin 2x$. Substituting into the differential equation we obtain $2B = 0$, $-2A = 1$, $-3C + 4D = 3$, and $-4C - 3D = 0$. Then $A = -\frac{1}{2}$, $B = 0$, $C = -\frac{9}{25}$, $D = \frac{12}{25}$, $y_p = -\frac{1}{2} \cos x - \frac{9}{25} \cos 2x + \frac{12}{25} \sin 2x$, and

$$y = c_1 e^{-x} + c_2 x e^{-x} - \frac{1}{2} \cos x - \frac{9}{25} \cos 2x + \frac{12}{25} \sin 2x.$$

20. From $m^2 + 2m - 24 = 0$ we find $m_1 = -6$ and $m_2 = 4$. Then $y_c = c_1 e^{-6x} + c_2 e^{4x}$ and we assume $y_p = A + (Bx^2 + Cx)e^{4x}$. Substituting into the differential equation we obtain $-24A = 16$, $2B + 10C = -2$, and $20B = -1$. Then $A = -\frac{2}{3}$, $B = -\frac{1}{20}$, $C = -\frac{19}{100}$, $y_p = -\frac{2}{3} - \left(\frac{1}{20}x^2 + \frac{19}{100}x\right)e^{4x}$, and

$$y = c_1 e^{-6x} + c_2 e^{4x} - \frac{2}{3} - \left(\frac{1}{20}x^2 + \frac{19}{100}x\right)e^{4x}.$$

21. From $m^3 - 6m^2 = 0$ we find $m_1 = m_2 = 0$ and $m_3 = 6$. Then $y_c = c_1 + c_2 x + c_3 e^{6x}$ and we assume $y_p = Ax^2 + B \cos x + C \sin x$. Substituting into the differential equation we obtain $-12A = 3$, $6B - C = -1$, and $B + 6C = 0$. Then $A = -\frac{1}{4}$, $B = -\frac{6}{37}$, $C = \frac{1}{37}$, $y_p = -\frac{1}{4}x^2 - \frac{6}{37} \cos x + \frac{1}{37} \sin x$, and

$$y = c_1 + c_2 x + c_3 e^{6x} - \frac{1}{4}x^2 - \frac{6}{37} \cos x + \frac{1}{37} \sin x.$$

3.4 Undetermined Coefficients

22. From $m^3 - 2m^2 - 4m + 8 = 0$ we find $m_1 = m_2 = 2$ and $m_3 = -2$. Then $y_c = c_1e^{2x} + c_2xe^{2x} + c_3e^{-2x}$ and we assume $y_p = (Ax^3 + Bx^2)e^{2x}$. Substituting into the differential equation we obtain $24A = 6$ and $6A + 8B = 0$. Then $A = \frac{1}{4}$, $B = -\frac{3}{16}$, $y_p = \left(\frac{1}{4}x^3 - \frac{3}{16}x^2\right)e^{2x}$, and

$$y = c_1e^{2x} + c_2xe^{2x} + c_3e^{-2x} + \left(\frac{1}{4}x^3 - \frac{3}{16}x^2\right)e^{2x}.$$

23. From $m^3 - 3m^2 + 3m - 1 = 0$ we find $m_1 = m_2 = m_3 = 1$. Then $y_c = c_1e^x + c_2xe^x + c_3x^2e^x$ and we assume $y_p = Ax + B + Cx^3e^x$. Substituting into the differential equation we obtain $-A = 1$, $3A - B = 0$, and $6C = -4$. Then $A = -1$, $B = -3$, $C = -\frac{2}{3}$, $y_p = -x - 3 - \frac{2}{3}x^3e^x$, and

$$y = c_1e^x + c_2xe^x + c_3x^2e^x - x - 3 - \frac{2}{3}x^3e^x.$$

24. From $m^3 - m^2 - 4m + 4 = 0$ we find $m_1 = 1$, $m_2 = 2$, and $m_3 = -2$. Then $y_c = c_1e^x + c_2e^{2x} + c_3e^{-2x}$ and we assume $y_p = A + Bxe^x + Cxe^{2x}$. Substituting into the differential equation we obtain $4A = 5$, $-3B = -1$, and $4C = 1$. Then $A = \frac{5}{4}$, $B = \frac{1}{3}$, $C = \frac{1}{4}$, $y_p = \frac{5}{4} + \frac{1}{3}xe^x + \frac{1}{4}xe^{2x}$, and

$$y = c_1e^x + c_2e^{2x} + c_3e^{-2x} + \frac{5}{4} + \frac{1}{3}xe^x + \frac{1}{4}xe^{2x}.$$

25. From $m^4 + 2m^2 + 1 = 0$ we find $m_1 = m_3 = i$ and $m_2 = m_4 = -i$. Then $y_c = c_1 \cos x + c_2 \sin x + c_3x \cos x + c_4x \sin x$ and we assume $y_p = Ax^2 + Bx + C$. Substituting into the differential equation we obtain $A = 1$, $B = -2$, and $4A + C = 1$. Then $A = 1$, $B = -2$, $C = -3$, $y_p = x^2 - 2x - 3$, and

$$y = c_1 \cos x + c_2 \sin x + c_3x \cos x + c_4x \sin x + x^2 - 2x - 3.$$

26. From $m^4 - m^2 = 0$ we find $m_1 = m_2 = 0$, $m_3 = 1$, and $m_4 = -1$. Then $y_c = c_1 + c_2x + c_3e^x + c_4e^{-x}$ and we assume $y_p = Ax^3 + Bx^2 + (Cx^2 + Dx)e^{-x}$. Substituting into the differential equation we obtain $-6A = 4$, $-2B = 0$, $10C - 2D = 0$, and $-4C = 2$. Then $A = -\frac{2}{3}$, $B = 0$, $C = -\frac{1}{2}$, $D = -\frac{5}{2}$, $y_p = -\frac{2}{3}x^3 - \left(\frac{1}{2}x^2 + \frac{5}{2}x\right)e^{-x}$, and

$$y = c_1 + c_2x + c_3e^x + c_4e^{-x} - \frac{2}{3}x^3 - \left(\frac{1}{2}x^2 + \frac{5}{2}x\right)e^{-x}.$$

27. We have $y_c = c_1 \cos 2x + c_2 \sin 2x$ and we assume $y_p = A$. Substituting into the differential equation we find $A = -\frac{1}{2}$. Thus $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{2}$. From the initial conditions we obtain $c_1 = 0$ and $c_2 = \sqrt{2}$, so

$$y = \sqrt{2} \sin 2x - \frac{1}{2}.$$

28. We have $y_c = c_1e^{-2x} + c_2e^{x/2}$ and we assume $y_p = Ax^2 + Bx + C$. Substituting into the differential equation we find $A = -7$, $B = -19$, and $C = -37$. Thus $y = c_1e^{-2x} + c_2e^{x/2} - 7x^2 - 19x - 37$. From the initial conditions we obtain $c_1 = -\frac{1}{5}$ and $c_2 = \frac{186}{5}$, so

$$y = -\frac{1}{5}e^{-2x} + \frac{186}{5}e^{x/2} - 7x^2 - 19x - 37.$$

29. We have $y_c = c_1e^{-x/5} + c_2$ and we assume $y_p = Ax^2 + Bx$. Substituting into the differential equation we find $A = -3$ and $B = 30$. Thus $y = c_1e^{-x/5} + c_2 - 3x^2 + 30x$. From the initial conditions we obtain $c_1 = 200$ and $c_2 = -200$, so

$$y = 200e^{-x/5} - 200 - 3x^2 + 30x.$$

30. We have $y_c = c_1 e^{-2x} + c_2 x e^{-2x}$ and we assume $y_p = (Ax^3 + Bx^2)e^{-2x}$. Substituting into the differential equation we find $A = \frac{1}{6}$ and $B = \frac{3}{2}$. Thus $y = c_1 e^{-2x} + c_2 x e^{-2x} + (\frac{1}{6}x^3 + \frac{3}{2}x^2)e^{-2x}$. From the initial conditions we obtain $c_1 = 2$ and $c_2 = 9$, so

$$y = 2e^{-2x} + 9xe^{-2x} + \left(\frac{1}{6}x^3 + \frac{3}{2}x^2\right)e^{-2x}.$$

31. We have $y_c = e^{-2x}(c_1 \cos x + c_2 \sin x)$ and we assume $y_p = Ae^{-4x}$. Substituting into the differential equation we find $A = 7$. Thus $y = e^{-2x}(c_1 \cos x + c_2 \sin x) + 7e^{-4x}$. From the initial conditions we obtain $c_1 = -10$ and $c_2 = 9$, so

$$y = e^{-2x}(-10 \cos x + 9 \sin x) + 7e^{-4x}.$$

32. We have $y_c = c_1 \cosh x + c_2 \sinh x$ and we assume $y_p = Ax \cosh x + Bx \sinh x$. Substituting into the differential equation we find $A = 0$ and $B = \frac{1}{2}$. Thus

$$y = c_1 \cosh x + c_2 \sinh x + \frac{1}{2}x \sinh x.$$

From the initial conditions we obtain $c_1 = 2$ and $c_2 = 12$, so

$$y = 2 \cosh x + 12 \sinh x + \frac{1}{2}x \sinh x.$$

33. We have $x_c = c_1 \cos \omega t + c_2 \sin \omega t$ and we assume $x_p = At \cos \omega t + Bt \sin \omega t$. Substituting into the differential equation we find $A = -F_0/2\omega$ and $B = 0$. Thus $x = c_1 \cos \omega t + c_2 \sin \omega t - (F_0/2\omega)t \cos \omega t$. From the initial conditions we obtain $c_1 = 0$ and $c_2 = F_0/2\omega^2$, so

$$x = (F_0/2\omega^2) \sin \omega t - (F_0/2\omega)t \cos \omega t.$$

34. We have $x_c = c_1 \cos \omega t + c_2 \sin \omega t$ and we assume $x_p = A \cos \gamma t + B \sin \gamma t$, where $\gamma \neq \omega$. Substituting into the differential equation we find $A = F_0/(\omega^2 - \gamma^2)$ and $B = 0$. Thus

$$x = c_1 \cos \omega t + c_2 \sin \omega t + \frac{F_0}{\omega^2 - \gamma^2} \cos \gamma t.$$

From the initial conditions we obtain $c_1 = -F_0/(\omega^2 - \gamma^2)$ and $c_2 = 0$, so

$$x = -\frac{F_0}{\omega^2 - \gamma^2} \cos \omega t + \frac{F_0}{\omega^2 - \gamma^2} \cos \gamma t.$$

35. We have $y_c = c_1 + c_2 e^x + c_3 x e^x$ and we assume $y_p = Ax + Bx^2 e^x + Ce^{5x}$. Substituting into the differential equation we find $A = 2$, $B = -12$, and $C = \frac{1}{2}$. Thus

$$y = c_1 + c_2 e^x + c_3 x e^x + 2x - 12x^2 e^x + \frac{1}{2}e^{5x}.$$

From the initial conditions we obtain $c_1 = 11$, $c_2 = -11$, and $c_3 = 9$, so

$$y = 11 - 11e^x + 9xe^x + 2x - 12x^2 e^x + \frac{1}{2}e^{5x}.$$

36. We have $y_c = c_1 e^{-2x} + e^x(c_2 \cos \sqrt{3}x + c_3 \sin \sqrt{3}x)$ and we assume $y_p = Ax + B + Cxe^{-2x}$. Substituting into the differential equation we find $A = \frac{1}{4}$, $B = -\frac{5}{8}$, and $C = \frac{2}{3}$. Thus

$$y = c_1 e^{-2x} + e^x(c_2 \cos \sqrt{3}x + c_3 \sin \sqrt{3}x) + \frac{1}{4}x - \frac{5}{8} + \frac{2}{3}xe^{-2x}.$$

From the initial conditions we obtain $c_1 = -\frac{23}{12}$, $c_2 = -\frac{59}{24}$, and $c_3 = \frac{17}{72}\sqrt{3}$, so

$$y = -\frac{23}{12}e^{-2x} + e^x\left(-\frac{59}{24}\cos \sqrt{3}x + \frac{17}{72}\sqrt{3}\sin \sqrt{3}x\right) + \frac{1}{4}x - \frac{5}{8} + \frac{2}{3}xe^{-2x}.$$

3.4 Undetermined Coefficients

37. We have $y_c = c_1 \cos x + c_2 \sin x$ and we assume $y_p = A^2 + Bx + C$. Substituting into the differential equation we find $A = 1$, $B = 0$, and $C = -1$. Thus $y = c_1 \cos x + c_2 \sin x + x^2 - 1$. From $y(0) = 5$ and $y(1) = 0$ we obtain

$$c_1 - 1 = 5$$

$$(\cos 1)c_1 + (\sin 1)c_2 = 0.$$

Solving this system we find $c_1 = 6$ and $c_2 = -6 \cot 1$. The solution of the boundary-value problem is

$$y = 6 \cos x - 6(\cot 1) \sin x + x^2 - 1.$$

38. We have $y_c = e^x(c_1 \cos x + c_2 \sin x)$ and we assume $y_p = Ax + B$. Substituting into the differential equation we find $A = 1$ and $B = 0$. Thus $y = e^x(c_1 \cos x + c_2 \sin x) + x$. From $y(0) = 0$ and $y(\pi) = \pi$ we obtain

$$c_1 = 0$$

$$\pi - e^\pi c_2 = \pi.$$

Solving this system we find $c_1 = 0$ and c_2 is any real number. The solution of the boundary-value problem is

$$y = c_2 e^x \sin x + x.$$

39. The general solution of the differential equation $y'' + 3y = 6x$ is $y = c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x + 2x$. The condition $y(0) = 0$ implies $c_1 = 0$ and so $y = c_2 \sin \sqrt{3}x + 2x$. The condition $y(1) + y'(1) = 0$ implies $c_2 \sin \sqrt{3} + 2 + c_2 \sqrt{3} \cos \sqrt{3} + 2 = 0$ so $c_2 = -4/(\sin \sqrt{3} + \sqrt{3} \cos \sqrt{3})$. The solution is

$$y = \frac{-4 \sin \sqrt{3}x}{\sin \sqrt{3} + \sqrt{3} \cos \sqrt{3}} + 2x.$$

40. Using the general solution $y = c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x + 2x$, the boundary conditions $y(0) + y'(0) = 0$, $y(1) = 0$ yield the system

$$c_1 + \sqrt{3}c_2 + 2 = 0$$

$$c_1 \cos \sqrt{3} + c_2 \sin \sqrt{3} + 2 = 0.$$

Solving gives

$$c_1 = \frac{2(-\sqrt{3} + \sin \sqrt{3})}{\sqrt{3} \cos \sqrt{3} - \sin \sqrt{3}} \quad \text{and} \quad c_2 = \frac{2(1 - \cos \sqrt{3})}{\sqrt{3} \cos \sqrt{3} - \sin \sqrt{3}}.$$

Thus,

$$y = \frac{2(-\sqrt{3} + \sin \sqrt{3}) \cos \sqrt{3}x}{\sqrt{3} \cos \sqrt{3} - \sin \sqrt{3}} + \frac{2(1 - \cos \sqrt{3}) \sin \sqrt{3}x}{\sqrt{3} \cos \sqrt{3} - \sin \sqrt{3}} + 2x.$$

41. We have $y_c = c_1 \cos 2x + c_2 \sin 2x$ and we assume $y_p = A \cos x + B \sin x$ on $[0, \pi/2]$. Substituting into the differential equation we find $A = 0$ and $B = \frac{1}{3}$. Thus $y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{3} \sin x$ on $[0, \pi/2]$. On $(\pi/2, \infty)$ we have $y = c_3 \cos 2x + c_4 \sin 2x$. From $y(0) = 1$ and $y'(0) = 2$ we obtain

$$c_1 = 1$$

$$\frac{1}{3} + 2c_2 = 2.$$

Solving this system we find $c_1 = 1$ and $c_2 = \frac{5}{6}$. Thus $y = \cos 2x + \frac{5}{6} \sin 2x + \frac{1}{3} \sin x$ on $[0, \pi/2]$. Now continuity of y at $x = \pi/2$ implies

$$\cos \pi + \frac{5}{6} \sin \pi + \frac{1}{3} \sin \frac{\pi}{2} = c_3 \cos \pi + c_4 \sin \pi$$

or $-1 + \frac{1}{3} = -c_3$. Hence $c_3 = \frac{2}{3}$. Continuity of y' at $x = \pi/2$ implies

$$-2 \sin \pi + \frac{5}{3} \cos \pi + \frac{1}{3} \cos \frac{\pi}{2} = -2c_3 \sin \pi + 2c_4 \cos \pi$$

or $-\frac{5}{3} = -2c_4$. Then $c_4 = \frac{5}{6}$ and the solution of the initial-value problem is

$$y(x) = \begin{cases} \cos 2x + \frac{5}{6} \sin 2x + \frac{1}{3} \sin x, & 0 \leq x \leq \pi/2 \\ \frac{2}{3} \cos 2x + \frac{5}{6} \sin 2x, & x > \pi/2. \end{cases}$$

42. We have $y_c = e^x(c_1 \cos 3x + c_2 \sin 3x)$ and we assume $y_p = A$ on $[0, \pi]$. Substituting into the differential equation we find $A = 2$. Thus, $y = e^x(c_1 \cos 3x + c_2 \sin 3x) + 2$ on $[0, \pi]$. On (π, ∞) we have $y = e^x(c_3 \cos 3x + c_4 \sin 3x)$. From $y(0) = 0$ and $y'(0) = 0$ we obtain

$$c_1 = -2, \quad c_1 + 3c_2 = 0.$$

Solving this system, we find $c_1 = -2$ and $c_2 = \frac{2}{3}$. Thus $y = e^x(-2 \cos 3x + \frac{2}{3} \sin 3x) + 2$ on $[0, \pi]$. Now, continuity of y at $x = \pi$ implies

$$e^\pi(-2 \cos 3\pi + \frac{2}{3} \sin 3\pi) + 2 = e^\pi(c_3 \cos 3\pi + c_4 \sin 3\pi)$$

or $2 + 2e^\pi = -c_3e^\pi$ or $c_3 = -2e^{-\pi}(1 + e^\pi)$. Continuity of y' at π implies

$$\frac{20}{3}e^\pi \sin 3\pi = e^\pi[(c_3 + 3c_4) \cos 3\pi + (-3c_3 + c_4) \sin 3\pi]$$

or $-c_3e^\pi - 3c_4e^\pi = 0$. Since $c_3 = -2e^{-\pi}(1 + e^\pi)$ we have $c_4 = \frac{2}{3}e^{-\pi}(1 + e^\pi)$. The solution of the initial-value problem is

$$y(x) = \begin{cases} e^x(-2 \cos 3x + \frac{2}{3} \sin 3x) + 2, & 0 \leq x \leq \pi \\ (1 + e^\pi)e^{x-\pi}(-2 \cos 3x + \frac{2}{3} \sin 3x), & x > \pi. \end{cases}$$

43. (a) From $y_p = Ae^{kx}$ we find $y'_p = Ake^{kx}$ and $y''_p = Ak^2e^{kx}$. Substituting into the differential equation we get

$$aAk^2e^{kx} + bAke^{kx} + cAe^{kx} = (ak^2 + bk + c)Ae^{kx} = e^{kx},$$

so $(ak^2 + bk + c)A = 1$. Since k is not a root of $am^2 + bm + c = 0$, $A = 1/(ak^2 + bk + c)$.

- (b) From $y_p = Axe^{kx}$ we find $y'_p = Akxe^{kx} + Ae^{kx}$ and $y''_p = Ak^2xe^{kx} + 2Ake^{kx}$. Substituting into the differential equation we get

$$\begin{aligned} aAk^2xe^{kx} + 2aAke^{kx} + bAkxe^{kx} + bAe^{kx} + cAxe^{kx} \\ = (ak^2 + bk + c)Axe^{kx} + (2ak + b)Ae^{kx} \\ = (0)Axe^{kx} + (2ak + b)Ae^{kx} = (2ak + b)Ae^{kx} = e^{kx} \end{aligned}$$

where $ak^2 + bk + c = 0$ because k is a root of the auxiliary equation. Now, the roots of the auxiliary equation are $-b/2a \pm \sqrt{b^2 - 4ac}/2a$, and since k is a root of multiplicity one, $k \neq -b/2a$ and $2ak + b \neq 0$. Thus $(2ak + b)A = 1$ and $A = 1/(2ak + b)$.

- (c) If k is a root of multiplicity two, then, as we saw in part (b), $k = -b/2a$ and $2ak + b = 0$. From $y_p = Ax^2e^{kx}$ we find $y'_p = Akx^2e^{kx} + 2Axe^{kx}$ and $y''_p = Ak^2x^2e^{kx} + 4Akxe^{kx} = 2Ae^{kx}$. Substituting into the differential equation, we get

$$\begin{aligned} aAk^2x^2e^{kx} + 4aAke^{kx} + 2aAe^{kx} + bAkx^2e^{kx} + 2bAxe^{kx} + cAx^2e^{kx} \\ = (ak^2 + bk + c)Ax^2e^{kx} + 2(2ak + b)Axe^{kx} + 2aAe^{kx} \\ = (0)Ax^2e^{kx} + 2(0)Axe^{kx} + 2aAe^{kx} = 2aAe^{kx} = e^{kx}. \end{aligned}$$

Since the differential equation is second order, $a \neq 0$ and $A = 1/(2a)$.

44. Using the double-angle formula for the cosine, we have

$$\sin x \cos 2x = \sin x(\cos^2 x - \sin^2 x) = \sin x(1 - 2\sin^2 x) = \sin x - 2\sin^3 x.$$

3.4 Undetermined Coefficients

Since $\sin x$ is a solution of the related homogeneous differential equation we look for a particular solution of the form $y_p = Ax \sin x + Bx \cos x + C \sin^3 x$. Substituting into the differential equation we obtain

$$2A \cos x + (6C - 2B) \sin x - 8C \sin^3 x = \sin x - 2 \sin^3 x.$$

Equating coefficients we find $A = 0$, $C = \frac{1}{4}$, and $B = \frac{1}{4}$. Thus, a particular solution is

$$y_p = \frac{1}{4}x \cos x + \frac{1}{4} \sin^3 x.$$

45. (a) $f(x) = e^x \sin x$. We see that $y_p \rightarrow \infty$ as $x \rightarrow \infty$ and $y_p \rightarrow 0$ as $x \rightarrow -\infty$.
(b) $f(x) = e^{-x}$. We see that $y_p \rightarrow \infty$ as $x \rightarrow \infty$ and $y_p \rightarrow \infty$ as $x \rightarrow -\infty$.
(c) $f(x) = \sin 2x$. We see that y_p is sinusoidal.
(d) $f(x) = 1$. We see that y_p is constant and simply translates y_c vertically.
46. The complementary function is $y_c = e^{2x}(c_1 \cos 2x + c_2 \sin 2x)$. We assume a particular solution of the form $y_p = (Ax^3 + Bx^2 + Cx)e^{2x} \cos 2x + (Dx^3 + Ex^2 + F)e^{2x} \sin 2x$. Substituting into the differential equation and using a CAS to simplify yields

$$\begin{aligned} & [12Dx^2 + (6A + 8E)x + (2B + 4F)]e^{2x} \cos 2x \\ & + [-12Ax^2 + (-8B + 6D)x + (-4C + 2E)]e^{2x} \sin 2x \\ & = (2x^2 - 3x)e^{2x} \cos 2x + (10x^2 - x - 1)e^{2x} \sin 2x. \end{aligned}$$

This gives the system of equations

$$\begin{aligned} 12D &= 2, & 6A + 8E &= -3, & 2B + 4F &= 0, \\ -12A &= 10, & -8B + 6D &= -1, & -4C + 2E &= -1, \end{aligned}$$

from which we find $A = -\frac{5}{6}$, $B = \frac{1}{4}$, $C = \frac{3}{8}$, $D = \frac{1}{6}$, $E = \frac{1}{4}$, and $F = -\frac{1}{8}$. Thus, a particular solution of the differential equation is

$$y_p = \left(-\frac{5}{6}x^3 + \frac{1}{4}x^2 + \frac{3}{8}x\right)e^{2x} \cos 2x + \left(\frac{1}{6}x^3 + \frac{1}{4}x^2 - \frac{1}{8}x\right)e^{2x} \sin 2x.$$

47. The complementary function is $y_c = c_1 \cos x + c_2 \sin x + c_3 x \cos x + c_4 x \sin x$. We assume a particular solution of the form $y_p = Ax^2 \cos x + Bx^3 \sin x$. Substituting into the differential equation and using a CAS to simplify yields

$$(-8A + 24B) \cos x + 3Bx \sin x = 2 \cos x - 3x \sin x.$$

This implies $-8A + 24B = 2$ and $-24B = -3$. Thus $B = \frac{1}{8}$, $A = \frac{1}{8}$, and $y_p = \frac{1}{8}x^2 \cos x + \frac{1}{8}x^3 \sin x$.

EXERCISES 3.5

Variation of Parameters

The particular solution, $y_p = u_1 y_1 + u_2 y_2$, in the following problems can take on a variety of forms, especially where trigonometric functions are involved. The validity of a particular form can best be checked by substituting it back into the differential equation.

1. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sec x$ we obtain

$$u_1' = -\frac{\sin x \sec x}{1} = -\tan x$$

$$u_2' = \frac{\cos x \sec x}{1} = 1.$$

Then $u_1 = \ln |\cos x|$, $u_2 = x$, and

$$y = c_1 \cos x + c_2 \sin x + \cos x \ln |\cos x| + x \sin x.$$

2. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \tan x$ we obtain

$$u_1' = -\sin x \tan x = \frac{\cos^2 x - 1}{\cos x} = \cos x - \sec x$$

$$u_2' = \sin x.$$

Then $u_1 = \sin x - \ln |\sec x + \tan x|$, $u_2 = -\cos x$, and

$$\begin{aligned} y &= c_1 \cos x + c_2 \sin x + \cos x (\sin x - \ln |\sec x + \tan x|) - \cos x \sin x \\ &= c_1 \cos x + c_2 \sin x - \cos x \ln |\sec x + \tan x|. \end{aligned}$$

3. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sin x$ we obtain

$$u_1' = -\sin^2 x$$

$$u_2' = \cos x \sin x.$$

Then

$$u_1 = \frac{1}{4} \sin 2x - \frac{1}{2} x = \frac{1}{2} \sin x \cos x - \frac{1}{2} x$$

$$u_2 = -\frac{1}{2} \cos^2 x.$$

3.5 Variation of Parameters

and

$$\begin{aligned} y &= c_1 \cos x + c_2 \sin x + \frac{1}{2} \sin x \cos^2 x - \frac{1}{2} x \cos x - \frac{1}{2} \cos^2 x \sin x \\ &= c_1 \cos x + c_2 \sin x - \frac{1}{2} x \cos x. \end{aligned}$$

4. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sec x \tan x$ we obtain

$$\begin{aligned} u_1' &= -\sin x (\sec x \tan x) = -\tan^2 x = 1 - \sec^2 x \\ u_2' &= \cos x (\sec x \tan x) = \tan x. \end{aligned}$$

Then $u_1 = x - \tan x$, $u_2 = -\ln |\cos x|$, and

$$\begin{aligned} y &= c_1 \cos x + c_2 \sin x + x \cos x - \sin x - \sin x \ln |\cos x| \\ &= c_1 \cos x + c_3 \sin x + x \cos x - \sin x \ln |\cos x|. \end{aligned}$$

5. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \cos^2 x$ we obtain

$$\begin{aligned} u_1' &= -\sin x \cos^2 x \\ u_2' &= \cos^3 x = \cos x (1 - \sin^2 x). \end{aligned}$$

Then $u_1 = \frac{1}{3} \cos^3 x$, $u_2 = \sin x - \frac{1}{3} \sin^3 x$, and

$$\begin{aligned} y &= c_1 \cos x + c_2 \sin x + \frac{1}{3} \cos^4 x + \sin^2 x - \frac{1}{3} \sin^4 x \\ &= c_1 \cos x + c_2 \sin x + \frac{1}{3} (\cos^2 x + \sin^2 x) (\cos^2 x - \sin^2 x) + \sin^2 x \\ &= c_1 \cos x + c_2 \sin x + \frac{1}{3} \cos^2 x + \frac{2}{3} \sin^2 x \\ &= c_1 \cos x + c_2 \sin x + \frac{1}{3} + \frac{1}{3} \sin^2 x. \end{aligned}$$

6. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sec^2 x$ we obtain

$$\begin{aligned} u_1' &= -\frac{\sin x}{\cos^2 x} \\ u_2' &= \sec x. \end{aligned}$$

Then

$$\begin{aligned} u_1 &= -\frac{1}{\cos x} = -\sec x \\ u_2 &= \ln |\sec x + \tan x| \end{aligned}$$

and

$$\begin{aligned} y &= c_1 \cos x + c_2 \sin x - \cos x \sec x + \sin x \ln |\sec x + \tan x| \\ &= c_1 \cos x + c_2 \sin x - 1 + \sin x \ln |\sec x + \tan x|. \end{aligned}$$

7. The auxiliary equation is $m^2 - 1 = 0$, so $y_c = c_1 e^x + c_2 e^{-x}$ and

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2.$$

Identifying $f(x) = \cosh x = \frac{1}{2}(e^{-x} + e^x)$ we obtain

$$\begin{aligned} u_1' &= \frac{1}{4}e^{-2x} + \frac{1}{4} \\ u_2' &= -\frac{1}{4} - \frac{1}{4}e^{2x}. \end{aligned}$$

Then

$$\begin{aligned} u_1 &= -\frac{1}{8}e^{-2x} + \frac{1}{4}x \\ u_2 &= -\frac{1}{8}e^{2x} - \frac{1}{4}x \end{aligned}$$

and

$$\begin{aligned} y &= c_1 e^x + c_2 e^{-x} - \frac{1}{8}e^{-x} + \frac{1}{4}x e^x - \frac{1}{8}e^x - \frac{1}{4}x e^{-x} \\ &= c_3 e^x + c_4 e^{-x} + \frac{1}{4}x(e^x - e^{-x}) \\ &= c_3 e^x + c_4 e^{-x} + \frac{1}{2}x \sinh x. \end{aligned}$$

8. The auxiliary equation is $m^2 - 1 = 0$, so $y_c = c_1 e^x + c_2 e^{-x}$ and

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2.$$

Identifying $f(x) = \sinh 2x$ we obtain

$$\begin{aligned} u_1' &= -\frac{1}{4}e^{-3x} + \frac{1}{4}e^x \\ u_2' &= \frac{1}{4}e^{-x} - \frac{1}{4}e^{3x}. \end{aligned}$$

Then

$$\begin{aligned} u_1 &= \frac{1}{12}e^{-3x} + \frac{1}{4}e^x \\ u_2 &= -\frac{1}{4}e^{-x} - \frac{1}{12}e^{3x}. \end{aligned}$$

and

$$\begin{aligned} y &= c_1 e^x + c_2 e^{-x} + \frac{1}{12}e^{-2x} + \frac{1}{4}e^{2x} - \frac{1}{4}e^{-2x} - \frac{1}{12}e^{2x} \\ &= c_1 e^x + c_2 e^{-x} + \frac{1}{6}(e^{2x} - e^{-2x}) \\ &= c_1 e^x + c_2 e^{-x} + \frac{1}{3} \sinh 2x. \end{aligned}$$

9. The auxiliary equation is $m^2 - 4 = 0$, so $y_c = c_1 e^{2x} + c_2 e^{-2x}$ and

$$W = \begin{vmatrix} e^{2x} & e^{-2x} \\ 2e^{2x} & -2e^{-2x} \end{vmatrix} = -4.$$

3.5 Variation of Parameters

Identifying $f(x) = e^{2x}/x$ we obtain $u'_1 = 1/4x$ and $u'_2 = -e^{4x}/4x$. Then

$$u_1 = \frac{1}{4} \ln |x|,$$

$$u_2 = -\frac{1}{4} \int_{x_0}^x \frac{e^{4t}}{t} dt$$

and

$$y = c_1 e^{2x} + c_2 e^{-2x} + \frac{1}{4} \left(e^{2x} \ln |x| - e^{-2x} \int_{x_0}^x \frac{e^{4t}}{t} dt \right), \quad x_0 > 0.$$

10. The auxiliary equation is $m^2 - 9 = 0$, so $y_c = c_1 e^{3x} + c_2 e^{-3x}$ and

$$W = \begin{vmatrix} e^{3x} & e^{-3x} \\ 3e^{3x} & -3e^{-3x} \end{vmatrix} = -6.$$

Identifying $f(x) = 9x/e^{3x}$ we obtain $u'_1 = \frac{3}{2}xe^{-6x}$ and $u'_2 = -\frac{3}{2}x$. Then

$$u_1 = -\frac{1}{24}e^{-6x} - \frac{1}{4}xe^{-6x},$$

$$u_2 = -\frac{3}{4}x^2$$

and

$$y = c_1 e^{3x} + c_2 e^{-3x} - \frac{1}{24}e^{-3x} - \frac{1}{4}xe^{-3x} - \frac{3}{4}x^2e^{-3x}$$

$$= c_1 e^{3x} + c_3 e^{-3x} - \frac{1}{4}xe^{-3x}(1 - 3x).$$

11. The auxiliary equation is $m^2 + 3m + 2 = (m+1)(m+2) = 0$, so $y_c = c_1 e^{-x} + c_2 e^{-2x}$ and

$$W = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -e^{-3x}.$$

Identifying $f(x) = 1/(1 + e^x)$ we obtain

$$u'_1 = \frac{e^x}{1 + e^x}$$

$$u'_2 = -\frac{e^{2x}}{1 + e^x} = \frac{e^x}{1 + e^x} - e^x.$$

Then $u_1 = \ln(1 + e^x)$, $u_2 = \ln(1 + e^x) - e^x$, and

$$y = c_1 e^{-x} + c_2 e^{-2x} + e^{-x} \ln(1 + e^x) + e^{-2x} \ln(1 + e^x) - e^{-x}$$

$$= c_3 e^{-x} + c_2 e^{-2x} + (1 + e^{-x})e^{-x} \ln(1 + e^x).$$

12. The auxiliary equation is $m^2 - 2m + 1 = (m-1)^2 = 0$, so $y_c = c_1 e^x + c_2 x e^x$ and

$$W = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix} = e^{2x}.$$

Identifying $f(x) = e^x/(1 + x^2)$ we obtain

$$u'_1 = -\frac{x e^x e^x}{e^{2x}(1 + x^2)} = -\frac{x}{1 + x^2}$$

$$u'_2 = \frac{e^x e^x}{e^{2x}(1 + x^2)} = \frac{1}{1 + x^2}.$$

Then $u_1 = -\frac{1}{2} \ln(1+x^2)$, $u_2 = \tan^{-1} x$, and

$$y = c_1 e^x + c_2 x e^x - \frac{1}{2} e^x \ln(1+x^2) + x e^x \tan^{-1} x.$$

13. The auxiliary equation is $m^2 + 3m + 2 = (m+1)(m+2) = 0$, so $y_c = c_1 e^{-x} + c_2 e^{-2x}$ and

$$W = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -e^{-3x}.$$

Identifying $f(x) = \sin e^x$ we obtain

$$u'_1 = \frac{e^{-2x} \sin e^x}{e^{-3x}} = e^x \sin e^x$$

$$u'_2 = \frac{e^{-x} \sin e^x}{-e^{-3x}} = -e^{2x} \sin e^x.$$

Then $u_1 = -\cos e^x$, $u_2 = e^x \cos x - \sin e^x$, and

$$y = c_1 e^{-x} + c_2 e^{-2x} - e^{-x} \cos e^x + e^{-x} \cos e^x - e^{-2x} \sin e^x$$

$$= c_1 e^{-x} + c_2 e^{-2x} - e^{-2x} \sin e^x.$$

14. The auxiliary equation is $m^2 - 2m + 1 = (m-1)^2 = 0$, so $y_c = c_1 e^t + c_2 t e^t$ and

$$W = \begin{vmatrix} e^t & t e^t \\ e^t & t e^t + e^t \end{vmatrix} = e^{2t}.$$

Identifying $f(t) = e^t \tan^{-1} t$ we obtain

$$u'_1 = -\frac{t e^t e^t \tan^{-1} t}{e^{2t}} = -t \tan^{-1} t$$

$$u'_2 = \frac{e^t e^t \tan^{-1} t}{e^{2t}} = \tan^{-1} t.$$

Then

$$u_1 = -\frac{1+t^2}{2} \tan^{-1} t + \frac{t}{2}$$

$$u_2 = t \tan^{-1} t - \frac{1}{2} \ln(1+t^2)$$

and

$$y = c_1 e^t + c_2 t e^t + \left(-\frac{1+t^2}{2} \tan^{-1} t + \frac{t}{2} \right) e^t + \left(t \tan^{-1} t - \frac{1}{2} \ln(1+t^2) \right) t e^t$$

$$= c_1 e^t + c_3 t e^t + \frac{1}{2} e^t [(t^2 - 1) \tan^{-1} t - \ln(1+t^2)].$$

15. The auxiliary equation is $m^2 + 2m + 1 = (m+1)^2 = 0$, so $y_c = c_1 e^{-t} + c_2 t e^{-t}$ and

$$W = \begin{vmatrix} e^{-t} & t e^{-t} \\ -e^{-t} & -t e^{-t} + e^{-t} \end{vmatrix} = e^{-2t}.$$

Identifying $f(t) = e^{-t} \ln t$ we obtain

$$u'_1 = -\frac{t e^{-t} e^{-t} \ln t}{e^{-2t}} = -t \ln t$$

$$u'_2 = \frac{e^{-t} e^{-t} \ln t}{e^{-2t}} = \ln t.$$

3.5 Variation of Parameters

Then

$$u_1 = -\frac{1}{2}t^2 \ln t + \frac{1}{4}t^2$$

$$u_2 = t \ln t - t$$

and

$$\begin{aligned} y &= c_1 e^{-t} + c_2 t e^{-t} - \frac{1}{2}t^2 e^{-t} \ln t + \frac{1}{4}t^2 e^{-t} + t^2 e^{-t} \ln t - t^2 e^{-t} \\ &= c_1 e^{-t} + c_2 t e^{-t} + \frac{1}{2}t^2 e^{-t} \ln t - \frac{3}{4}t^2 e^{-t}. \end{aligned}$$

16. The auxiliary equation is $2m^2 + 2m + 1 = 0$, so $y_c = e^{-x/2}[c_1 \cos(x/2) + c_2 \sin(x/2)]$ and

$$W = \begin{vmatrix} e^{-x/2} \cos \frac{x}{2} & e^{-x/2} \sin \frac{x}{2} \\ -\frac{1}{2}e^{-x/2} \cos \frac{x}{2} - \frac{1}{2}e^{-x/2} \sin \frac{x}{2} & \frac{1}{2}e^{-x/2} \cos \frac{x}{2} - \frac{1}{2}e^{-x/2} \sin \frac{x}{2} \end{vmatrix} = \frac{1}{2}e^{-x}.$$

Identifying $f(x) = 2\sqrt{x}$ we obtain

$$\begin{aligned} u_1' &= -\frac{e^{-x/2} \sin(x/2) 2\sqrt{x}}{e^{-x/2}} = -4e^{x/2} \sqrt{x} \sin \frac{x}{2} \\ u_2' &= -\frac{e^{-x/2} \cos(x/2) 2\sqrt{x}}{e^{-x/2}} = 4e^{x/2} \sqrt{x} \cos \frac{x}{2}. \end{aligned}$$

Then

$$\begin{aligned} u_1 &= -4 \int_{x_0}^x e^{t/2} \sqrt{t} \sin \frac{t}{2} dt \\ u_2 &= 4 \int_{x_0}^x e^{t/2} \sqrt{t} \cos \frac{t}{2} dt \end{aligned}$$

and

$$y = e^{-x/2} \left(c_1 \cos \frac{x}{2} + c_2 \sin \frac{x}{2} \right) - 4e^{-x/2} \cos \frac{x}{2} \int_{x_0}^x e^{t/2} \sqrt{t} \sin \frac{t}{2} dt + 4e^{-x/2} \sin \frac{x}{2} \int_{x_0}^x e^{t/2} \sqrt{t} \cos \frac{t}{2} dt.$$

17. The auxiliary equation is $3m^2 - 6m + 6 = 0$, so $y_c = e^x(c_1 \cos x + c_2 \sin x)$ and

$$W = \begin{vmatrix} e^x \cos x & e^x \sin x \\ e^x \cos x - e^x \sin x & e^x \cos x + e^x \sin x \end{vmatrix} = e^{2x}.$$

Identifying $f(x) = \frac{1}{3}e^x \sec x$ we obtain

$$\begin{aligned} u_1' &= -\frac{(e^x \sin x)(e^x \sec x)/3}{e^{2x}} = -\frac{1}{3} \tan x \\ u_2' &= \frac{(e^x \cos x)(e^x \sec x)/3}{e^{2x}} = \frac{1}{3}. \end{aligned}$$

Then $u_1 = \frac{1}{3} \ln(\cos x)$, $u_2 = \frac{1}{3}x$, and

$$y = c_1 e^x \cos x + c_2 e^x \sin x + \frac{1}{3} \ln(\cos x) e^x \cos x + \frac{1}{3} x e^x \sin x.$$

18. The auxiliary equation is $4m^2 - 4m + 1 = (2m - 1)^2 = 0$, so $y_c = c_1 e^{x/2} + c_2 x e^{x/2}$ and

$$W = \begin{vmatrix} e^{x/2} & x e^{x/2} \\ \frac{1}{2} e^{x/2} & \frac{1}{2} x e^{x/2} + e^{x/2} \end{vmatrix} = e^x.$$

Identifying $f(x) = \frac{1}{4}e^{x/2}\sqrt{1-x^2}$ we obtain

$$u'_1 = -\frac{xe^{x/2}e^{x/2}\sqrt{1-x^2}}{4e^x} = -\frac{1}{4}x\sqrt{1-x^2}$$

$$u'_2 = \frac{e^{x/2}e^{x/2}\sqrt{1-x^2}}{4e^x} = \frac{1}{4}\sqrt{1-x^2}.$$

To find u_1 and u_2 we use the substitution $v = 1 - x^2$ and the trig substitution $x = \sin \theta$, respectively:

$$u_1 = \frac{1}{12}(1-x^2)^{3/2}$$

$$u_2 = \frac{x}{8}\sqrt{1-x^2} + \frac{1}{8}\sin^{-1} x.$$

Thus

$$y = c_1e^{x/2} + c_2xe^{x/2} + \frac{1}{12}e^{x/2}(1-x^2)^{3/2} + \frac{1}{8}x^2e^{x/2}\sqrt{1-x^2} + \frac{1}{8}xe^{x/2}\sin^{-1} x.$$

19. The auxiliary equation is $4m^2 - 1 = (2m - 1)(2m + 1) = 0$, so $y_c = c_1e^{x/2} + c_2e^{-x/2}$ and

$$W = \begin{vmatrix} e^{x/2} & e^{-x/2} \\ \frac{1}{2}e^{x/2} & -\frac{1}{2}e^{-x/2} \end{vmatrix} = -1.$$

Identifying $f(x) = xe^{x/2}/4$ we obtain $u'_1 = x/4$ and $u'_2 = -xe^x/4$. Then $u_1 = x^2/8$ and $u_2 = -xe^x/4 + e^x/4$. Thus

$$\begin{aligned} y &= c_1e^{x/2} + c_2e^{-x/2} + \frac{1}{8}x^2e^{x/2} - \frac{1}{4}xe^{x/2} + \frac{1}{4}e^{x/2} \\ &= c_3e^{x/2} + c_2e^{-x/2} + \frac{1}{8}x^2e^{x/2} - \frac{1}{4}xe^{x/2} \end{aligned}$$

and

$$y' = \frac{1}{2}c_3e^{x/2} - \frac{1}{2}c_2e^{-x/2} + \frac{1}{16}x^2e^{x/2} + \frac{1}{8}xe^{x/2} - \frac{1}{4}e^{x/2}.$$

The initial conditions imply

$$c_3 + c_2 = 1$$

$$\frac{1}{2}c_3 - \frac{1}{2}c_2 - \frac{1}{4} = 0.$$

Thus $c_3 = 3/4$ and $c_2 = 1/4$, and

$$y = \frac{3}{4}e^{x/2} + \frac{1}{4}e^{-x/2} + \frac{1}{8}x^2e^{x/2} - \frac{1}{4}xe^{x/2}.$$

20. The auxiliary equation is $2m^2 + m - 1 = (2m - 1)(m + 1) = 0$, so $y_c = c_1e^{x/2} + c_2e^{-x}$ and

$$W = \begin{vmatrix} e^{x/2} & e^{-x} \\ \frac{1}{2}e^{x/2} & -e^{-x} \end{vmatrix} = -\frac{3}{2}e^{-x/2}.$$

Identifying $f(x) = (x+1)/2$ we obtain

$$u'_1 = \frac{1}{3}e^{-x/2}(x+1)$$

$$u'_2 = -\frac{1}{3}e^x(x+1).$$

Then

$$u_1 = -e^{-x/2}\left(\frac{2}{3}x - 2\right)$$

$$u_2 = -\frac{1}{3}xe^x.$$

3.5 Variation of Parameters

Thus

$$y = c_1 e^{x/2} + c_2 e^{-x} - x - 2$$

and

$$y' = \frac{1}{2}c_1 e^{x/2} - c_2 e^{-x} - 1.$$

The initial conditions imply

$$c_1 - c_2 - 2 = 1$$

$$\frac{1}{2}c_1 - c_2 - 1 = 0.$$

Thus $c_1 = 8/3$ and $c_2 = 1/3$, and

$$y = \frac{8}{3}e^{x/2} + \frac{1}{3}e^{-x} - x - 2.$$

- 21.** The auxiliary equation is $m^2 + 2m - 8 = (m - 2)(m + 4) = 0$, so $y_c = c_1 e^{2x} + c_2 e^{-4x}$ and

$$W = \begin{vmatrix} e^{2x} & e^{-4x} \\ 2e^{2x} & -4e^{-4x} \end{vmatrix} = -6e^{-2x}.$$

Identifying $f(x) = 2e^{-2x} - e^{-x}$ we obtain

$$u_1' = \frac{1}{3}e^{-4x} - \frac{1}{6}e^{-3x}$$

$$u_2' = \frac{1}{6}e^{3x} - \frac{1}{3}e^{2x}.$$

Then

$$u_1 = -\frac{1}{12}e^{-4x} + \frac{1}{18}e^{-3x}$$

$$u_2 = \frac{1}{18}e^{3x} - \frac{1}{6}e^{2x}.$$

Thus

$$\begin{aligned} y &= c_1 e^{2x} + c_2 e^{-4x} - \frac{1}{12}e^{-2x} + \frac{1}{18}e^{-x} + \frac{1}{18}e^{-x} - \frac{1}{6}e^{-2x} \\ &= c_1 e^{2x} + c_2 e^{-4x} - \frac{1}{4}e^{-2x} + \frac{1}{9}e^{-x} \end{aligned}$$

and

$$y' = 2c_1 e^{2x} - 4c_2 e^{-4x} + \frac{1}{2}e^{-2x} - \frac{1}{9}e^{-x}.$$

The initial conditions imply

$$c_1 + c_2 - \frac{5}{36} = 1$$

$$2c_1 - 4c_2 + \frac{7}{18} = 0.$$

Thus $c_1 = 25/36$ and $c_2 = 4/9$, and

$$y = \frac{25}{36}e^{2x} + \frac{4}{9}e^{-4x} - \frac{1}{4}e^{-2x} + \frac{1}{9}e^{-x}.$$

- 22.** The auxiliary equation is $m^2 - 4m + 4 = (m - 2)^2 = 0$, so $y_c = c_1 e^{2x} + c_2 x e^{2x}$ and

$$W = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & 2x e^{2x} + e^{2x} \end{vmatrix} = e^{4x}.$$

Identifying $f(x) = (12x^2 - 6x)e^{2x}$ we obtain

$$u_1' = 6x^2 - 12x^3$$

$$u_2' = 12x^2 - 6x.$$

Then

$$u_1 = 2x^3 - 3x^4$$

$$u_2 = 4x^3 - 3x^2.$$

Thus

$$\begin{aligned} y &= c_1 e^{2x} + c_2 x e^{2x} + (2x^3 - 3x^4) e^{2x} + (4x^3 - 3x^2) x e^{2x} \\ &= c_1 e^{2x} + c_2 x e^{2x} + e^{2x} (x^4 - x^3) \end{aligned}$$

and

$$y' = 2c_1 e^{2x} + c_2 (2x e^{2x} + e^{2x}) + e^{2x} (4x^3 - 3x^2) + 2e^{2x} (x^4 - x^3).$$

The initial conditions imply

$$c_1 = 1$$

$$2c_1 + c_2 = 0.$$

Thus $c_1 = 1$ and $c_2 = -2$, and

$$y = e^{2x} - 2x e^{2x} + e^{2x} (x^4 - x^3) = e^{2x} (x^4 - x^3 - 2x + 1).$$

23. Write the equation in the form

$$y'' + \frac{1}{x} y' + \left(1 - \frac{1}{4x^2}\right) y = x^{-1/2}$$

and identify $f(x) = x^{-1/2}$. From $y_1 = x^{-1/2} \cos x$ and $y_2 = x^{-1/2} \sin x$ we compute

$$W(y_1, y_2) = \begin{vmatrix} x^{-1/2} \cos x & x^{-1/2} \sin x \\ -x^{-1/2} \sin x - \frac{1}{2} x^{-3/2} \cos x & x^{-1/2} \cos x - \frac{1}{2} x^{-3/2} \sin x \end{vmatrix} = \frac{1}{x}.$$

Now

$$u_1' = -\sin x \quad \text{so} \quad u_1 = \cos x,$$

and

$$u_2' = \cos x \quad \text{so} \quad u_2 = \sin x.$$

Thus a particular solution is

$$y_p = x^{-1/2} \cos^2 x + x^{-1/2} \sin^2 x,$$

and the general solution is

$$\begin{aligned} y &= c_1 x^{-1/2} \cos x + c_2 x^{-1/2} \sin x + x^{-1/2} \cos^2 x + x^{-1/2} \sin^2 x \\ &= c_1 x^{-1/2} \cos x + c_2 x^{-1/2} \sin x + x^{-1/2}. \end{aligned}$$

24. Write the equation in the form

$$y'' + \frac{1}{x} y' + \frac{1}{x^2} y = \frac{\sec(\ln x)}{x^2}$$

and identify $f(x) = \sec(\ln x)/x^2$. From $y_1 = \cos(\ln x)$ and $y_2 = \sin(\ln x)$ we compute

$$W = \begin{vmatrix} \cos(\ln x) & \sin(\ln x) \\ -\frac{\sin(\ln x)}{x} & \frac{\cos(\ln x)}{x} \end{vmatrix} = \frac{1}{x}.$$

3.5 Variation of Parameters

Now

$$u_1' = -\frac{\tan(\ln x)}{x} \quad \text{so} \quad u_1 = \ln |\cos(\ln x)|,$$

and

$$u_2' = \frac{1}{x} \quad \text{so} \quad u_2 = \ln x.$$

Thus, a particular solution is

$$y_p = \cos(\ln x) \ln |\cos(\ln x)| + (\ln x) \sin(\ln x),$$

and the general solution is

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x) + \cos(\ln x) \ln |\cos(\ln x)| + (\ln x) \sin(\ln x).$$

25. The auxiliary equation is $m^3 + m = m(m^2 + 1) = 0$, so $y_c = c_1 + c_2 \cos x + c_3 \sin x$ and

$$W = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix} = 1.$$

Identifying $f(x) = \tan x$ we obtain

$$u_1' = W_1 = \begin{vmatrix} 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ \tan x & -\cos x & -\sin x \end{vmatrix} = \tan x$$

$$u_2' = W_2 = \begin{vmatrix} 1 & 0 & \sin x \\ 0 & 0 & \cos x \\ 0 & \tan x & -\sin x \end{vmatrix} = -\sin x$$

$$u_3' = W_3 = \begin{vmatrix} 1 & \cos x & 0 \\ 0 & -\sin x & 0 \\ 0 & -\cos x & \tan x \end{vmatrix} = -\sin x \tan x = \frac{\cos^2 x - 1}{\cos x} = \cos x - \sec x.$$

Then

$$u_1 = -\ln |\cos x|$$

$$u_2 = \cos x$$

$$u_3 = \sin x - \ln |\sec x + \tan x|$$

and

$$\begin{aligned} y &= c_1 + c_2 \cos x + c_3 \sin x - \ln |\cos x| + \cos^2 x \\ &\quad + \sin^2 x - \sin x \ln |\sec x + \tan x| \\ &= c_4 + c_2 \cos x + c_3 \sin x - \ln |\cos x| - \sin x \ln |\sec x + \tan x| \end{aligned}$$

for $-\pi/2 < x < \pi/2$.

26. The auxiliary equation is $m^3 + 4m = m(m^2 + 4) = 0$, so $y_c = c_1 + c_2 \cos 2x + c_3 \sin 2x$ and

$$W = \begin{vmatrix} 1 & \cos 2x & \sin 2x \\ 0 & -2 \sin 2x & 2 \cos 2x \\ 0 & -4 \cos 2x & -4 \sin 2x \end{vmatrix} = 8.$$

Identifying $f(x) = \sec 2x$ we obtain

$$\begin{aligned} u'_1 &= \frac{1}{8}W_1 = \frac{1}{8} \begin{vmatrix} 0 & \cos 2x & \sin 2x \\ 0 & -2 \sin 2x & 2 \cos 2x \\ \sec 2x & -4 \cos 2x & -4 \sin 2x \end{vmatrix} = \frac{1}{4} \sec 2x \\ u'_2 &= \frac{1}{8}W_2 = \frac{1}{8} \begin{vmatrix} 1 & 0 & \sin 2x \\ 0 & 0 & 2 \cos 2x \\ 0 & \sec 2x & -4 \sin 2x \end{vmatrix} = -\frac{1}{4} \\ u'_3 &= \frac{1}{8}W_3 = \frac{1}{8} \begin{vmatrix} 1 & \cos 2x & 0 \\ 0 & -2 \sin 2x & 0 \\ 0 & -4 \cos 2x & \sec 2x \end{vmatrix} = -\frac{1}{4} \tan 2x. \end{aligned}$$

Then

$$\begin{aligned} u_1 &= \frac{1}{8} \ln |\sec 2x + \tan 2x| \\ u_2 &= -\frac{1}{4}x \\ u_3 &= \frac{1}{8} \ln |\cos 2x| \end{aligned}$$

and

$$y = c_1 + c_2 \cos 2x + c_3 \sin 2x + \frac{1}{8} \ln |\sec 2x + \tan 2x| - \frac{1}{4}x \cos 2x + \frac{1}{8} \sin 2x \ln |\cos 2x|$$

for $-\pi/4 < x < \pi/4$.

- 27.** The auxiliary equation is $3m^2 - 6m + 30 = 0$, which has roots $1 \pm 3i$, so $y_c = e^x(c_1 \cos 3x + c_2 \sin 3x)$. We consider first the differential equation $3y'' - 6y' + 30y = 15 \sin x$, which can be solved using undetermined coefficients. Letting $y_{p1} = A \cos x + B \sin x$ and substituting into the differential equation we get

$$(27A - 6B) \cos x + (6A + 27B) \sin x = 15 \sin x.$$

Then

$$27A - 6B = 0 \quad \text{and} \quad 6A + 27B = 15,$$

so $A = \frac{2}{17}$ and $B = \frac{9}{17}$. Thus, $y_{p1} = \frac{2}{17} \cos x + \frac{9}{17} \sin x$. Next, we consider the differential equation $3y'' - 6y' + 30y = 15 \sin 3x$, for which a particular solution y_{p2} can be found using variation of parameters. The Wronskian is

$$W = \begin{vmatrix} e^x \cos 3x & e^x \sin 3x \\ e^x \cos 3x - 3e^x \sin 3x & 3e^x \cos 3x + e^x \sin 3x \end{vmatrix} = 3e^{2x}.$$

Identifying $f(x) = \frac{1}{3}e^x \tan x$ we obtain

$$u'_1 = -\frac{1}{9} \sin 3x \tan 3x = -\frac{1}{9} \left(\frac{\sin^2 3x}{\cos 3x} \right) = -\frac{1}{9} \left(\frac{1 - \cos^2 3x}{\cos 3x} \right) = -\frac{1}{9} (\sec 3x - \cos 3x)$$

so

$$u_1 = -\frac{1}{27} \ln |\sec 3x + \tan 3x| + \frac{1}{27} \sin 3x.$$

Next

$$u'_2 = \frac{1}{9} \sin 3x \quad \text{so} \quad u_2 = -\frac{1}{27} \cos 3x.$$

Thus

$$\begin{aligned} y_{p2} &= -\frac{1}{27} e^x \cos 3x (\ln |\sec 3x + \tan 3x| - \sin 3x) - \frac{1}{27} e^x \sin 3x \cos 3x \\ &= -\frac{1}{27} e^x (\cos 3x) \ln |\sec 3x + \tan 3x| \end{aligned}$$

3.5 Variation of Parameters

and the general solution of the original differential equation is

$$y = e^x(c_1 \cos 3x + c_2 \sin 3x) + y_{p_1}(x) + y_{p_2}(x).$$

28. The auxiliary equation is $m^2 - 2m + 1 = (m - 1)^2 = 0$, which has repeated root 1, so $y_c = c_1 e^x + c_2 x e^x$. We consider first the differential equation $y'' - 2y' + y = 4x^2 - 3$, which can be solved using undetermined coefficients. Letting $y_{p_1} = Ax^2 + Bx + C$ and substituting into the differential equation we get

$$Ax^2 + (-4A + B)x + (2A - 2B + C) = 4x^2 - 3.$$

Then

$$A = 4, \quad -4A + B = 0, \quad \text{and} \quad 2A - 2B + C = -3,$$

so $A = 4$, $B = 16$, and $C = 21$. Thus, $y_{p_1} = 4x^2 + 16x + 21$. Next we consider the differential equation $y'' - 2y' + y = x^{-1}e^x$, for which a particular solution y_{p_2} can be found using variation of parameters. The Wronskian is

$$W = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix} = e^{2x}.$$

Identifying $f(x) = e^x/x$ we obtain $u'_1 = -1$ and $u'_2 = 1/x$. Then $u_1 = -x$ and $u_2 = \ln x$, so that

$$y_{p_2} = -x e^x + x e^x \ln x,$$

and the general solution of the original differential equation is

$$\begin{aligned} y = y_c + y_{p_1} + y_{p_2} &= c_1 e^x + c_2 x e^x + 4x^2 + 16x + 21 - x e^x + x e^x \ln x \\ &= c_1 e^x + c_3 x e^x + 4x^2 + 16x + 21 + x e^x \ln x \end{aligned}.$$

29. The interval of definition for Problem 1 is $(-\pi/2, \pi/2)$, for Problem 7 is $(-\infty, \infty)$, for Problem 9 is $(0, \infty)$, and for Problem 18 is $(-1, 1)$. In Problem 24 the general solution is

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x) + \cos(\ln x) \ln |\cos(\ln x)| + (\ln x) \sin(\ln x)$$

for $-\pi/2 < \ln x < \pi/2$ or $e^{-\pi/2} < x < e^{\pi/2}$. The bounds on $\ln x$ are due to the presence of $\sec(\ln x)$ in the differential equation.

30. We are given that $y_1 = x^2$ is a solution of $x^4 y'' + x^3 y' - 4x^2 y = 0$. To find a second solution we use reduction of order. Let $y = x^2 u(x)$. Then the product rule gives

$$y' = x^2 u' + 2xu \quad \text{and} \quad y'' = x^2 u'' + 4xu' + 2u,$$

so

$$x^4 y'' + x^3 y' - 4x^2 y = x^5(xu'' + 5u') = 0.$$

Letting $w = u'$, this becomes $xw' + 5w = 0$. Separating variables and integrating we have

$$\frac{dw}{w} = -\frac{5}{x} dx \quad \text{and} \quad \ln |w| = -5 \ln x + c.$$

Thus, $w = x^{-5}$ and $u = -\frac{1}{4}x^{-4}$. A second solution is then $y_2 = x^2 x^{-4} = 1/x^2$, and the general solution of the homogeneous differential equation is $y_c = c_1 x^2 + c_2/x^2$. To find a particular solution, y_p , we use variation of parameters. The Wronskian is

$$W = \begin{vmatrix} x^2 & 1/x^2 \\ 2x & -2/x^3 \end{vmatrix} = -\frac{4}{x}.$$

Identifying $f(x) = 1/x^4$ we obtain $u'_1 = \frac{1}{4}x^{-5}$ and $u'_2 = -\frac{1}{4}x^{-1}$. Then $u_1 = -\frac{1}{16}x^{-4}$ and $u_2 = -\frac{1}{4}\ln x$, so

$$y_p = -\frac{1}{16}x^{-4}x^2 - \frac{1}{4}(\ln x)x^{-2} = -\frac{1}{16}x^{-2} - \frac{1}{4}x^{-2}\ln x.$$

The general solution is

$$y = c_1 x^2 + \frac{c_2}{x^2} - \frac{1}{16x^2} - \frac{1}{4x^2} \ln x.$$

EXERCISES 3.6

Cauchy-Euler Equation

1. The auxiliary equation is $m^2 - m - 2 = (m + 1)(m - 2) = 0$ so that $y = c_1 x^{-1} + c_2 x^2$.
2. The auxiliary equation is $4m^2 - 4m + 1 = (2m - 1)^2 = 0$ so that $y = c_1 x^{1/2} + c_2 x^{1/2} \ln x$.
3. The auxiliary equation is $m^2 = 0$ so that $y = c_1 + c_2 \ln x$.
4. The auxiliary equation is $m^2 - 4m = m(m - 4) = 0$ so that $y = c_1 + c_2 x^4$.
5. The auxiliary equation is $m^2 + 4 = 0$ so that $y = c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)$.
6. The auxiliary equation is $m^2 + 4m + 3 = (m + 1)(m + 3) = 0$ so that $y = c_1 x^{-1} + c_2 x^{-3}$.
7. The auxiliary equation is $m^2 - 4m - 2 = 0$ so that $y = c_1 x^{2-\sqrt{6}} + c_2 x^{2+\sqrt{6}}$.
8. The auxiliary equation is $m^2 + 2m - 4 = 0$ so that $y = c_1 x^{-1+\sqrt{5}} + c_2 x^{-1-\sqrt{5}}$.
9. The auxiliary equation is $25m^2 + 1 = 0$ so that $y = c_1 \cos\left(\frac{1}{5} \ln x\right) + c_2 \sin\left(\frac{1}{5} \ln x\right)$.
10. The auxiliary equation is $4m^2 - 1 = (2m - 1)(2m + 1) = 0$ so that $y = c_1 x^{1/2} + c_2 x^{-1/2}$.
11. The auxiliary equation is $m^2 + 4m + 4 = (m + 2)^2 = 0$ so that $y = c_1 x^{-2} + c_2 x^{-2} \ln x$.
12. The auxiliary equation is $m^2 + 7m + 6 = (m + 1)(m + 6) = 0$ so that $y = c_1 x^{-1} + c_2 x^{-6}$.
13. The auxiliary equation is $3m^2 + 3m + 1 = 0$ so that

$$y = x^{-1/2} \left[c_1 \cos\left(\frac{\sqrt{3}}{6} \ln x\right) + c_2 \sin\left(\frac{\sqrt{3}}{6} \ln x\right) \right].$$

14. The auxiliary equation is $m^2 - 8m + 41 = 0$ so that $y = x^4 [c_1 \cos(5 \ln x) + c_2 \sin(5 \ln x)]$.
15. Assuming that $y = x^m$ and substituting into the differential equation we obtain

$$m(m-1)(m-2) - 6 = m^3 - 3m^2 + 2m - 6 = (m-3)(m^2 + 2) = 0.$$

Thus

$$y = c_1 x^3 + c_2 \cos(\sqrt{2} \ln x) + c_3 \sin(\sqrt{2} \ln x).$$

16. Assuming that $y = x^m$ and substituting into the differential equation we obtain

$$m(m-1)(m-2) + m - 1 = m^3 - 3m^2 + 3m - 1 = (m-1)^3 = 0.$$

Thus

$$y = c_1 x + c_2 x \ln x + c_3 x (\ln x)^2.$$

3.6 Cauchy-Euler Equation

17. Assuming that $y = x^m$ and substituting into the differential equation we obtain

$$m(m-1)(m-2)(m-3) + 6m(m-1)(m-2) = m^4 - 7m^2 + 6m = m(m-1)(m-2)(m+3) = 0.$$

Thus

$$y = c_1 + c_2x + c_3x^2 + c_4x^{-3}.$$

18. Assuming that $y = x^m$ and substituting into the differential equation we obtain

$$m(m-1)(m-2)(m-3) + 6m(m-1)(m-2) + 9m(m-1) + 3m + 1 = m^4 + 2m^2 + 1 = (m^2 + 1)^2 = 0.$$

Thus

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x) + c_3(\ln x) \cos(\ln x) + c_4(\ln x) \sin(\ln x).$$

19. The auxiliary equation is $m^2 - 5m = m(m-5) = 0$ so that $y_c = c_1 + c_2x^5$ and

$$W(1, x^5) = \begin{vmatrix} 1 & x^5 \\ 0 & 5x^4 \end{vmatrix} = 5x^4.$$

Identifying $f(x) = x^3$ we obtain $u'_1 = -\frac{1}{5}x^4$ and $u'_2 = 1/5x$. Then $u_1 = -\frac{1}{25}x^5$, $u_2 = \frac{1}{5} \ln x$, and

$$y = c_1 + c_2x^5 - \frac{1}{25}x^5 + \frac{1}{5}x^5 \ln x = c_1 + c_3x^5 + \frac{1}{5}x^5 \ln x.$$

20. The auxiliary equation is $2m^2 + 3m + 1 = (2m+1)(m+1) = 0$ so that $y_c = c_1x^{-1} + c_2x^{-1/2}$ and

$$W(x^{-1}, x^{-1/2}) = \begin{vmatrix} x^{-1} & x^{-1/2} \\ -x^{-2} & -\frac{1}{2}x^{-3/2} \end{vmatrix} = \frac{1}{2}x^{-5/2}.$$

Identifying $f(x) = \frac{1}{2} - \frac{1}{2x}$ we obtain $u'_1 = x - x^2$ and $u'_2 = x^{3/2} - x^{1/2}$. Then $u_1 = \frac{1}{2}x^2 - \frac{1}{3}x^3$, $u_2 = \frac{2}{5}x^{5/2} - \frac{2}{3}x^{3/2}$, and

$$y = c_1x^{-1} + c_2x^{-1/2} + \frac{1}{2}x - \frac{1}{3}x^2 + \frac{2}{5}x^2 - \frac{2}{3}x = c_1x^{-1} + c_2x^{-1/2} - \frac{1}{6}x + \frac{1}{15}x^2.$$

21. The auxiliary equation is $m^2 - 2m + 1 = (m-1)^2 = 0$ so that $y_c = c_1x + c_2x \ln x$ and

$$W(x, x \ln x) = \begin{vmatrix} x & x \ln x \\ 1 & 1 + \ln x \end{vmatrix} = x.$$

Identifying $f(x) = 2/x$ we obtain $u'_1 = -2 \ln x/x$ and $u'_2 = 2/x$. Then $u_1 = -(\ln x)^2$, $u_2 = 2 \ln x$, and

$$\begin{aligned} y &= c_1x + c_2x \ln x - x(\ln x)^2 + 2x(\ln x)^2 \\ &= c_1x + c_2x \ln x + x(\ln x)^2, \quad x > 0. \end{aligned}$$

22. The auxiliary equation is $m^2 - 3m + 2 = (m-1)(m-2) = 0$ so that $y_c = c_1x + c_2x^2$ and

$$W(x, x^2) = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = x^2.$$

Identifying $f(x) = x^2e^x$ we obtain $u'_1 = -x^2e^x$ and $u'_2 = xe^x$. Then $u_1 = -x^2e^x + 2xe^x - 2e^x$, $u_2 = xe^x - e^x$, and

$$\begin{aligned} y &= c_1x + c_2x^2 - x^3e^x + 2x^2e^x - 2xe^x + x^3e^x - x^2e^x \\ &= c_1x + c_2x^2 + x^2e^x - 2xe^x. \end{aligned}$$

23. The auxiliary equation $m(m-1) + m - 1 = m^2 - 1 = 0$ has roots $m_1 = -1$, $m_2 = 1$, so $y_c = c_1x^{-1} + c_2x$. With $y_1 = x^{-1}$, $y_2 = x$, and the identification $f(x) = \ln x/x^2$, we get

$$W = 2x^{-1}, \quad W_1 = -\ln x/x, \quad \text{and} \quad W_2 = \ln x/x^3.$$

Then $u'_1 = W_1/W = -(\ln x)/2$, $u'_2 = W_2/W = (\ln x)/2x^2$, and integration by parts gives

$$\begin{aligned} u_1 &= \frac{1}{2}x - \frac{1}{2}x \ln x \\ u_2 &= -\frac{1}{2}x^{-1} \ln x - \frac{1}{2}x^{-1}, \end{aligned}$$

so

$$y_p = u_1 y_1 + u_2 y_2 = \left(\frac{1}{2}x - \frac{1}{2}x \ln x \right) x^{-1} + \left(-\frac{1}{2}x^{-1} \ln x - \frac{1}{2}x^{-1} \right) x = -\ln x$$

and

$$y = y_c + y_p = c_1 x^{-1} + c_2 x - \ln x, \quad x > 0.$$

24. The auxiliary equation $m(m-1) + m - 1 = m^2 - 1 = 0$ has roots $m_1 = -1$, $m_2 = 1$, so $y_c = c_1 x^{-1} + c_2 x$. With $y_1 = x^{-1}$, $y_2 = x$, and the identification $f(x) = 1/x^2(x+1)$, we get

$$W = 2x^{-1}, \quad W_1 = -1/x(x+1), \quad \text{and} \quad W_2 = 1/x^3(x+1).$$

Then $u'_1 = W_1/W = -1/2(x+1)$, $u'_2 = W_2/W = 1/2x^2(x+1)$, and integration (by partial fractions for u'_2) gives

$$\begin{aligned} u_1 &= -\frac{1}{2} \ln(x+1) \\ u_2 &= -\frac{1}{2}x^{-1} - \frac{1}{2} \ln x + \frac{1}{2} \ln(x+1), \end{aligned}$$

so

$$\begin{aligned} y_p &= u_1 y_1 + u_2 y_2 = \left[-\frac{1}{2} \ln(x+1) \right] x^{-1} + \left[-\frac{1}{2}x^{-1} - \frac{1}{2} \ln x + \frac{1}{2} \ln(x+1) \right] x \\ &= -\frac{1}{2} - \frac{1}{2}x \ln x + \frac{1}{2}x \ln(x+1) - \frac{\ln(x+1)}{2x} = -\frac{1}{2} + \frac{1}{2}x \ln \left(1 + \frac{1}{x} \right) - \frac{\ln(x+1)}{2x} \end{aligned}$$

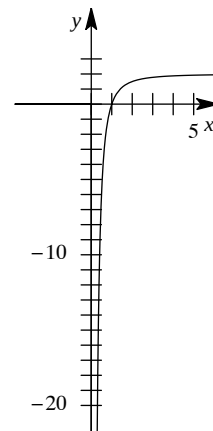
and

$$y = y_c + y_p = c_1 x^{-1} + c_2 x - \frac{1}{2} + \frac{1}{2}x \ln \left(1 + \frac{1}{x} \right) - \frac{\ln(x+1)}{2x}, \quad x > 0.$$

25. The auxiliary equation is $m^2 + 2m = m(m+2) = 0$, so that $y = c_1 + c_2 x^{-2}$ and $y' = -2c_2 x^{-3}$. The initial conditions imply

$$\begin{aligned} c_1 + c_2 &= 0 \\ -2c_2 &= 4. \end{aligned}$$

Thus, $c_1 = 2$, $c_2 = -2$, and $y = 2 - 2x^{-2}$. The graph is given to the right.



3.6 Cauchy-Euler Equation

26. The auxiliary equation is $m^2 - 6m + 8 = (m - 2)(m - 4) = 0$, so that

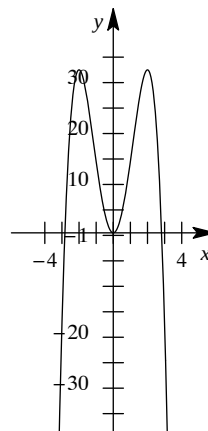
$$y = c_1 x^2 + c_2 x^4 \quad \text{and} \quad y' = 2c_1 x + 4c_2 x^3.$$

The initial conditions imply

$$4c_1 + 16c_2 = 32$$

$$4c_1 + 32c_2 = 0.$$

Thus, $c_1 = 16$, $c_2 = -2$, and $y = 16x^2 - 2x^4$. The graph is given to the right.



27. The auxiliary equation is $m^2 + 1 = 0$, so that

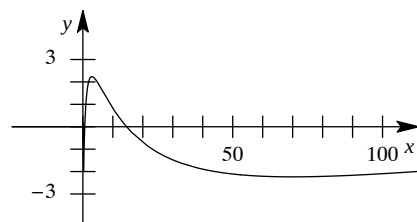
$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x)$$

and

$$y' = -c_1 \frac{1}{x} \sin(\ln x) + c_2 \frac{1}{x} \cos(\ln x).$$

The initial conditions imply $c_1 = 1$ and $c_2 = 2$. Thus

$y = \cos(\ln x) + 2 \sin(\ln x)$. The graph is given to the right.

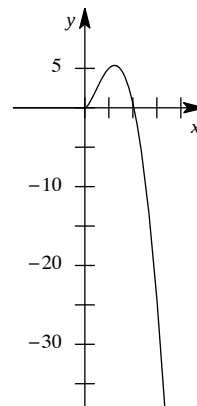


28. The auxiliary equation is $m^2 - 4m + 4 = (m - 2)^2 = 0$, so that

$$y = c_1 x^2 + c_2 x^2 \ln x \quad \text{and} \quad y' = 2c_1 x + c_2(x + 2x \ln x).$$

The initial conditions imply $c_1 = 5$ and $c_2 + 10 = 3$. Thus $y = 5x^2 - 7x^2 \ln x$.

The graph is given to the right.



29. The auxiliary equation is $m^2 = 0$ so that $y_c = c_1 + c_2 \ln x$ and

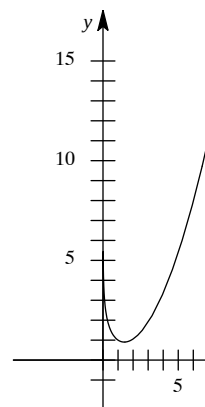
$$W(1, \ln x) = \begin{vmatrix} 1 & \ln x \\ 0 & 1/x \end{vmatrix} = \frac{1}{x}.$$

Identifying $f(x) = 1$ we obtain $u'_1 = -x \ln x$ and $u'_2 = x$. Then

$u_1 = \frac{1}{4}x^2 - \frac{1}{2}x^2 \ln x$, $u_2 = \frac{1}{2}x^2$, and

$$y = c_1 + c_2 \ln x + \frac{1}{4}x^2 - \frac{1}{2}x^2 \ln x + \frac{1}{2}x^2 \ln x = c_1 + c_2 \ln x + \frac{1}{4}x^2.$$

The initial conditions imply $c_1 + \frac{1}{4} = 1$ and $c_2 + \frac{1}{2} = -\frac{1}{2}$. Thus, $c_1 = \frac{3}{4}$, $c_2 = -1$, and $y = \frac{3}{4} - \ln x + \frac{1}{4}x^2$. The graph is given to the right.



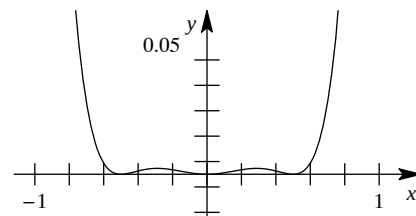
30. The auxiliary equation is $m^2 - 6m + 8 = (m - 2)(m - 4) = 0$, so that $y_c = c_1x^2 + c_2x^4$ and

$$W = \begin{vmatrix} x^2 & x^4 \\ 2x & 4x^3 \end{vmatrix} = 2x^5.$$

Identifying $f(x) = 8x^4$ we obtain $u'_1 = -4x^3$ and $u'_2 = 4x$. Then $u_1 = -x^4$, $u_2 = 2x^2$, and $y = c_1x^2 + c_2x^4 + x^6$. The initial conditions imply

$$\begin{aligned} \frac{1}{4}c_1 + \frac{1}{16}c_2 &= -\frac{1}{64} \\ c_1 + \frac{1}{2}c_2 &= -\frac{3}{16}. \end{aligned}$$

Thus $c_1 = \frac{1}{16}$, $c_2 = -\frac{1}{2}$, and $y = \frac{1}{16}x^2 - \frac{1}{2}x^4 + x^6$. The graph is given above.



31. Substituting $x = e^t$ into the differential equation we obtain

$$\frac{d^2y}{dt^2} + 8\frac{dy}{dt} - 20y = 0.$$

The auxiliary equation is $m^2 + 8m - 20 = (m + 10)(m - 2) = 0$ so that

$$y = c_1e^{-10t} + c_2e^{2t} = c_1x^{-10} + c_2x^2.$$

32. Substituting $x = e^t$ into the differential equation we obtain

$$\frac{d^2y}{dt^2} - 10\frac{dy}{dt} + 25y = 0.$$

The auxiliary equation is $m^2 - 10m + 25 = (m - 5)^2 = 0$ so that

$$y = c_1e^{5t} + c_2te^{5t} = c_1x^5 + c_2x^5 \ln x.$$

33. Substituting $x = e^t$ into the differential equation we obtain

$$\frac{d^2y}{dt^2} + 9\frac{dy}{dt} + 8y = e^{2t}.$$

The auxiliary equation is $m^2 + 9m + 8 = (m + 1)(m + 8) = 0$ so that $y_c = c_1e^{-t} + c_2e^{-8t}$. Using undetermined coefficients we try $y_p = Ae^{2t}$. This leads to $30Ae^{2t} = e^{2t}$, so that $A = 1/30$ and

$$y = c_1e^{-t} + c_2e^{-8t} + \frac{1}{30}e^{2t} = c_1x^{-1} + c_2x^{-8} + \frac{1}{30}x^2.$$

34. Substituting $x = e^t$ into the differential equation we obtain

$$\frac{d^2y}{dt^2} - 5\frac{dy}{dt} + 6y = 2t.$$

The auxiliary equation is $m^2 - 5m + 6 = (m - 2)(m - 3) = 0$ so that $y_c = c_1e^{2t} + c_2e^{3t}$. Using undetermined coefficients we try $y_p = At + B$. This leads to $(-5A + 6B) + 6At = 2t$, so that $A = 1/3$, $B = 5/18$, and

$$y = c_1e^{2t} + c_2e^{3t} + \frac{1}{3}t + \frac{5}{18} = c_1x^2 + c_2x^3 + \frac{1}{3}\ln x + \frac{5}{18}.$$

35. Substituting $x = e^t$ into the differential equation we obtain

$$\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 13y = 4 + 3e^t.$$

3.6 Cauchy-Euler Equation

The auxiliary equation is $m^2 - 4m + 13 = 0$ so that $y_c = e^{2t}(c_1 \cos 3t + c_2 \sin 3t)$. Using undetermined coefficients we try $y_p = A + Be^t$. This leads to $13A + 10Be^t = 4 + 3e^t$, so that $A = 4/13$, $B = 3/10$, and

$$\begin{aligned} y &= e^{2t}(c_1 \cos 3t + c_2 \sin 3t) + \frac{4}{13} + \frac{3}{10}e^t \\ &= x^2 [c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x)] + \frac{4}{13} + \frac{3}{10}x. \end{aligned}$$

36. From

$$\frac{d^2 y}{dx^2} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

it follows that

$$\begin{aligned} \frac{d^3 y}{dx^3} &= \frac{1}{x^2} \frac{d}{dx} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) - \frac{2}{x^3} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \\ &= \frac{1}{x^2} \frac{d}{dx} \left(\frac{d^2 y}{dt^2} \right) - \frac{1}{x^2} \frac{d}{dx} \left(\frac{dy}{dt} \right) - \frac{2}{x^3} \frac{d^2 y}{dt^2} + \frac{2}{x^3} \frac{dy}{dt} \\ &= \frac{1}{x^2} \frac{d^3 y}{dt^3} \left(\frac{1}{x} \right) - \frac{1}{x^2} \frac{d^2 y}{dt^2} \left(\frac{1}{x} \right) - \frac{2}{x^3} \frac{d^2 y}{dt^2} + \frac{2}{x^3} \frac{dy}{dt} \\ &= \frac{1}{x^3} \left(\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} \right). \end{aligned}$$

Substituting into the differential equation we obtain

$$\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} - 3 \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + 6 \frac{dy}{dt} - 6y = 3 + 3t$$

or

$$\frac{d^3 y}{dt^3} - 6 \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} - 6y = 3 + 3t.$$

The auxiliary equation is $m^3 - 6m^2 + 11m - 6 = (m-1)(m-2)(m-3) = 0$ so that $y_c = c_1 e^t + c_2 e^{2t} + c_3 e^{3t}$. Using undetermined coefficients we try $y_p = A + Bt$. This leads to $(11B - 6A) - 6Bt = 3 + 3t$, so that $A = -17/12$, $B = -1/2$, and

$$y = c_1 e^t + c_2 e^{2t} + c_3 e^{3t} - \frac{17}{12} - \frac{1}{2}t = c_1 x + c_2 x^2 + c_3 x^3 - \frac{17}{12} - \frac{1}{2} \ln x.$$

In the next two problems we use the substitution $t = -x$ since the initial conditions are on the interval $(-\infty, 0)$. In this case

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = -\frac{dy}{dx}$$

and

$$\frac{d^2 y}{dt^2} = \frac{d}{dt} \left(\frac{dy}{dt} \right) = \frac{d}{dt} \left(-\frac{dy}{dx} \right) = -\frac{d}{dt} \left(y' \right) = -\frac{dy'}{dx} \frac{dx}{dt} = -\frac{d^2 y}{dx^2} \frac{dx}{dt} = \frac{d^2 y}{dx^2}.$$

37. The differential equation and initial conditions become

$$4t^2 \frac{d^2 y}{dt^2} + y = 0; \quad y(t) \Big|_{t=1} = 2, \quad y'(t) \Big|_{t=1} = -4.$$

The auxiliary equation is $4m^2 - 4m + 1 = (2m-1)^2 = 0$, so that

$$y = c_1 t^{1/2} + c_2 t^{1/2} \ln t \quad \text{and} \quad y' = \frac{1}{2} c_1 t^{-1/2} + c_2 \left(t^{-1/2} + \frac{1}{2} t^{-1/2} \ln t \right).$$

The initial conditions imply $c_1 = 2$ and $1 + c_2 = -4$. Thus

$$y = 2t^{1/2} - 5t^{1/2} \ln t = 2(-x)^{1/2} - 5(-x)^{1/2} \ln(-x), \quad x < 0.$$

38. The differential equation and initial conditions become

$$t^2 \frac{d^2 y}{dt^2} - 4t \frac{dy}{dt} + 6y = 0; \quad y(t) \Big|_{t=2} = 8, \quad y'(t) \Big|_{t=2} = 0.$$

The auxiliary equation is $m^2 - 5m + 6 = (m - 2)(m - 3) = 0$, so that

$$y = c_1 t^2 + c_2 t^3 \quad \text{and} \quad y' = 2c_1 t + 3c_2 t^2.$$

The initial conditions imply

$$4c_1 + 8c_2 = 8$$

$$4c_1 + 12c_2 = 0$$

from which we find $c_1 = 6$ and $c_2 = -2$. Thus

$$y = 6t^2 - 2t^3 = 6x^2 + 2x^3, \quad x < 0.$$

39. Letting $u = x + 2$ we obtain $dy/dx = dy/du$ and, using the Chain Rule,

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{du} \right) = \frac{d^2 y}{du^2} \frac{du}{dx} = \frac{d^2 y}{du^2} (1) = \frac{d^2 y}{du^2}.$$

Substituting into the differential equation we obtain

$$u^2 \frac{d^2 y}{du^2} + u \frac{dy}{du} + y = 0.$$

The auxiliary equation is $m^2 + 1 = 0$ so that

$$y = c_1 \cos(\ln u) + c_2 \sin(\ln u) = c_1 \cos[\ln(x + 2)] + c_2 \sin[\ln(x + 2)].$$

40. If $1 - i$ is a root of the auxiliary equation then so is $1 + i$, and the auxiliary equation is

$$(m - 2)[m - (1 + i)][m - (1 - i)] = m^3 - 4m^2 + 6m - 4 = 0.$$

We need $m^3 - 4m^2 + 6m - 4$ to have the form $m(m - 1)(m - 2) + bm(m - 1) + cm + d$. Expanding this last expression and equating coefficients we get $b = -1$, $c = 3$, and $d = -4$. Thus, the differential equation is

$$x^3 y''' - x^2 y'' + 3xy' - 4y = 0.$$

41. For $x^2 y'' = 0$ the auxiliary equation is $m(m - 1) = 0$ and the general solution is $y = c_1 + c_2 x$. The initial conditions imply $c_1 = y_0$ and $c_2 = y_1$, so $y = y_0 + y_1 x$. The initial conditions are satisfied for all real values of y_0 and y_1 .

For $x^2 y'' - 2xy' + 2y = 0$ the auxiliary equation is $m^2 - 3m + 2 = (m - 1)(m - 2) = 0$ and the general solution is $y = c_1 x + c_2 x^2$. The initial condition $y(0) = y_0$ implies $0 = y_0$ and the condition $y'(0) = y_1$ implies $c_1 = y_1$. Thus, the initial conditions are satisfied for $y_0 = 0$ and for all real values of y_1 .

For $x^2 y'' - 4xy' + 6y = 0$ the auxiliary equation is $m^2 - 5m + 6 = (m - 2)(m - 3) = 0$ and the general solution is $y = c_1 x^2 + c_2 x^3$. The initial conditions imply $y(0) = 0 = y_0$ and $y'(0) = 0$. Thus, the initial conditions are satisfied only for $y_0 = y_1 = 0$.

42. The function $y(x) = -\sqrt{x} \cos(\ln x)$ is defined for $x > 0$ and has x -intercepts where $\ln x = \pi/2 + k\pi$ for k an integer or where $x = e^{\pi/2 + k\pi}$. Solving $\pi/2 + k\pi = 0.5$ we get $k \approx -0.34$, so $e^{\pi/2 + k\pi} < 0.5$ for all negative integers and the graph has infinitely many x -intercepts in the interval $(0, 0.5)$.

3.6 Cauchy-Euler Equation

43. The auxiliary equation is $2m(m-1)(m-2) - 10.98m(m-1) + 8.5m + 1.3 = 0$, so that $m_1 = -0.053299$, $m_2 = 1.81164$, $m_3 = 6.73166$, and

$$y = c_1 x^{-0.053299} + c_2 x^{1.81164} + c_3 x^{6.73166}.$$

44. The auxiliary equation is $m(m-1)(m-2) + 4m(m-1) + 5m - 9 = 0$, so that $m_1 = 1.40819$ and the two complex roots are $-1.20409 \pm 2.22291i$. The general solution of the differential equation is

$$y = c_1 x^{1.40819} + x^{-1.20409} [c_2 \cos(2.22291 \ln x) + c_3 \sin(2.22291 \ln x)].$$

45. The auxiliary equation is $m(m-1)(m-2)(m-3) + 6m(m-1)(m-2) + 3m(m-1) - 3m + 4 = 0$, so that $m_1 = m_2 = \sqrt{2}$ and $m_3 = m_4 = -\sqrt{2}$. The general solution of the differential equation is

$$y = c_1 x^{\sqrt{2}} + c_2 x^{\sqrt{2}} \ln x + c_3 x^{-\sqrt{2}} + c_4 x^{-\sqrt{2}} \ln x.$$

46. The auxiliary equation is $m(m-1)(m-2)(m-3) - 6m(m-1)(m-2) + 33m(m-1) - 105m + 169 = 0$, so that $m_1 = m_2 = 3 + 2i$ and $m_3 = m_4 = 3 - 2i$. The general solution of the differential equation is

$$y = x^3 [c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)] + x^3 \ln x [c_3 \cos(2 \ln x) + c_4 \sin(2 \ln x)].$$

47. The auxiliary equation

$$m(m-1)(m-2) - m(m-1) - 2m + 6 = m^3 - 4m^2 + m + 6 = 0$$

has roots $m_1 = -1$, $m_2 = 2$, and $m_3 = 3$, so $y_c = c_1 x^{-1} + c_2 x^2 + c_3 x^3$. With $y_1 = x^{-1}$, $y_2 = x^2$, $y_3 = x^3$, and the identification $f(x) = 1/x$, we get from (10) of Section 4.6 in the text

$$W_1 = x^3, \quad W_2 = -4, \quad W_3 = 3/x, \quad \text{and} \quad W = 12x.$$

Then $u'_1 = W_1/W = x^2/12$, $u'_2 = W_2/W = -1/3x$, $u'_3 = 1/4x^2$, and integration gives

$$u_1 = \frac{x^3}{36}, \quad u_2 = -\frac{1}{3} \ln x, \quad \text{and} \quad u_3 = -\frac{1}{4x},$$

so

$$y_p = u_1 y_1 + u_2 y_2 + u_3 y_3 = \frac{x^3}{36} x^{-1} + x^2 \left(-\frac{1}{3} \ln x \right) + x^3 \left(-\frac{1}{4x} \right) = -\frac{2}{9} x^2 - \frac{1}{3} x^2 \ln x,$$

and

$$y = y_c + y_p = c_1 x^{-1} + c_2 x^2 + c_3 x^3 - \frac{2}{9} x^2 - \frac{1}{3} x^2 \ln x, \quad x > 0.$$

EXERCISES 3.7

Nonlinear Equations

1. We have $y'_1 = y''_1 = e^x$, so

$$(y''_1)^2 = (e^x)^2 = e^{2x} = y_1^2.$$

Also, $y'_2 = -\sin x$ and $y''_2 = -\cos x$, so

$$(y''_2)^2 = (-\cos x)^2 = \cos^2 x = y_2^2.$$

However, if $y = c_1 y_1 + c_2 y_2$, we have $(y'')^2 = (c_1 e^x - c_2 \cos x)^2$ and $y^2 = (c_1 e^x + c_2 \cos x)^2$. Thus $(y'')^2 \neq y^2$.

2. We have $y'_1 = y''_1 = 0$, so

$$y_1 y''_1 = 1 \cdot 0 = 0 = \frac{1}{2}(0)^2 = \frac{1}{2}(y'_1)^2.$$

Also, $y'_2 = 2x$ and $y''_2 = 2$, so

$$y_2 y''_2 = x^2(2) = 2x^2 = \frac{1}{2}(2x)^2 = \frac{1}{2}(y'_2)^2.$$

However, if $y = c_1 y_1 + c_2 y_2$, we have $yy'' = (c_1 \cdot 1 + c_2 x^2)(c_1 \cdot 0 + 2c_2) = 2c_2(c_1 + c_2 x^2)$ and $\frac{1}{2}(y')^2 = \frac{1}{2}[c_1 \cdot 0 + c_2(2x)]^2 = 2c_2^2 x^2$. Thus $yy'' \neq \frac{1}{2}(y')^2$.

3. Let $u = y'$ so that $u' = y''$. The equation becomes $u' = -u - 1$ which is separable. Thus

$$\frac{du}{u^2 + 1} = -dx \implies \tan^{-1} u = -x + c_1 \implies y' = \tan(c_1 - x) \implies y = \ln |\cos(c_1 - x)| + c_2.$$

4. Let $u = y'$ so that $u' = y''$. The equation becomes $u' = 1 + u^2$. Separating variables we obtain

$$\frac{du}{1 + u^2} = dx \implies \tan^{-1} u = x + c_1 \implies u = \tan(x + c_1) \implies y = -\ln |\cos(x + c_1)| + c_2.$$

5. Let $u = y'$ so that $u' = y''$. The equation becomes $x^2 u' + u^2 = 0$. Separating variables we obtain

$$\begin{aligned} \frac{du}{u^2} &= -\frac{dx}{x^2} \implies -\frac{1}{u} = \frac{1}{x} + c_1 = \frac{c_1 x + 1}{x} \implies u = -\frac{1}{c_1} \left(\frac{x}{x + 1/c_1} \right) = \frac{1}{c_1} \left(\frac{1}{c_1 x + 1} - 1 \right) \\ &\implies y = \frac{1}{c_1^2} \ln |c_1 x + 1| - \frac{1}{c_1} x + c_2. \end{aligned}$$

6. Let $u = y'$ so that $y'' = u du/dy$. The equation becomes $(y + 1)u du/dy = u^2$. Separating variables we obtain

$$\begin{aligned} \frac{du}{u} &= \frac{dy}{y + 1} \implies \ln |u| = \ln |y + 1| + \ln c_1 \implies u = c_1(y + 1) \\ &\implies \frac{dy}{dx} = c_1(y + 1) \implies \frac{dy}{y + 1} = c_1 dx \\ &\implies \ln |y + 1| = c_1 x + c_2 \implies y + 1 = c_3 e^{c_1 x}. \end{aligned}$$

7. Let $u = y'$ so that $y'' = u du/dy$. The equation becomes $u du/dy + 2yu^3 = 0$. Separating variables we obtain

$$\begin{aligned} \frac{du}{u^2} + 2y dy &= 0 \implies -\frac{1}{u} + y^2 = c \implies u = \frac{1}{y^2 + c_1} \implies y' = \frac{1}{y^2 + c_1} \\ &\implies (y^2 + c_1) dy = dx \implies \frac{1}{3} y^3 + c_1 y = x + c_2. \end{aligned}$$

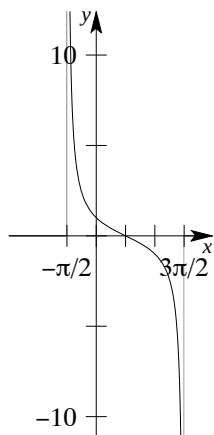
3.7 Nonlinear Equations

8. Let $u = y'$ so that $y'' = u du/dy$. The equation becomes $y^2 u du/dy = u$. Separating variables we obtain

$$\begin{aligned} du = \frac{dy}{y^2} &\implies u = -\frac{1}{y} + c_1 \implies y' = \frac{c_1 y - 1}{y} \implies \frac{y}{c_1 y - 1} dy = dx \\ &\implies \frac{1}{c_1} \left(1 + \frac{1}{c_1 y - 1} \right) dy = dx \text{ (for } c_1 \neq 0) \implies \frac{1}{c_1} y + \frac{1}{c_1^2} \ln |y - 1| = x + c_2. \end{aligned}$$

If $c_1 = 0$, then $y dy = -dx$ and another solution is $\frac{1}{2}y^2 = -x + c_2$.

9. (a)



- (b) Let $u = y'$ so that $y'' = u du/dy$. The equation becomes $u du/dy + yu = 0$. Separating variables we obtain

$$du = -y dy \implies u = -\frac{1}{2}y^2 + c_1 \implies y' = -\frac{1}{2}y^2 + c_1.$$

When $x = 0$, $y = 1$ and $y' = -1$ so $-1 = -1/2 + c_1$ and $c_1 = -1/2$. Then

$$\begin{aligned} \frac{dy}{dx} = -\frac{1}{2}y^2 - \frac{1}{2} &\implies \frac{dy}{y^2 + 1} = -\frac{1}{2} dx \implies \tan^{-1} y = -\frac{1}{2}x + c_2 \\ &\implies y = \tan \left(-\frac{1}{2}x + c_2 \right). \end{aligned}$$

When $x = 0$, $y = 1$ so $1 = \tan c_2$ and $c_2 = \pi/4$. The solution of the initial-value problem is

$$y = \tan \left(\frac{\pi}{4} - \frac{1}{2}x \right).$$

The graph is shown in part (a).

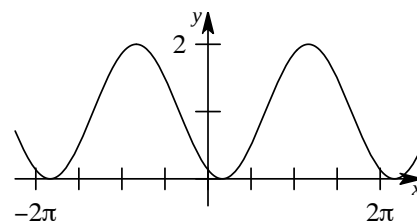
- (c) The interval of definition is $-\pi/2 < \pi/4 - x/2 < \pi/2$ or $-\pi/2 < x < 3\pi/2$.

10. Let $u = y'$ so that $u' = y''$. The equation becomes $(u')^2 + u^2 = 1$ which results in $u' = \pm\sqrt{1-u^2}$. To solve $u' = \sqrt{1-u^2}$ we separate variables:

$$\begin{aligned} \frac{du}{\sqrt{1-u^2}} = dx &\implies \sin^{-1} u = x + c_1 \implies u = \sin(x + c_1) \\ &\implies y' = \sin(x + c_1). \end{aligned}$$

When $x = \pi/2$, $y' = \sqrt{3}/2$, so $\sqrt{3}/2 = \sin(\pi/2 + c_1)$ and $c_1 = -\pi/6$. Thus

$$y' = \sin \left(x - \frac{\pi}{6} \right) \implies y = -\cos \left(x - \frac{\pi}{6} \right) + c_2.$$



When $x = \pi/2$, $y = 1/2$, so $1/2 = -\cos(\pi/2 - \pi/6) + c_2 = -1/2 + c_2$ and $c_2 = 1$. The solution of the initial-value problem is $y = 1 - \cos(x - \pi/6)$.

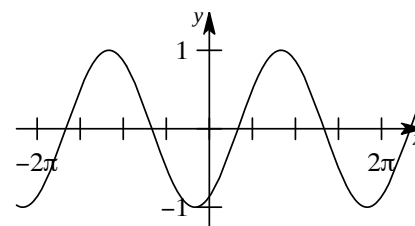
To solve $u' = -\sqrt{1-u^2}$ we separate variables:

$$\begin{aligned}\frac{du}{\sqrt{1-u^2}} &= -dx \implies \cos^{-1} u = x + c_1 \\ \implies u &= \cos(x + c_1) \implies y' = \cos(x + c_1).\end{aligned}$$

When $x = \pi/2$, $y' = \sqrt{3}/2$, so $\sqrt{3}/2 = \cos(\pi/2 + c_1)$ and $c_1 = -\pi/3$. Thus

$$y' = \cos\left(x - \frac{\pi}{3}\right) \implies y = \sin\left(x - \frac{\pi}{3}\right) + c_2.$$

When $x = \pi/2$, $y = 1/2$, so $1/2 = \sin(\pi/2 - \pi/3) + c_2 = 1/2 + c_2$ and $c_2 = 0$. The solution of the initial-value problem is $y = \sin(x - \pi/3)$.



11. Let $u = y'$ so that $u' = y''$. The equation becomes $u' - (1/x)u = (1/x)u^3$, which is Bernoulli. Using $w = u^{-2}$ we obtain $dw/dx + (2/x)w = -2/x$. An integrating factor is x^2 , so

$$\begin{aligned}\frac{d}{dx}[x^2w] &= -2x \implies x^2w = -x^2 + c_1 \implies w = -1 + \frac{c_1}{x^2} \\ \implies u^{-2} &= -1 + \frac{c_1}{x^2} \implies u = \frac{x}{\sqrt{c_1 - x^2}} \\ \implies \frac{dy}{dx} &= \frac{x}{\sqrt{c_1 - x^2}} \implies y = -\sqrt{c_1 - x^2} + c_2 \\ \implies c_1 - x^2 &= (c_2 - y)^2 \implies x^2 + (c_2 - y)^2 = c_1.\end{aligned}$$

12. Let $u = y'$ so that $u' = y''$. The equation becomes $u' - (1/x)u = u^2$, which is a Bernoulli differential equation. Using the substitution $w = u^{-1}$ we obtain $dw/dx + (1/x)w = -1$. An integrating factor is x , so

$$\frac{d}{dx}[xw] = -x \implies w = -\frac{1}{2}x + \frac{1}{x}c \implies \frac{1}{u} = \frac{c_1 - x^2}{2x} \implies u = \frac{2x}{c_1 - x^2} \implies y = -\ln|c_1 - x^2| + c_2.$$

In Problems 13-16 the thinner curve is obtained using a numerical solver, while the thicker curve is the graph of the Taylor polynomial.

13. We look for a solution of the form

$$y(x) = y(0) + y'(0)x + \frac{1}{2!}y''(0)x^2 + \frac{1}{3!}y'''(0)x^3 + \frac{1}{4!}y^{(4)}(0)x^4 + \frac{1}{5!}y^{(5)}(0)x^5.$$

From $y''(x) = x + y^2$ we compute

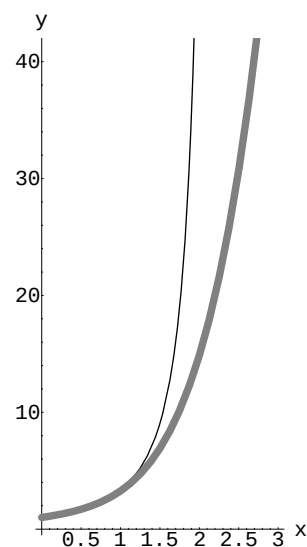
$$\begin{aligned}y'''(x) &= 1 + 2yy' \\ y^{(4)}(x) &= 2yy'' + 2(y')^2 \\ y^{(5)}(x) &= 2yy''' + 6y'y''.\end{aligned}$$

Using $y(0) = 1$ and $y'(0) = 1$ we find

$$y''(0) = 1, \quad y'''(0) = 3, \quad y^{(4)}(0) = 4, \quad y^{(5)}(0) = 12.$$

An approximate solution is

$$y(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \frac{1}{6}x^4 + \frac{1}{10}x^5.$$



3.7 Nonlinear Equations

14. We look for a solution of the form

$$y(x) = y(0) + y'(0)x + \frac{1}{2!}y''(0)x^2 + \frac{1}{3!}y'''(0)x^3 + \frac{1}{4!}y^{(4)}(0)x^4 + \frac{1}{5!}y^{(5)}(0)x^5.$$

From $y''(x) = 1 - y^2$ we compute

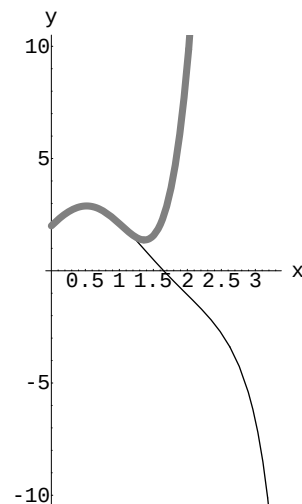
$$\begin{aligned}y'''(x) &= -2yy' \\y^{(4)}(x) &= -2yy'' - 2(y')^2 \\y^{(5)}(x) &= -2yy''' - 6y'y''.\end{aligned}$$

Using $y(0) = 2$ and $y'(0) = 3$ we find

$$y''(0) = -3, \quad y'''(0) = -12, \quad y^{(4)}(0) = -6, \quad y^{(5)}(0) = 102.$$

An approximate solution is

$$y(x) = 2 + 3x - \frac{3}{2}x^2 - 2x^3 - \frac{1}{4}x^4 + \frac{17}{20}x^5.$$



15. We look for a solution of the form

$$y(x) = y(0) + y'(0)x + \frac{1}{2!}y''(0)x^2 + \frac{1}{3!}y'''(0)x^3 + \frac{1}{4!}y^{(4)}(0)x^4 + \frac{1}{5!}y^{(5)}(0)x^5.$$

From $y''(x) = x^2 + y^2 - 2y'$ we compute

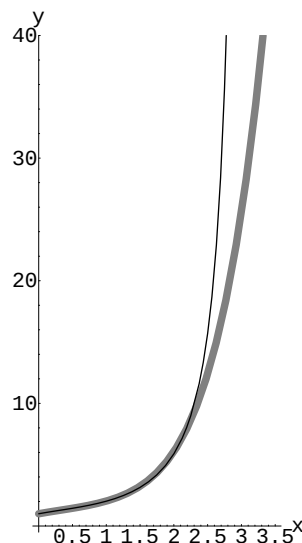
$$\begin{aligned}y'''(x) &= 2x + 2yy' - 2y'' \\y^{(4)}(x) &= 2 + 2(y')^2 + 2yy'' - 2y''' \\y^{(5)}(x) &= 6y'y'' + 2yy''' - 2y^{(4)}.\end{aligned}$$

Using $y(0) = 1$ and $y'(0) = 1$ we find

$$y''(0) = -1, \quad y'''(0) = 4, \quad y^{(4)}(0) = -6, \quad y^{(5)}(0) = 14.$$

An approximate solution is

$$y(x) = 1 + x - \frac{1}{2}x^2 + \frac{2}{3}x^3 - \frac{1}{4}x^4 + \frac{7}{60}x^5.$$

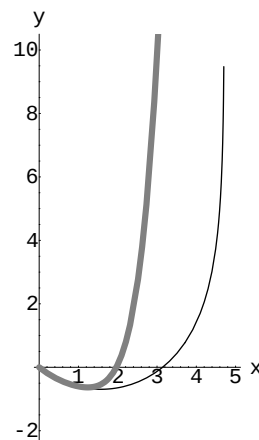


16. We look for a solution of the form

$$\begin{aligned}y(x) &= y(0) + y'(0)x + \frac{1}{2!}y''(0)x^2 + \frac{1}{3!}y'''(0)x^3 + \frac{1}{4!}y^{(4)}(0)x^4 \\&\quad + \frac{1}{5!}y^{(5)}(0)x^5 + \frac{1}{6!}y^{(6)}(0)x^6.\end{aligned}$$

From $y''(x) = e^y$ we compute

$$\begin{aligned}y'''(x) &= e^y y' \\y^{(4)}(x) &= e^y (y')^2 + e^y y'' \\y^{(5)}(x) &= e^y (y')^3 + 3e^y y' y'' + e^y y''' \\y^{(6)}(x) &= e^y (y')^4 + 6e^y (y')^2 y'' + 3e^y (y'')^2 + 4e^y y' y''' + e^y y^{(4)}.\end{aligned}$$



Using $y(0) = 0$ and $y'(0) = -1$ we find

$$y''(0) = 1, \quad y'''(0) = -1, \quad y^{(4)}(0) = 2, \quad y^{(5)}(0) = -5, \quad y^{(6)}(0) = 16.$$

An approximate solution is

$$y(x) = -x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{24}x^5 + \frac{1}{45}x^6.$$

17. We need to solve $[1 + (y')^2]^{3/2} = y''$. Let $u = y'$ so that $u' = y''$. The equation becomes $(1 + u^2)^{3/2} = u'$ or $(1 + u^2)^{3/2} = du/dx$. Separating variables and using the substitution $u = \tan \theta$ we have

$$\begin{aligned} \frac{du}{(1 + u^2)^{3/2}} = dx &\implies \int \frac{\sec^2 \theta}{(1 + \tan^2 \theta)^{3/2}} d\theta = x \implies \int \frac{\sec^2 \theta}{\sec^3 \theta} d\theta = x \\ &\implies \int \cos \theta d\theta = x \implies \sin \theta = x \implies \frac{u}{\sqrt{1 + u^2}} = x \\ &\implies \frac{y'}{\sqrt{1 + (y')^2}} = x \implies (y')^2 = x^2 [1 + (y')^2] = \frac{x^2}{1 - x^2} \\ &\implies y' = \frac{x}{\sqrt{1 - x^2}} \quad (\text{for } x > 0) \implies y = -\sqrt{1 - x^2}. \end{aligned}$$

18. When $y = \sin x$, $y' = \cos x$, $y'' = -\sin x$, and

$$(y'')^2 - y^2 = \sin^2 x - \sin^2 x = 0.$$

When $y = e^{-x}$, $y' = -e^{-x}$, $y'' = e^{-x}$, and

$$(y'')^2 - y^2 = e^{-2x} - e^{-2x} = 0.$$

From $(y'')^2 - y^2 = 0$ we have $y'' = \pm y$, which can be treated as two linear equations. Since linear combinations of solutions of linear homogeneous differential equations are also solutions, we see that $y = c_1 e^x + c_2 e^{-x}$ and $y = c_3 \cos x + c_4 \sin x$ must satisfy the differential equation. However, linear combinations that involve both exponential and trigonometric functions will not be solutions since the differential equation is not linear and each type of function satisfies a different linear differential equation that is part of the original differential equation.

19. Letting $u = y''$, separating variables, and integrating we have

$$\frac{du}{dx} = \sqrt{1 + u^2}, \quad \frac{du}{\sqrt{1 + u^2}} = dx, \quad \text{and} \quad \sinh^{-1} u = x + c_1.$$

Then

$$u = y'' = \sinh(x + c_1), \quad y' = \cosh(x + c_1) + c_2, \quad \text{and} \quad y = \sinh(x + c_1) + c_2 x + c_3.$$

20. If the constant $-c_1^2$ is used instead of c_1^2 , then, using partial fractions,

$$y = - \int \frac{dx}{x^2 - c_1^2} = - \frac{1}{2c_1} \int \left(\frac{1}{x - c_1} - \frac{1}{x + c_1} \right) dx = \frac{1}{2c_1} \ln \left| \frac{x + c_1}{x - c_1} \right| + c_2.$$

Alternatively, the inverse hyperbolic tangent can be used.

21. Let $u = dx/dt$ so that $d^2x/dt^2 = u du/dx$. The equation becomes $u du/dx = -k^2/x^2$. Separating variables we obtain

$$u du = -\frac{k^2}{x^2} dx \implies \frac{1}{2}u^2 = \frac{k^2}{x} + c \implies \frac{1}{2}v^2 = \frac{k^2}{x} + c.$$

When $t = 0$, $x = x_0$ and $v = 0$ so $0 = (k^2/x_0) + c$ and $c = -k^2/x_0$. Then

$$\frac{1}{2}v^2 = k^2 \left(\frac{1}{x} - \frac{1}{x_0} \right) \quad \text{and} \quad \frac{dx}{dt} = -k\sqrt{2} \sqrt{\frac{x_0 - x}{xx_0}}.$$

3.7 Nonlinear Equations

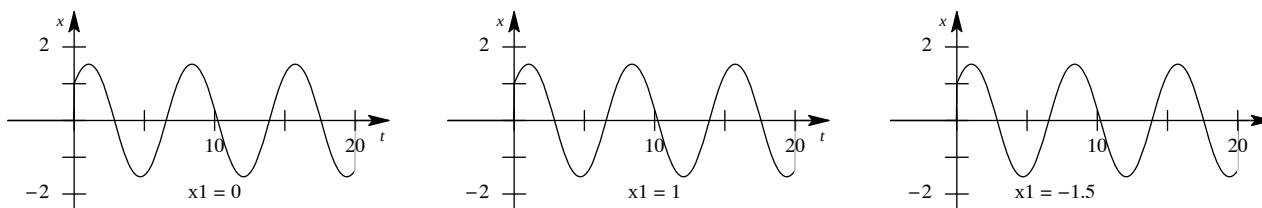
Separating variables we have

$$-\sqrt{\frac{xx_0}{x_0-x}} dx = k\sqrt{2} dt \implies t = -\frac{1}{k}\sqrt{\frac{x_0}{2}} \int \sqrt{\frac{x}{x_0-x}} dx.$$

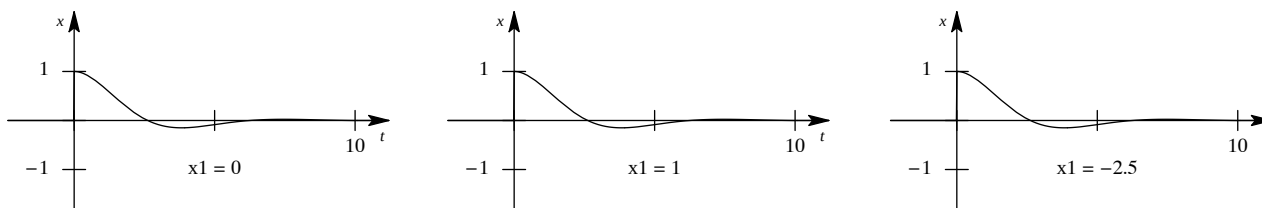
Using *Mathematica* to integrate we obtain

$$\begin{aligned} t &= -\frac{1}{k}\sqrt{\frac{x_0}{2}} \left[-\sqrt{x(x_0-x)} - \frac{x_0}{2} \tan^{-1} \frac{(x_0-2x)}{2x} \sqrt{\frac{x}{x_0-x}} \right] \\ &= \frac{1}{k}\sqrt{\frac{x_0}{2}} \left[\sqrt{x(x_0-x)} + \frac{x_0}{2} \tan^{-1} \frac{x_0-2x}{2\sqrt{x(x_0-x)}} \right]. \end{aligned}$$

22.



For $d^2x/dt^2 + \sin x = 0$ the motion appears to be periodic with amplitude 1 when $x_1 = 0$. The amplitude and period are larger for larger magnitudes of x_1 .



For $d^2x/dt^2 + dx/dt + \sin x = 0$ the motion appears to be periodic with decreasing amplitude. The dx/dt term could be said to have a damping effect.

EXERCISES 3.8

Linear Models: Initial-Value Problems

1. From $\frac{1}{8}x'' + 16x = 0$ we obtain

$$x = c_1 \cos 8\sqrt{2}t + c_2 \sin 8\sqrt{2}t$$

so that the period of motion is $2\pi/8\sqrt{2} = \sqrt{2}\pi/8$ seconds.

2. From $20x'' + kx = 0$ we obtain

$$x = c_1 \cos \frac{1}{2}\sqrt{\frac{k}{5}}t + c_2 \sin \frac{1}{2}\sqrt{\frac{k}{5}}t$$

so that the frequency $2/\pi = \frac{1}{4}\sqrt{k/5}\pi$ and $k = 320$ N/m. If $80x'' + 320x = 0$ then $x = c_1 \cos 2t + c_2 \sin 2t$ so that the frequency is $2/2\pi = 1/\pi$ cycles/s.

3. From $\frac{3}{4}x'' + 72x = 0$, $x(0) = -1/4$, and $x'(0) = 0$ we obtain $x = -\frac{1}{4} \cos 4\sqrt{6}t$.

4. From $\frac{3}{4}x'' + 72x = 0$, $x(0) = 0$, and $x'(0) = 2$ we obtain $x = \frac{\sqrt{6}}{12} \sin 4\sqrt{6}t$.
5. From $\frac{5}{8}x'' + 40x = 0$, $x(0) = 1/2$, and $x'(0) = 0$ we obtain $x = \frac{1}{2} \cos 8t$.
- (a) $x(\pi/12) = -1/4$, $x(\pi/8) = -1/2$, $x(\pi/6) = -1/4$, $x(\pi/4) = 1/2$, $x(9\pi/32) = \sqrt{2}/4$.
- (b) $x' = -4 \sin 8t$ so that $x'(3\pi/16) = 4$ ft/s directed downward.
- (c) If $x = \frac{1}{2} \cos 8t = 0$ then $t = (2n+1)\pi/16$ for $n = 0, 1, 2, \dots$.
6. From $50x'' + 200x = 0$, $x(0) = 0$, and $x'(0) = -10$ we obtain $x = -5 \sin 2t$ and $x' = -10 \cos 2t$.
7. From $20x'' + 20x = 0$, $x(0) = 0$, and $x'(0) = -10$ we obtain $x = -10 \sin t$ and $x' = -10 \cos t$.
- (a) The 20 kg mass has the larger amplitude.
- (b) 20 kg: $x'(\pi/4) = -5\sqrt{2}$ m/s, $x'(\pi/2) = 0$ m/s; 50 kg: $x'(\pi/4) = 0$ m/s, $x'(\pi/2) = 10$ m/s
- (c) If $-5 \sin 2t = -10 \sin t$ then $2 \sin t(\cos t - 1) = 0$ so that $t = n\pi$ for $n = 0, 1, 2, \dots$, placing both masses at the equilibrium position. The 50 kg mass is moving upward; the 20 kg mass is moving upward when n is even and downward when n is odd.

8. From $x'' + 16x = 0$, $x(0) = -1$, and $x'(0) = -2$ we obtain

$$x = -\cos 4t - \frac{1}{2} \sin 4t = \frac{\sqrt{5}}{2} \cos(4t - 3.605).$$

The period is $\pi/2$ seconds and the amplitude is $\sqrt{5}/2$ feet. In 4π seconds it will make 8 complete cycles.

9. From $\frac{1}{4}x'' + x = 0$, $x(0) = 1/2$, and $x'(0) = 3/2$ we obtain

$$x = \frac{1}{2} \cos 2t + \frac{3}{4} \sin 2t = \frac{\sqrt{13}}{4} \sin(2t + 0.588).$$

10. From $1.6x'' + 40x = 0$, $x(0) = -1/3$, and $x'(0) = 5/4$ we obtain

$$x = -\frac{1}{3} \cos 5t + \frac{1}{4} \sin 5t = \frac{5}{12} \sin(5t - 0.927).$$

If $x = 5/24$ then $t = \frac{1}{5} \left(\frac{\pi}{6} + 0.927 + 2n\pi \right)$ and $t = \frac{1}{5} \left(\frac{5\pi}{6} + 0.927 + 2n\pi \right)$ for $n = 0, 1, 2, \dots$.

11. From $2x'' + 200x = 0$, $x(0) = -2/3$, and $x'(0) = 5$ we obtain
- (a) $x = -\frac{2}{3} \cos 10t + \frac{1}{2} \sin 10t = \frac{5}{6} \sin(10t - 0.927)$.
- (b) The amplitude is $5/6$ ft and the period is $2\pi/10 = \pi/5$.
- (c) $3\pi = \pi k/5$ and $k = 15$ cycles.
- (d) If $x = 0$ and the weight is moving downward for the second time, then $10t - 0.927 = 2\pi$ or $t = 0.721$ s.
- (e) If $x' = \frac{25}{3} \cos(10t - 0.927) = 0$ then $10t - 0.927 = \pi/2 + n\pi$ or $t = (2n+1)\pi/20 + 0.0927$ for $n = 0, 1, 2, \dots$.
- (f) $x(3) = -0.597$ ft
- (g) $x'(3) = -5.814$ ft/s
- (h) $x''(3) = 59.702$ ft/s²
- (i) If $x = 0$ then $t = \frac{1}{10}(0.927 + n\pi)$ for $n = 0, 1, 2, \dots$. The velocity at these times is $x' = \pm 8.33$ ft/s.
- (j) If $x = 5/12$ then $t = \frac{1}{10}(\pi/6 + 0.927 + 2n\pi)$ and $t = \frac{1}{10}(5\pi/6 + 0.927 + 2n\pi)$ for $n = 0, 1, 2, \dots$.
- (k) If $x = 5/12$ and $x' < 0$ then $t = \frac{1}{10}(5\pi/6 + 0.927 + 2n\pi)$ for $n = 0, 1, 2, \dots$.
12. From $x'' + 9x = 0$, $x(0) = -1$, and $x'(0) = -\sqrt{3}$ we obtain

$$x = -\cos 3t - \frac{\sqrt{3}}{3} \sin 3t = \frac{2}{\sqrt{3}} \sin \left(3t + \frac{4\pi}{3} \right)$$

and $x' = 2\sqrt{3} \cos(3t + 4\pi/3)$. If $x' = 3$ then $t = -7\pi/18 + 2n\pi/3$ and $t = -\pi/2 + 2n\pi/3$ for $n = 1, 2, 3, \dots$.

3.8 Linear Models: Initial-Value Problems

13. From $k_1 = 40$ and $k_2 = 120$ we compute the effective spring constant $k = 4(40)(120)/160 = 120$. Now, $m = 20/32$ so $k/m = 120(32)/20 = 192$ and $x'' + 192x = 0$. Using $x(0) = 0$ and $x'(0) = 2$ we obtain $x(t) = \frac{\sqrt{3}}{12} \sin 8\sqrt{3}t$.
14. Let m be the mass and k_1 and k_2 the spring constants. Then $k = 4k_1k_2/(k_1 + k_2)$ is the effective spring constant of the system. Since the initial mass stretches one spring $\frac{1}{3}$ foot and another spring $\frac{1}{2}$ foot, using $F = ks$, we have $\frac{1}{3}k_1 = \frac{1}{2}k_2$ or $2k_1 = 3k_2$. The given period of the combined system is $2\pi/\omega = \pi/15$, so $\omega = 30$. Since a mass weighing 8 pounds is $\frac{1}{4}$ slug, we have from $w^2 = k/m$

$$30^2 = \frac{k}{1/4} = 4k \quad \text{or} \quad k = 225.$$

We now have the system of equations

$$\frac{4k_1k_2}{k_1 + k_2} = 225$$

$$2k_1 = 3k_2.$$

Solving the second equation for k_1 and substituting in the first equation, we obtain

$$\frac{4(3k_2/2)k_2}{3k_2/2 + k_2} = \frac{12k_2^2}{5k_2} = \frac{12k_2}{5} = 225.$$

Thus, $k_2 = 375/4$ and $k_1 = 1125/8$. Finally, the weight of the first mass is

$$32m = \frac{k_1}{3} = \frac{1125/8}{3} = \frac{375}{8} \approx 46.88 \text{ lb.}$$

15. For large values of t the differential equation is approximated by $x'' = 0$. The solution of this equation is the linear function $x = c_1t + c_2$. Thus, for large time, the restoring force will have decayed to the point where the spring is incapable of returning the mass, and the spring will simply keep on stretching.
16. As t becomes larger the spring constant increases; that is, the spring is stiffening. It would seem that the oscillations would become periodic and the spring would oscillate more rapidly. It is likely that the amplitudes of the oscillations would decrease as t increases.
17. (a) above (b) heading upward
18. (a) below (b) from rest
19. (a) below (b) heading upward
20. (a) above (b) heading downward
21. From $\frac{1}{8}x'' + x' + 2x = 0$, $x(0) = -1$, and $x'(0) = 8$ we obtain $x = 4te^{-4t} - e^{-4t}$ and $x' = 8e^{-4t} - 16te^{-4t}$. If $x = 0$ then $t = 1/4$ second. If $x' = 0$ then $t = 1/2$ second and the extreme displacement is $x = e^{-2}$ feet.
22. From $\frac{1}{4}x'' + \sqrt{2}x' + 2x = 0$, $x(0) = 0$, and $x'(0) = 5$ we obtain $x = 5te^{-2\sqrt{2}t}$ and $x' = 5e^{-2\sqrt{2}t}(1 - 2\sqrt{2}t)$. If $x' = 0$ then $t = \sqrt{2}/4$ second and the extreme displacement is $x = 5\sqrt{2}e^{-1}/4$ feet.
23. (a) From $x'' + 10x' + 16x = 0$, $x(0) = 1$, and $x'(0) = 0$ we obtain $x = \frac{4}{3}e^{-2t} - \frac{1}{3}e^{-8t}$.
 (b) From $x'' + x' + 16x = 0$, $x(0) = 1$, and $x'(0) = -12$ then $x = -\frac{2}{3}e^{-2t} + \frac{5}{3}e^{-8t}$.
24. (a) $x = \frac{1}{3}e^{-8t}(4e^{6t} - 1)$ is not zero for $t \geq 0$; the extreme displacement is $x(0) = 1$ meter.
 (b) $x = \frac{1}{3}e^{-8t}(5 - 2e^{6t}) = 0$ when $t = \frac{1}{6} \ln \frac{5}{2} \approx 0.153$ second; if $x' = \frac{4}{3}e^{-8t}(e^{6t} - 10) = 0$ then $t = \frac{1}{6} \ln 10 \approx 0.384$ second and the extreme displacement is $x = -0.232$ meter.

25. (a) From $0.1x'' + 0.4x' + 2x = 0$, $x(0) = -1$, and $x'(0) = 0$ we obtain $x = e^{-2t} [-\cos 4t - \frac{1}{2} \sin 4t]$.

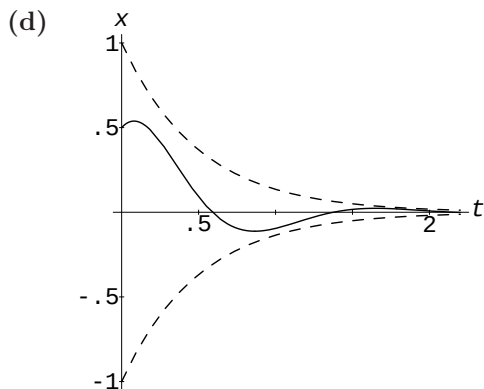
(b) $x = \frac{\sqrt{5}}{2} e^{-2t} \sin(4t + 4.25)$

(c) If $x = 0$ then $4t + 4.25 = 2\pi, 3\pi, 4\pi, \dots$ so that the first time heading upward is $t = 1.294$ seconds.

26. (a) From $\frac{1}{4}x'' + x' + 5x = 0$, $x(0) = 1/2$, and $x'(0) = 1$ we obtain $x = e^{-2t} (\frac{1}{2} \cos 4t + \frac{1}{2} \sin 4t)$.

(b) $x = \frac{1}{\sqrt{2}} e^{-2t} \sin \left(4t + \frac{\pi}{4} \right)$.

(c) If $x = 0$ then $4t + \pi/4 = \pi, 2\pi, 3\pi, \dots$ so that the times heading downward are $t = (7 + 8n)\pi/16$ for $n = 0, 1, 2, \dots$.



27. From $\frac{5}{16}x'' + \beta x' + 5x = 0$ we find that the roots of the auxiliary equation are $m = -\frac{8}{5}\beta \pm \frac{4}{5}\sqrt{4\beta^2 - 25}$.

(a) If $4\beta^2 - 25 > 0$ then $\beta > 5/2$.

(b) If $4\beta^2 - 25 = 0$ then $\beta = 5/2$.

(c) If $4\beta^2 - 25 < 0$ then $0 < \beta < 5/2$.

28. From $0.75x'' + \beta x' + 6x = 0$ and $\beta > 3\sqrt{2}$ we find that the roots of the auxiliary equation are $m = -\frac{2}{3}\beta \pm \frac{2}{3}\sqrt{\beta^2 - 18}$ and

$$x = e^{-2\beta t/3} \left[c_1 \cosh \frac{2}{3}\sqrt{\beta^2 - 18}t + c_2 \sinh \frac{2}{3}\sqrt{\beta^2 - 18}t \right].$$

If $x(0) = 0$ and $x'(0) = -2$ then $c_1 = 0$ and $c_2 = -3/\sqrt{\beta^2 - 18}$.

29. If $\frac{1}{2}x'' + \frac{1}{2}x' + 6x = 10 \cos 3t$, $x(0) = -2$, and $x'(0) = 0$ then

$$x_c = e^{-t/2} \left(c_1 \cos \frac{\sqrt{47}}{2}t + c_2 \sin \frac{\sqrt{47}}{2}t \right)$$

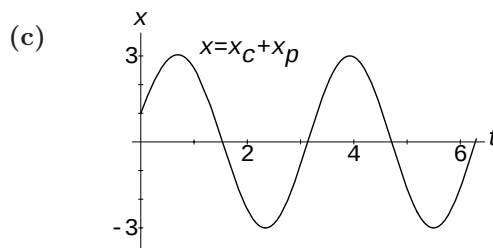
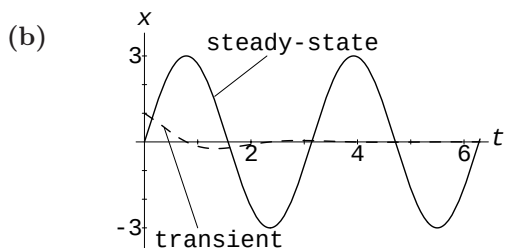
and $x_p = \frac{10}{3}(\cos 3t + \sin 3t)$ so that the equation of motion is

$$x = e^{-t/2} \left(-\frac{4}{3} \cos \frac{\sqrt{47}}{2}t - \frac{64}{3\sqrt{47}} \sin \frac{\sqrt{47}}{2}t \right) + \frac{10}{3}(\cos 3t + \sin 3t).$$

30. (a) If $x'' + 2x' + 5x = 12 \cos 2t + 3 \sin 2t$, $x(0) = 1$, and $x'(0) = 5$ then $x_c = e^{-t}(c_1 \cos 2t + c_2 \sin 2t)$ and $x_p = 3 \sin 2t$ so that the equation of motion is

$$x = e^{-t} \cos 2t + 3 \sin 2t.$$

3.8 Linear Models: Initial-Value Problems



31. From $x'' + 8x' + 16x = 8 \sin 4t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 e^{-4t} + c_2 t e^{-4t}$ and $x_p = -\frac{1}{4} \cos 4t$ so that the equation of motion is

$$x = \frac{1}{4} e^{-4t} + t e^{-4t} - \frac{1}{4} \cos 4t.$$

32. From $x'' + 8x' + 16x = e^{-t} \sin 4t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 e^{-4t} + c_2 t e^{-4t}$ and $x_p = -\frac{24}{625} e^{-t} \cos 4t - \frac{7}{625} e^{-t} \sin 4t$ so that

$$x = \frac{1}{625} e^{-4t} (24 + 100t) - \frac{1}{625} e^{-t} (24 \cos 4t + 7 \sin 4t).$$

As $t \rightarrow \infty$ the displacement $x \rightarrow 0$.

33. From $2x'' + 32x = 68e^{-2t} \cos 4t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos 4t + c_2 \sin 4t$ and $x_p = \frac{1}{2} e^{-2t} \cos 4t - 2e^{-2t} \sin 4t$ so that

$$x = -\frac{1}{2} \cos 4t + \frac{9}{4} \sin 4t + \frac{1}{2} e^{-2t} \cos 4t - 2e^{-2t} \sin 4t.$$

34. Since $x = \frac{\sqrt{85}}{4} \sin(4t - 0.219) - \frac{\sqrt{17}}{2} e^{-2t} \sin(4t - 2.897)$, the amplitude approaches $\sqrt{85}/4$ as $t \rightarrow \infty$.

35. (a) By Hooke's law the external force is $F(t) = kh(t)$ so that $mx'' + \beta x' + kx = kh(t)$.

- (b) From $\frac{1}{2}x'' + 2x' + 4x = 20 \cos t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = e^{-2t}(c_1 \cos 2t + c_2 \sin 2t)$ and $x_p = \frac{56}{13} \cos t + \frac{32}{13} \sin t$ so that

$$x = e^{-2t} \left(-\frac{56}{13} \cos 2t - \frac{72}{13} \sin 2t \right) + \frac{56}{13} \cos t + \frac{32}{13} \sin t.$$

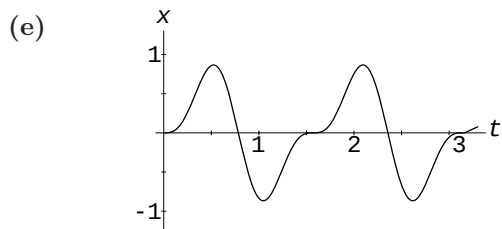
36. (a) From $100x'' + 1600x = 1600 \sin 8t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos 4t + c_2 \sin 4t$ and $x_p = -\frac{1}{3} \sin 8t$ so that by a trig identity

$$x = \frac{2}{3} \sin 4t - \frac{1}{3} \sin 8t = \frac{2}{3} \sin 4t - \frac{2}{3} \sin 4t \cos 4t.$$

- (b) If $x = \frac{1}{3} \sin 4t(2 - 2 \cos 4t) = 0$ then $t = n\pi/4$ for $n = 0, 1, 2, \dots$.

- (c) If $x' = \frac{8}{3} \cos 4t - \frac{8}{3} \cos 8t = \frac{8}{3} (1 - \cos 4t)(1 + 2 \cos 4t) = 0$ then $t = \pi/3 + n\pi/2$ and $t = \pi/6 + n\pi/2$ for $n = 0, 1, 2, \dots$ at the extreme values. *Note:* There are many other values of t for which $x' = 0$.

- (d) $x(\pi/6 + n\pi/2) = \sqrt{3}/2$ cm and $x(\pi/3 + n\pi/2) = -\sqrt{3}/2$ cm.



37. From $x'' + 4x = -5 \sin 2t + 3 \cos 2t$, $x(0) = -1$, and $x'(0) = 1$ we obtain $x_c = c_1 \cos 2t + c_2 \sin 2t$, $x_p = \frac{3}{4}t \sin 2t + \frac{5}{4}t \cos 2t$, and

$$x = -\cos 2t - \frac{1}{8} \sin 2t + \frac{3}{4}t \sin 2t + \frac{5}{4}t \cos 2t.$$

38. From $x'' + 9x = 5 \sin 3t$, $x(0) = 2$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos 3t + c_2 \sin 3t$, $x_p = -\frac{5}{6}t \cos 3t$, and

$$x = 2 \cos 3t + \frac{5}{18} \sin 3t - \frac{5}{6}t \cos 3t.$$

39. (a) From $x'' + \omega^2 x = F_0 \cos \gamma t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos \omega t + c_2 \sin \omega t$ and $x_p = (F_0 \cos \gamma t)/(\omega^2 - \gamma^2)$ so that

$$x = -\frac{F_0}{\omega^2 - \gamma^2} \cos \omega t + \frac{F_0}{\omega^2 - \gamma^2} \cos \gamma t.$$

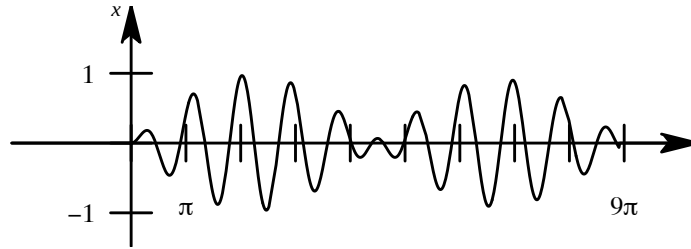
(b) $\lim_{\gamma \rightarrow \omega} \frac{F_0}{\omega^2 - \gamma^2} (\cos \gamma t - \cos \omega t) = \lim_{\gamma \rightarrow \omega} \frac{-F_0 t \sin \gamma t}{-2\gamma} = \frac{F_0}{2\omega} t \sin \omega t.$

40. From $x'' + \omega^2 x = F_0 \cos \omega t$, $x(0) = 0$, and $x'(0) = 0$ we obtain $x_c = c_1 \cos \omega t + c_2 \sin \omega t$ and $x_p = (F_0 t/2\omega) \sin \omega t$ so that $x = (F_0 t/2\omega) \sin \omega t$.

41. (a) From $\cos(u - v) = \cos u \cos v + \sin u \sin v$ and $\cos(u + v) = \cos u \cos v - \sin u \sin v$ we obtain $\sin u \sin v = \frac{1}{2}[\cos(u - v) - \cos(u + v)]$. Letting $u = \frac{1}{2}(\gamma - \omega)t$ and $v = \frac{1}{2}(\gamma + \omega)t$, the result follows.

(b) If $\epsilon = \frac{1}{2}(\gamma - \omega)$ then $\gamma \approx \omega$ so that $x = (F_0/2\epsilon\gamma) \sin \epsilon t \sin \gamma t$.

42. See the article "Distinguished Oscillations of a Forced Harmonic Oscillator" by T.G. Procter in *The College Mathematics Journal*, March, 1995. In this article the author illustrates that for $F_0 = 1$, $\lambda = 0.01$, $\gamma = 22/9$, and $\omega = 2$ the system exhibits beats oscillations on the interval $[0, 9\pi]$, but that this phenomenon is transient as $t \rightarrow \infty$.



43. (a) The general solution of the homogeneous equation is

$$\begin{aligned} x_c(t) &= c_1 e^{-\lambda t} \cos(\sqrt{\omega^2 - \lambda^2} t) + c_2 e^{-\lambda t} \sin(\sqrt{\omega^2 - \lambda^2} t) \\ &= A e^{-\lambda t} \sin[\sqrt{\omega^2 - \lambda^2} t + \phi], \end{aligned}$$

where $A = \sqrt{c_1^2 + c_2^2}$, $\sin \phi = c_1/A$, and $\cos \phi = c_2/A$. Now

$$x_p(t) = \frac{F_0(\omega^2 - \gamma^2)}{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2} \sin \gamma t + \frac{F_0(-2\lambda\gamma)}{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2} \cos \gamma t = A \sin(\gamma t + \theta),$$

where

$$\sin \theta = \frac{\frac{F_0(-2\lambda\gamma)}{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}{\frac{F_0}{\sqrt{\omega^2 - \gamma^2 + 4\lambda^2\gamma^2}}} = \frac{-2\lambda\gamma}{\sqrt{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}$$

and

3.8 Linear Models: Initial-Value Problems

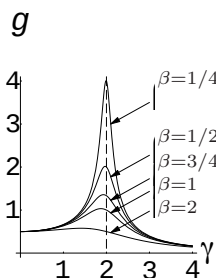
$$\cos \theta = \frac{\frac{F_0(\omega^2 - \gamma^2)}{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}{\frac{F_0}{\sqrt{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}} = \frac{\omega^2 - \gamma^2}{\sqrt{(\omega^2 - \gamma^2)^2 + 4\lambda^2\gamma^2}}.$$

- (b) If $g'(\gamma) = 0$ then $\gamma(\gamma^2 + 2\lambda^2 - \omega^2) = 0$ so that $\gamma = 0$ or $\gamma = \sqrt{\omega^2 - 2\lambda^2}$. The first derivative test shows that g has a maximum value at $\gamma = \sqrt{\omega^2 - 2\lambda^2}$. The maximum value of g is

$$g\left(\sqrt{\omega^2 - 2\lambda^2}\right) = F_0/2\lambda\sqrt{\omega^2 - \lambda^2}.$$

- (c) We identify $\omega^2 = k/m = 4$, $\lambda = \beta/2$, and $\gamma_1 = \sqrt{\omega^2 - 2\lambda^2} = \sqrt{4 - \beta^2/2}$. As $\beta \rightarrow 0$, $\gamma_1 \rightarrow 2$ and the resonance curve grows without bound at $\gamma_1 = 2$. That is, the system approaches pure resonance.

β	γ_1	g
2.00	1.41	0.58
1.00	1.87	1.03
0.75	1.93	1.36
0.50	1.97	2.02
0.25	1.99	4.01



44. (a) For $n = 2$, $\sin^2 \gamma t = \frac{1}{2}(1 - \cos 2\gamma t)$. The system is in pure resonance when $2\gamma_1/2\pi = \omega/2\pi$, or when $\gamma_1 = \omega/2$.

- (b) Note that

$$\sin^3 \gamma t = \sin \gamma t \sin^2 \gamma t = \frac{1}{2}[\sin \gamma t - \sin \gamma t \cos 2\gamma t].$$

Now

$$\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$$

so

$$\sin \gamma t \cos 2\gamma t = \frac{1}{2}[\sin 3\gamma t - \sin \gamma t]$$

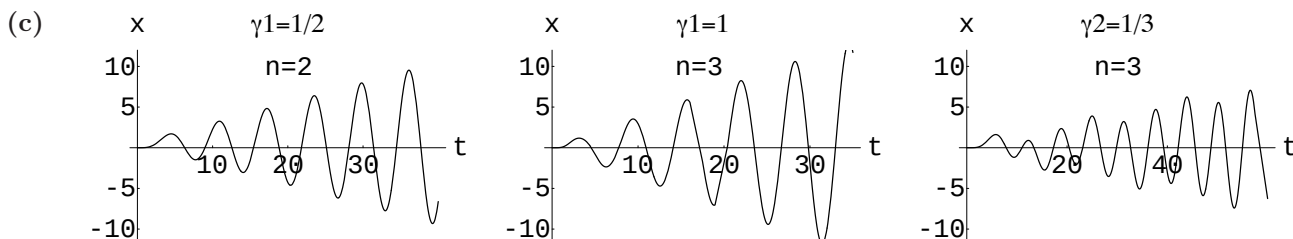
and

$$\sin^3 \gamma t = \frac{3}{4} \sin \gamma t - \frac{1}{4} \sin 3\gamma t.$$

Thus

$$x'' + \omega^2 x = \frac{3}{4} \sin \gamma t - \frac{1}{4} \sin 3\gamma t.$$

The frequency of free vibration is $\omega/2\pi$. Thus, when $\gamma_1/2\pi = \omega/2\pi$ or $\gamma_1 = \omega$, and when $3\gamma_2/2\pi = \omega/2\pi$ or $3\gamma_2 = \omega$ or $\gamma_3 = \omega/3$, the system will be in pure resonance.



3.8 Linear Models: Initial-Value Problems

45. Solving $\frac{1}{20}q'' + 2q' + 100q = 0$ we obtain $q(t) = e^{-20t}(c_1 \cos 40t + c_2 \sin 40t)$. The initial conditions $q(0) = 5$ and $q'(0) = 0$ imply $c_1 = 5$ and $c_2 = 5/2$. Thus

$$q(t) = e^{-20t} \left(5 \cos 40t + \frac{5}{2} \sin 40t \right) = \sqrt{25 + 25/4} e^{-20t} \sin(40t + 1.1071)$$

and $q(0.01) \approx 4.5676$ coulombs. The charge is zero for the first time when $40t + 1.1071 = \pi$ or $t \approx 0.0509$ second.

46. Solving $\frac{1}{4}q'' + 20q' + 300q = 0$ we obtain $q(t) = c_1 e^{-20t} + c_2 e^{-60t}$. The initial conditions $q(0) = 4$ and $q'(0) = 0$ imply $c_1 = 6$ and $c_2 = -2$. Thus

$$q(t) = 6e^{-20t} - 2e^{-60t}.$$

Setting $q = 0$ we find $e^{40t} = 1/3$ which implies $t < 0$. Therefore the charge is not 0 for $t \geq 0$.

47. Solving $\frac{5}{3}q'' + 10q' + 30q = 300$ we obtain $q(t) = e^{-3t}(c_1 \cos 3t + c_2 \sin 3t) + 10$. The initial conditions $q(0) = q'(0) = 0$ imply $c_1 = c_2 = -10$. Thus

$$q(t) = 10 - 10e^{-3t}(\cos 3t + \sin 3t) \quad \text{and} \quad i(t) = 60e^{-3t} \sin 3t.$$

Solving $i(t) = 0$ we see that the maximum charge occurs when $t = \pi/3$ and $q(\pi/3) \approx 10.432$.

48. Solving $q'' + 100q' + 2500q = 30$ we obtain $q(t) = c_1 e^{-50t} + c_2 t e^{-50t} + 0.012$. The initial conditions $q(0) = 0$ and $q'(0) = 2$ imply $c_1 = -0.012$ and $c_2 = 1.4$. Thus, using $i(t) = q'(t)$ we get

$$q(t) = -0.012e^{-50t} + 1.4te^{-50t} + 0.012 \quad \text{and} \quad i(t) = 2e^{-50t} - 70te^{-50t}.$$

Solving $i(t) = 0$ we see that the maximum charge occurs when $t = 1/35$ second and $q(1/35) \approx 0.01871$ coulomb.

49. Solving $q'' + 2q' + 4q = 0$ we obtain $q_c = e^{-t}(\cos \sqrt{3}t + \sin \sqrt{3}t)$. The steady-state charge has the form $q_p = A \cos t + B \sin t$. Substituting into the differential equation we find

$$(3A + 2B) \cos t + (3B - 2A) \sin t = 50 \cos t.$$

Thus, $A = 150/13$ and $B = 100/13$. The steady-state charge is

$$q_p(t) = \frac{150}{13} \cos t + \frac{100}{13} \sin t$$

and the steady-state current is

$$i_p(t) = -\frac{150}{13} \sin t + \frac{100}{13} \cos t.$$

50. From

$$i_p(t) = \frac{E_0}{Z} \left(\frac{R}{Z} \sin \gamma t - \frac{X}{Z} \cos \gamma t \right)$$

and $Z = \sqrt{X^2 + R^2}$ we see that the amplitude of $i_p(t)$ is

$$A = \sqrt{\frac{E_0^2 R^2}{Z^4} + \frac{E_0^2 X^2}{Z^4}} = \frac{E_0}{Z^2} \sqrt{R^2 + X^2} = \frac{E_0}{Z}.$$

51. The differential equation is $\frac{1}{2}q'' + 20q' + 1000q = 100 \sin 60t$. To use Example 10 in the text we identify $E_0 = 100$ and $\gamma = 60$. Then

$$X = L\gamma - \frac{1}{c\gamma} = \frac{1}{2}(60) - \frac{1}{0.001(60)} \approx 13.3333,$$

$$Z = \sqrt{X^2 + R^2} = \sqrt{X^2 + 400} \approx 24.0370,$$

and

3.8 Linear Models: Initial-Value Problems

$$\frac{E_0}{Z} = \frac{100}{Z} \approx 4.1603.$$

From Problem 50, then

$$i_p(t) \approx 4.1603 \sin(60t + \phi)$$

where $\sin \phi = -X/Z$ and $\cos \phi = R/Z$. Thus $\tan \phi = -X/R \approx -0.6667$ and ϕ is a fourth quadrant angle. Now $\phi \approx -0.5880$ and

$$i_p(t) = 4.1603 \sin(60t - 0.5880).$$

52. Solving $\frac{1}{2}q'' + 20q' + 1000q = 0$ we obtain $q_c(t) = e^{-20t}(c_1 \cos 40t + c_2 \sin 40t)$. The steady-state charge has the form $q_p(t) = A \sin 60t + B \cos 60t + C \sin 40t + D \cos 40t$. Substituting into the differential equation we find

$$\begin{aligned} & (-1600A - 2400B) \sin 60t + (2400A - 1600B) \cos 60t \\ & + (400C - 1600D) \sin 40t + (1600C + 400D) \cos 40t \\ & = 200 \sin 60t + 400 \cos 40t. \end{aligned}$$

Equating coefficients we obtain $A = -1/26$, $B = -3/52$, $C = 4/17$, and $D = 1/17$. The steady-state charge is

$$q_p(t) = -\frac{1}{26} \sin 60t - \frac{3}{52} \cos 60t + \frac{4}{17} \sin 40t + \frac{1}{17} \cos 40t$$

and the steady-state current is

$$i_p(t) = -\frac{30}{13} \cos 60t + \frac{45}{13} \sin 60t + \frac{160}{17} \cos 40t - \frac{40}{17} \sin 40t.$$

53. Solving $\frac{1}{2}q'' + 10q' + 100q = 150$ we obtain $q(t) = e^{-10t}(c_1 \cos 10t + c_2 \sin 10t) + 3/2$. The initial conditions $q(0) = 1$ and $q'(0) = 0$ imply $c_1 = c_2 = -1/2$. Thus

$$q(t) = -\frac{1}{2}e^{-10t}(\cos 10t + \sin 10t) + \frac{3}{2}.$$

As $t \rightarrow \infty$, $q(t) \rightarrow 3/2$.

54. In Problem 50 it is shown that the amplitude of the steady-state current is E_0/Z , where $Z = \sqrt{X^2 + R^2}$ and $X = L\gamma - 1/C\gamma$. Since E_0 is constant the amplitude will be a maximum when Z is a minimum. Since R is constant, Z will be a minimum when $X = 0$. Solving $L\gamma - 1/C\gamma = 0$ for γ we obtain $\gamma = 1/\sqrt{LC}$. The maximum amplitude will be E_0/R .
55. By Problem 50 the amplitude of the steady-state current is E_0/Z , where $Z = \sqrt{X^2 + R^2}$ and $X = L\gamma - 1/C\gamma$. Since E_0 is constant the amplitude will be a maximum when Z is a minimum. Since R is constant, Z will be a minimum when $X = 0$. Solving $L\gamma - 1/C\gamma = 0$ for C we obtain $C = 1/L\gamma^2$.
56. Solving $0.1q'' + 10q = 100 \sin \gamma t$ we obtain

$$q(t) = c_1 \cos 10t + c_2 \sin 10t + q_p(t)$$

where $q_p(t) = A \sin \gamma t + B \cos \gamma t$. Substituting $q_p(t)$ into the differential equation we find

$$(100 - \gamma^2)A \sin \gamma t + (100 - \gamma^2)B \cos \gamma t = 100 \sin \gamma t.$$

Equating coefficients we obtain $A = 100/(100 - \gamma^2)$ and $B = 0$. Thus, $q_p(t) = \frac{100}{100 - \gamma^2} \sin \gamma t$. The initial conditions $q(0) = q'(0) = 0$ imply $c_1 = 0$ and $c_2 = -10\gamma/(100 - \gamma^2)$. The charge is

$$q(t) = \frac{10}{100 - \gamma^2}(10 \sin \gamma t - \gamma \sin 10t)$$

and the current is

$$i(t) = \frac{100\gamma}{100 - \gamma^2}(\cos \gamma t - \cos 10t).$$

57. In an LC -series circuit there is no resistor, so the differential equation is

$$L \frac{d^2 q}{dt^2} + \frac{1}{C} q = E(t).$$

Then $q(t) = c_1 \cos(t/\sqrt{LC}) + c_2 \sin(t/\sqrt{LC}) + q_p(t)$ where $q_p(t) = A \sin \gamma t + B \cos \gamma t$. Substituting $q_p(t)$ into the differential equation we find

$$\left(\frac{1}{C} - L\gamma^2\right) A \sin \gamma t + \left(\frac{1}{C} - L\gamma^2\right) B \cos \gamma t = E_0 \cos \gamma t.$$

Equating coefficients we obtain $A = 0$ and $B = E_0 C / (1 - LC\gamma^2)$. Thus, the charge is

$$q(t) = c_1 \cos \frac{1}{\sqrt{LC}} t + c_2 \sin \frac{1}{\sqrt{LC}} t + \frac{E_0 C}{1 - LC\gamma^2} \cos \gamma t.$$

The initial conditions $q(0) = q_0$ and $q'(0) = i_0$ imply $c_1 = q_0 - E_0 C / (1 - LC\gamma^2)$ and $c_2 = i_0 \sqrt{LC}$. The current is $i(t) = q'(t)$ or

$$\begin{aligned} i(t) &= -\frac{c_1}{\sqrt{LC}} \sin \frac{1}{\sqrt{LC}} t + \frac{c_2}{\sqrt{LC}} \cos \frac{1}{\sqrt{LC}} t - \frac{E_0 C \gamma}{1 - LC\gamma^2} \sin \gamma t \\ &= i_0 \cos \frac{1}{\sqrt{LC}} t - \frac{1}{\sqrt{LC}} \left(q_0 - \frac{E_0 C}{1 - LC\gamma^2} \right) \sin \frac{1}{\sqrt{LC}} t - \frac{E_0 C \gamma}{1 - LC\gamma^2} \sin \gamma t. \end{aligned}$$

58. When the circuit is in resonance the form of $q_p(t)$ is $q_p(t) = At \cos kt + Bt \sin kt$ where $k = 1/\sqrt{LC}$. Substituting $q_p(t)$ into the differential equation we find

$$q_p'' + k^2 q_p = -2kA \sin kt + 2kB \cos kt = \frac{E_0}{L} \cos kt.$$

Equating coefficients we obtain $A = 0$ and $B = E_0 / 2kL$. The charge is

$$q(t) = c_1 \cos kt + c_2 \sin kt + \frac{E_0}{2kL} t \sin kt.$$

The initial conditions $q(0) = q_0$ and $q'(0) = i_0$ imply $c_1 = q_0$ and $c_2 = i_0/k$. The current is

$$\begin{aligned} i(t) &= -c_1 k \sin kt + c_2 k \cos kt + \frac{E_0}{2kL} (kt \cos kt + \sin kt) \\ &= \left(\frac{E_0}{2kL} - q_0 k \right) \sin kt + i_0 \cos kt + \frac{E_0}{2L} t \cos kt. \end{aligned}$$

EXERCISES 3.9

Linear Models: Boundary-Value Problems

1. (a) The general solution is

$$y(x) = c_1 + c_2 x + c_3 x^2 + c_4 x^3 + \frac{w_0}{24EI} x^4.$$

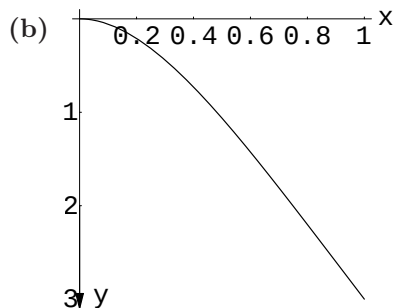
3.9 Linear Models: Boundary-Value Problems

The boundary conditions are $y(0) = 0$, $y'(0) = 0$, $y''(L) = 0$, $y'''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_2 = 0$. The conditions at $x = L$ give the system

$$\begin{aligned} 2c_3 + 6c_4L + \frac{w_0}{2EI}L^2 &= 0 \\ 6c_4 + \frac{w_0}{EI}L &= 0. \end{aligned}$$

Solving, we obtain $c_3 = w_0L^2/4EI$ and $c_4 = -w_0L/6EI$. The deflection is

$$y(x) = \frac{w_0}{24EI}(6L^2x^2 - 4Lx^3 + x^4).$$



2. (a) The general solution is

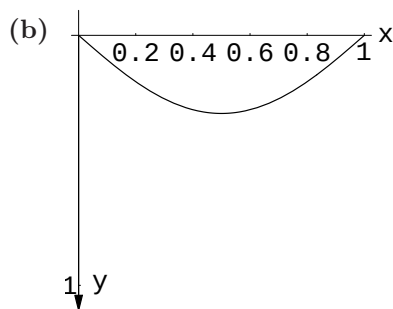
$$y(x) = c_1 + c_2x + c_3x^2 + c_4x^3 + \frac{w_0}{24EI}x^4.$$

The boundary conditions are $y(0) = 0$, $y''(0) = 0$, $y(L) = 0$, $y''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_3 = 0$. The conditions at $x = L$ give the system

$$\begin{aligned} c_2L + c_4L^3 + \frac{w_0}{24EI}L^4 &= 0 \\ 6c_4L + \frac{w_0}{2EI}L^2 &= 0. \end{aligned}$$

Solving, we obtain $c_2 = w_0L^3/24EI$ and $c_4 = -w_0L/12EI$. The deflection is

$$y(x) = \frac{w_0}{24EI}(L^3x - 2Lx^3 + x^4).$$



3. (a) The general solution is

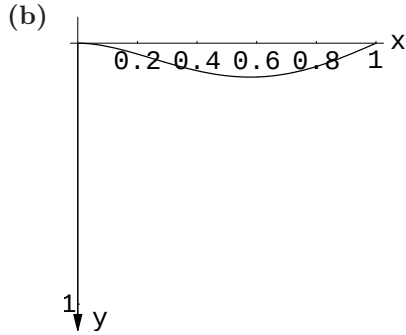
$$y(x) = c_1 + c_2x + c_3x^2 + c_4x^3 + \frac{w_0}{24EI}x^4.$$

The boundary conditions are $y(0) = 0$, $y'(0) = 0$, $y(L) = 0$, $y''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_2 = 0$. The conditions at $x = L$ give the system

$$\begin{aligned} c_3L^2 + c_4L^3 + \frac{w_0}{24EI}L^4 &= 0 \\ 2c_3 + 6c_4L + \frac{w_0}{2EI}L^2 &= 0. \end{aligned}$$

Solving, we obtain $c_3 = w_0 L^2 / 16EI$ and $c_4 = -5w_0 L / 48EI$. The deflection is

$$y(x) = \frac{w_0}{48EI} (3L^2 x^2 - 5Lx^3 + 2x^4).$$



4. (a) The general solution is

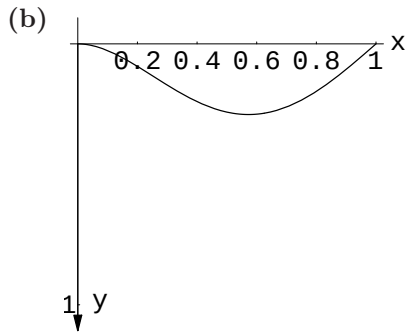
$$y(x) = c_1 + c_2 x + c_3 x^2 + c_4 x^3 + \frac{w_0 L^4}{EI \pi^4} \sin \frac{\pi}{L} x.$$

The boundary conditions are $y(0) = 0$, $y'(0) = 0$, $y(L) = 0$, $y''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_2 = -w_0 L^3 / EI \pi^3$. The conditions at $x = L$ give the system

$$\begin{aligned} c_3 L^2 + c_4 L^3 + \frac{w_0}{EI \pi^3} L^4 &= 0 \\ 2c_3 + 6c_4 L &= 0. \end{aligned}$$

Solving, we obtain $c_3 = 3w_0 L^2 / 2EI \pi^3$ and $c_4 = -w_0 L / 2EI \pi^3$. The deflection is

$$y(x) = \frac{w_0 L}{2EI \pi^3} \left(-2L^2 x + 3Lx^2 - x^3 + \frac{2L^3}{\pi} \sin \frac{\pi}{L} x \right).$$



- (c) Using a CAS we find the maximum deflection to be 0.270806 when $x = 0.572536$.

5. (a) The general solution is

$$y(x) = c_1 + c_2 x + c_3 x^2 + c_4 x^3 + \frac{w_0}{120EI} x^5.$$

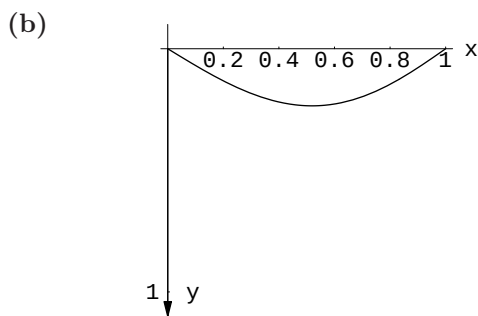
The boundary conditions are $y(0) = 0$, $y''(0) = 0$, $y(L) = 0$, $y''(L) = 0$. The first two conditions give $c_1 = 0$ and $c_3 = 0$. The conditions at $x = L$ give the system

$$\begin{aligned} c_2 L + c_4 L^3 + \frac{w_0}{120EI} L^5 &= 0 \\ 6c_4 L + \frac{w_0}{6EI} L^3 &= 0. \end{aligned}$$

Solving, we obtain $c_2 = 7w_0 L^4 / 360EI$ and $c_4 = -w_0 L^2 / 36EI$. The deflection is

$$y(x) = \frac{w_0}{360EI} (7L^4 x - 10L^2 x^3 + 3x^5).$$

3.9 Linear Models: Boundary-Value Problems



(c) Using a CAS we find the maximum deflection to be 0.234799 when $x = 0.51933$.

6. (a) $y_{\max} = y(L) = w_0 L^4 / 8EI$
 - (b) Replacing both L and x by $L/2$ in $y(x)$ we obtain $w_0 L^4 / 128EI$, which is $1/16$ of the maximum deflection when the length of the beam is L .
 - (c) $y_{\max} = y(L/2) = 5w_0 L^4 / 384EI$
 - (d) The maximum deflection in Example 1 is $y(L/2) = (w_0 / 24EI) L^4 / 16 = w_0 L^4 / 384EI$, which is $1/5$ of the maximum displacement of the beam in part c.
7. The general solution of the differential equation is

$$y = c_1 \cosh \sqrt{\frac{P}{EI}} x + c_2 \sinh \sqrt{\frac{P}{EI}} x + \frac{w_0}{2P} x^2 + \frac{w_0 EI}{P^2}.$$

Setting $y(0) = 0$ we obtain $c_1 = -w_0 EI / P^2$, so that

$$y = -\frac{w_0 EI}{P^2} \cosh \sqrt{\frac{P}{EI}} x + c_2 \sinh \sqrt{\frac{P}{EI}} x + \frac{w_0}{2P} x^2 + \frac{w_0 EI}{P^2}.$$

Setting $y'(L) = 0$ we find

$$c_2 = \left(\sqrt{\frac{P}{EI}} \frac{w_0 EI}{P^2} \sinh \sqrt{\frac{P}{EI}} L - \frac{w_0 L}{P} \right) / \left(\sqrt{\frac{P}{EI}} \cosh \sqrt{\frac{P}{EI}} L \right).$$

8. The general solution of the differential equation is

$$y = c_1 \cos \sqrt{\frac{P}{EI}} x + c_2 \sin \sqrt{\frac{P}{EI}} x + \frac{w_0}{2P} x^2 + \frac{w_0 EI}{P^2}.$$

Setting $y(0) = 0$ we obtain $c_1 = -w_0 EI / P^2$, so that

$$y = -\frac{w_0 EI}{P^2} \cos \sqrt{\frac{P}{EI}} x + c_2 \sin \sqrt{\frac{P}{EI}} x + \frac{w_0}{2P} x^2 + \frac{w_0 EI}{P^2}.$$

Setting $y'(L) = 0$ we find

$$c_2 = \left(-\sqrt{\frac{P}{EI}} \frac{w_0 EI}{P^2} \sin \sqrt{\frac{P}{EI}} L - \frac{w_0 L}{P} \right) / \left(\sqrt{\frac{P}{EI}} \cos \sqrt{\frac{P}{EI}} L \right).$$

9. This is Example 2 in the text with $L = \pi$. The eigenvalues are $\lambda_n = n^2 \pi^2 / \pi^2 = n^2$, $n = 1, 2, 3, \dots$ and the corresponding eigenfunctions are $y_n = \sin(n\pi x / \pi) = \sin nx$, $n = 1, 2, 3, \dots$.
10. This is Example 2 in the text with $L = \pi/4$. The eigenvalues are $\lambda_n = n^2 \pi^2 / (\pi/4)^2 = 16n^2$, $n = 1, 2, 3, \dots$ and the eigenfunctions are $y_n = \sin(n\pi x / (\pi/4)) = \sin 4nx$, $n = 1, 2, 3, \dots$.

11. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = \alpha^2 > 0$ we have

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x.$$

Now

$$y'(x) = -c_1 \alpha \sin \alpha x + c_2 \alpha \cos \alpha x$$

and $y'(0) = 0$ implies $c_2 = 0$, so

$$y(L) = c_1 \cos \alpha L = 0$$

gives

$$\alpha L = \frac{(2n-1)\pi}{2} \quad \text{or} \quad \lambda = \alpha^2 = \frac{(2n-1)^2 \pi^2}{4L^2}, \quad n = 1, 2, 3, \dots$$

The eigenvalues $(2n-1)^2 \pi^2 / 4L^2$ correspond to the eigenfunctions $\cos \frac{(2n-1)\pi}{2L} x$ for $n = 1, 2, 3, \dots$

12. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = \alpha^2 > 0$ we have

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x.$$

Since $y(0) = 0$ implies $c_1 = 0$, $y = c_2 \sin \alpha x$. Now

$$y'\left(\frac{\pi}{2}\right) = c_2 \alpha \cos \alpha \frac{\pi}{2} = 0$$

gives

$$\alpha \frac{\pi}{2} = \frac{(2n-1)\pi}{2} \quad \text{or} \quad \lambda = \alpha^2 = (2n-1)^2, \quad n = 1, 2, 3, \dots$$

The eigenvalues $\lambda_n = (2n-1)^2$ correspond to the eigenfunctions $y_n = \sin(2n-1)x$.

13. For $\lambda = -\alpha^2 < 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = 0$ we have $y = c_1 x + c_2$. Now $y' = c_1$ and $y'(0) = 0$ implies $c_1 = 0$. Then $y = c_2$ and $y'(\pi) = 0$. Thus, $\lambda = 0$ is an eigenvalue with corresponding eigenfunction $y = 1$.

For $\lambda = \alpha^2 > 0$ we have

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x.$$

Now

$$y'(x) = -c_1 \alpha \sin \alpha x + c_2 \alpha \cos \alpha x$$

and $y'(0) = 0$ implies $c_2 = 0$, so

$$y'(\pi) = -c_1 \alpha \sin \alpha \pi = 0$$

gives

$$\alpha \pi = n\pi \quad \text{or} \quad \lambda = \alpha^2 = n^2, \quad n = 1, 2, 3, \dots$$

The eigenvalues n^2 correspond to the eigenfunctions $\cos nx$ for $n = 0, 1, 2, \dots$

14. For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = \alpha^2 > 0$ we have

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x.$$

Now $y(-\pi) = y(\pi) = 0$ implies

$$c_1 \cos \alpha \pi - c_2 \sin \alpha \pi = 0$$

$$c_1 \cos \alpha \pi + c_2 \sin \alpha \pi = 0.$$

(1)

This homogeneous system will have a nontrivial solution when

$$\begin{vmatrix} \cos \alpha \pi & -\sin \alpha \pi \\ \cos \alpha \pi & \sin \alpha \pi \end{vmatrix} = 2 \sin \alpha \pi \cos \alpha \pi = \sin 2\alpha \pi = 0.$$

3.9 Linear Models: Boundary-Value Problems

Then

$$2\alpha\pi = n\pi \quad \text{or} \quad \lambda = \alpha^2 = \frac{n^2}{4}; \quad n = 1, 2, 3, \dots$$

When $n = 2k - 1$ is odd, the eigenvalues are $(2k - 1)^2/4$. Since $\cos(2k - 1)\pi/2 = 0$ and $\sin(2k - 1)\pi/2 \neq 0$, we see from either equation in (1) that $c_2 = 0$. Thus, the eigenfunctions corresponding to the eigenvalues $(2k - 1)^2/4$ are $y = \cos(2k - 1)x/2$ for $k = 1, 2, 3, \dots$. Similarly, when $n = 2k$ is even, the eigenvalues are k^2 with corresponding eigenfunctions $y = \sin kx$ for $k = 1, 2, 3, \dots$.

15. The auxiliary equation has solutions

$$m = \frac{1}{2} \left(-2 \pm \sqrt{4 - 4(\lambda + 1)} \right) = -1 \pm \alpha.$$

For $\lambda = -\alpha^2 < 0$ we have

$$y = e^{-x} (c_1 \cosh \alpha x + c_2 \sinh \alpha x).$$

The boundary conditions imply

$$y(0) = c_1 = 0$$

$$y(5) = c_2 e^{-5} \sinh 5\alpha = 0$$

so $c_1 = c_2 = 0$ and the only solution of the boundary-value problem is $y = 0$.

For $\lambda = 0$ we have

$$y = c_1 e^{-x} + c_2 x e^{-x}$$

and the only solution of the boundary-value problem is $y = 0$.

For $\lambda = \alpha^2 > 0$ we have

$$y = e^{-x} (c_1 \cos \alpha x + c_2 \sin \alpha x).$$

Now $y(0) = 0$ implies $c_1 = 0$, so

$$y(5) = c_2 e^{-5} \sin 5\alpha = 0$$

gives

$$5\alpha = n\pi \quad \text{or} \quad \lambda = \alpha^2 = \frac{n^2\pi^2}{25}, \quad n = 1, 2, 3, \dots$$

The eigenvalues $\lambda_n = \frac{n^2\pi^2}{25}$ correspond to the eigenfunctions $y_n = e^{-x} \sin \frac{n\pi}{5}x$ for $n = 1, 2, 3, \dots$.

16. For $\lambda < -1$ the only solution of the boundary-value problem is $y = 0$. For $\lambda = -1$ we have $y = c_1x + c_2$. Now $y' = c_1$ and $y'(0) = 0$ implies $c_1 = 0$. Then $y = c_2$ and $y'(1) = 0$. Thus, $\lambda = -1$ is an eigenvalue with corresponding eigenfunction $y = 1$.

For $\lambda > -1$ or $\lambda + 1 = \alpha^2 > 0$ we have

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x.$$

Now

$$y' = -c_1\alpha \sin \alpha x + c_2\alpha \cos \alpha x$$

and $y'(0) = 0$ implies $c_2 = 0$, so

$$y'(1) = -c_1\alpha \sin \alpha = 0$$

gives

$$\alpha = n\pi, \quad \lambda + 1 = \alpha^2 = n^2\pi^2, \quad \text{or} \quad \lambda = n^2\pi^2 - 1, \quad n = 1, 2, 3, \dots$$

The eigenvalues $n^2\pi^2 - 1$ correspond to the eigenfunctions $\cos n\pi x$ for $n = 0, 1, 2, \dots$.

17. For $\lambda = \alpha^2 > 0$ a general solution of the given differential equation is

$$y = c_1 \cos(\alpha \ln x) + c_2 \sin(\alpha \ln x).$$

Since $\ln 1 = 0$, the boundary condition $y(1) = 0$ implies $c_1 = 0$. Therefore

$$y = c_2 \sin(\alpha \ln x).$$

Using $\ln e^\pi = \pi$ we find that $y(e^\pi) = 0$ implies

$$c_2 \sin \alpha \pi = 0$$

or $\alpha \pi = n\pi$, $n = 1, 2, 3, \dots$. The eigenvalues and eigenfunctions are, in turn,

$$\lambda = \alpha^2 = n^2, \quad n = 1, 2, 3, \dots \quad \text{and} \quad y = \sin(n \ln x).$$

For $\lambda \leq 0$ the only solution of the boundary-value problem is $y = 0$.

18. For $\lambda = 0$ the general solution is $y = c_1 + c_2 \ln x$. Now $y' = c_2/x$, so $y'(e^{-1}) = c_2 e = 0$ implies $c_2 = 0$. Then $y = c_1$ and $y(1) = 0$ gives $c_1 = 0$. Thus $y(x) = 0$.

For $\lambda = -\alpha^2 < 0$, $y = c_1 x^{-\alpha} + c_2 x^\alpha$. The boundary conditions give $c_2 = c_1 e^{2\alpha}$ and $c_1 = 0$, so that $c_2 = 0$ and $y(x) = 0$.

For $\lambda = \alpha^2 > 0$, $y = c_1 \cos(\alpha \ln x) + c_2 \sin(\alpha \ln x)$. From $y(1) = 0$ we obtain $c_1 = 0$ and $y = c_2 \sin(\alpha \ln x)$. Now $y' = c_2(\alpha/x) \cos(\alpha \ln x)$, so $y'(e^{-1}) = c_2 e \alpha \cos \alpha = 0$ implies $\cos \alpha = 0$ or $\alpha = (2n-1)\pi/2$ and $\lambda = \alpha^2 = (2n-1)^2 \pi^2/4$ for $n = 1, 2, 3, \dots$. The corresponding eigenfunctions are

$$y_n = \sin \left(\frac{2n-1}{2} \pi \ln x \right).$$

19. For $\lambda = \alpha^4$, $\alpha > 0$, the general solution of the boundary-value problem

$$y^{(4)} - \lambda y = 0, \quad y(0) = 0, \quad y''(0) = 0, \quad y(1) = 0, \quad y''(1) = 0$$

is

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x + c_3 \cosh \alpha x + c_4 \sinh \alpha x.$$

The boundary conditions $y(0) = 0$, $y''(0) = 0$ give $c_1 + c_3 = 0$ and $-c_1 \alpha^2 + c_3 \alpha^2 = 0$, from which we conclude $c_1 = c_3 = 0$. Thus, $y = c_2 \sin \alpha x + c_4 \sinh \alpha x$. The boundary conditions $y(1) = 0$, $y''(1) = 0$ then give

$$\begin{aligned} c_2 \sin \alpha + c_4 \sinh \alpha &= 0 \\ -c_2 \alpha^2 \sin \alpha + c_4 \alpha^2 \sinh \alpha &= 0. \end{aligned}$$

In order to have nonzero solutions of this system, we must have the determinant of the coefficients equal zero, that is,

$$\begin{vmatrix} \sin \alpha & \sinh \alpha \\ -\alpha^2 \sin \alpha & \alpha^2 \sinh \alpha \end{vmatrix} = 0 \quad \text{or} \quad 2\alpha^2 \sinh \alpha \sin \alpha = 0.$$

But since $\alpha > 0$, the only way that this is satisfied is to have $\sin \alpha = 0$ or $\alpha = n\pi$. The system is then satisfied by choosing $c_2 \neq 0$, $c_4 = 0$, and $\alpha = n\pi$. The eigenvalues and corresponding eigenfunctions are then

$$\lambda_n = \alpha^4 = (n\pi)^4, \quad n = 1, 2, 3, \dots \quad \text{and} \quad y = \sin n\pi x.$$

20. For $\lambda = \alpha^4$, $\alpha > 0$, the general solution of the differential equation is

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x + c_3 \cosh \alpha x + c_4 \sinh \alpha x.$$

3.9 Linear Models: Boundary-Value Problems

The boundary conditions $y'(0) = 0$, $y'''(0) = 0$ give $c_2\alpha + c_4\alpha = 0$ and $-c_2\alpha^3 + c_4\alpha^3 = 0$ from which we conclude $c_2 = c_4 = 0$. Thus, $y = c_1 \cos \alpha x + c_3 \cosh \alpha x$. The boundary conditions $y(\pi) = 0$, $y''(\pi) = 0$ then give

$$\begin{aligned} c_2 \cos \alpha \pi + c_4 \cosh \alpha \pi &= 0 \\ -c_2 \lambda^2 \cos \alpha \pi + c_4 \lambda^2 \cosh \alpha \pi &= 0. \end{aligned}$$

The determinant of the coefficients is $2\alpha^2 \cosh \alpha \cos \alpha = 0$. But since $\alpha > 0$, the only way that this is satisfied is to have $\cos \alpha \pi = 0$ or $\alpha = (2n-1)/2$, $n = 1, 2, 3, \dots$. The eigenvalues and corresponding eigenfunctions are

$$\lambda_n = \alpha^4 = \left(\frac{2n-1}{2}\right)^4, \quad n = 1, 2, 3, \dots \quad \text{and} \quad y = \cos\left(\frac{2n-1}{2}\right)x.$$

21. If restraints are put on the column at $x = L/4$, $x = L/2$, and $x = 3L/4$, then the critical load will be P_4 .



22. (a) The general solution of the differential equation is

$$y = c_1 \cos \sqrt{\frac{P}{EI}} x + c_2 \sin \sqrt{\frac{P}{EI}} x + \delta.$$

Since the column is embedded at $x = 0$, the boundary conditions are $y(0) = y'(0) = 0$. If $\delta = 0$ this implies that $c_1 = c_2 = 0$ and $y(x) = 0$. That is, there is no deflection.

- (b) If $\delta \neq 0$, the boundary conditions give, in turn, $c_1 = -\delta$ and $c_2 = 0$. Then

$$y = \delta \left(1 - \cos \sqrt{\frac{P}{EI}} x \right).$$

In order to satisfy the boundary condition $y(L) = \delta$ we must have

$$\delta = \delta \left(1 - \cos \sqrt{\frac{P}{EI}} L \right) \quad \text{or} \quad \cos \sqrt{\frac{P}{EI}} L = 0.$$

This gives $\sqrt{P/EI} L = n\pi/2$ for $n = 1, 2, 3, \dots$. The smallest value of P_n , the Euler load, is then

$$\sqrt{\frac{P_1}{EI}} L = \frac{\pi}{2} \quad \text{or} \quad P_1 = \frac{1}{4} \left(\frac{\pi^2 EI}{L^2} \right).$$

23. If $\lambda = \alpha^2 = P/EI$, then the solution of the differential equation is

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x + c_3 x + c_4.$$

The conditions $y(0) = 0$, $y''(0) = 0$ yield, in turn, $c_1 + c_4 = 0$ and $c_1 = 0$. With $c_1 = 0$ and $c_4 = 0$ the solution is $y = c_2 \sin \alpha x + c_3 x$. The conditions $y(L) = 0$, $y''(L) = 0$, then yield

$$c_2 \sin \alpha L + c_3 L = 0 \quad \text{and} \quad c_2 \sin \alpha L = 0.$$

Hence, nontrivial solutions of the problem exist only if $\sin \alpha L = 0$. From this point on, the analysis is the same as in Example 3 in the text.

24. (a) The boundary-value problem is

$$\frac{d^4 y}{dx^4} + \lambda \frac{d^2 y}{dx^2} = 0, \quad y(0) = 0, y''(0) = 0, \quad y(L) = 0, y'(L) = 0,$$

where $\lambda = \alpha^2 = P/EI$. The solution of the differential equation is $y = c_1 \cos \alpha x + c_2 \sin \alpha x + c_3 x + c_4$ and the conditions $y(0) = 0, y''(0) = 0$ yield $c_1 = 0$ and $c_4 = 0$. Next, by applying $y(L) = 0, y'(L) = 0$ to $y = c_2 \sin \alpha x + c_3 x$ we get the system of equations

$$\begin{aligned} c_2 \sin \alpha L + c_3 L &= 0 \\ \alpha c_2 \cos \alpha L + c_3 &= 0. \end{aligned}$$

To obtain nontrivial solutions c_2, c_3 , we must have the determinant of the coefficients equal to zero:

$$\begin{vmatrix} \sin \alpha L & L \\ \alpha \cos \alpha L & 1 \end{vmatrix} = 0 \quad \text{or} \quad \tan \beta = \beta,$$

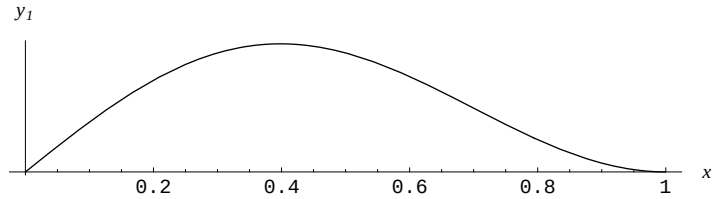
where $\beta = \alpha L$. If β_n denotes the positive roots of the last equation, then the eigenvalues are found from $\beta_n = \alpha_n L = \sqrt{\lambda_n} L$ or $\lambda_n = (\beta_n/L)^2$. From $\lambda = P/EI$ we see that the critical loads are $P_n = \beta_n^2 EI/L^2$. With the aid of a CAS we find that the first positive root of $\tan \beta = \beta$ is (approximately) $\beta_1 = 4.4934$, and so the Euler load is (approximately) $P_1 = 20.1907 EI/L^2$. Finally, if we use $c_3 = -c_2 \alpha \cos \alpha L$, then the deflection curves are

$$y_n(x) = c_2 \sin \alpha_n x + c_3 x = c_2 \left[\sin \left(\frac{\beta_n}{L} x \right) - \left(\frac{\beta_n}{L} \cos \beta_n \right) x \right].$$

- (b) With $L = 1$ and c_2 appropriately chosen, the general shape of the first buckling mode,

$$y_1(x) = c_2 \left[\sin \left(\frac{4.4934}{L} x \right) - \left(\frac{4.4934}{L} \cos(4.4934) \right) x \right],$$

is shown below.



25. The general solution is

$$y = c_1 \cos \sqrt{\frac{\rho}{T}} \omega x + c_2 \sin \sqrt{\frac{\rho}{T}} \omega x.$$

From $y(0) = 0$ we obtain $c_1 = 0$. Setting $y(L) = 0$ we find $\sqrt{\rho/T} \omega L = n\pi, n = 1, 2, 3, \dots$. Thus, critical speeds are $\omega_n = n\pi\sqrt{T/L}\sqrt{\rho}, n = 1, 2, 3, \dots$. The corresponding deflection curves are

$$y(x) = c_2 \sin \frac{n\pi}{L} x, \quad n = 1, 2, 3, \dots,$$

where $c_2 \neq 0$.

26. (a) When $T(x) = x^2$ the given differential equation is the Cauchy-Euler equation

$$x^2 y'' + 2xy' + \rho\omega^2 y = 0.$$

The solutions of the auxiliary equation

$$m(m-1) + 2m + \rho\omega^2 = m^2 + m + \rho\omega^2 = 0$$

3.9 Linear Models: Boundary-Value Problems

are

$$m_1 = -\frac{1}{2} - \frac{1}{2}\sqrt{4\rho\omega^2 - 1}i, \quad m_2 = -\frac{1}{2} + \frac{1}{2}\sqrt{4\rho\omega^2 - 1}i$$

when $\rho\omega^2 > 0.25$. Thus

$$y = c_1 x^{-1/2} \cos(\lambda \ln x) + c_2 x^{-1/2} \sin(\lambda \ln x)$$

where $\lambda = \frac{1}{2}\sqrt{4\rho\omega^2 - 1}$. Applying $y(1) = 0$ gives $c_1 = 0$ and consequently

$$y = c_2 x^{-1/2} \sin(\lambda \ln x).$$

The condition $y(e) = 0$ requires $c_2 e^{-1/2} \sin \lambda = 0$. We obtain a nontrivial solution when $\lambda_n = n\pi$, $n = 1, 2, 3, \dots$. But

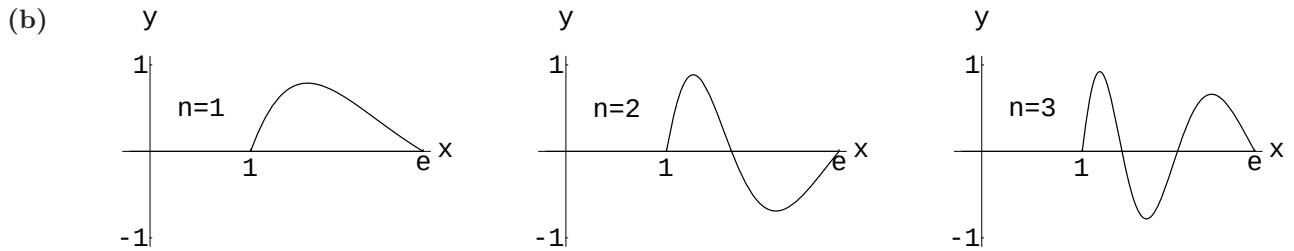
$$\lambda_n = \frac{1}{2}\sqrt{4\rho\omega_n^2 - 1} = n\pi.$$

Solving for ω_n gives

$$\omega_n = \frac{1}{2}\sqrt{(4n^2\pi^2 + 1)/\rho}.$$

The corresponding solutions are

$$y_n(x) = c_2 x^{-1/2} \sin(n\pi \ln x).$$



27. The auxiliary equation is $m^2 + m = m(m+1) = 0$ so that $u(r) = c_1 r^{-1} + c_2$. The boundary conditions $u(a) = u_0$ and $u(b) = u_1$ yield the system $c_1 a^{-1} + c_2 = u_0$, $c_1 b^{-1} + c_2 = u_1$. Solving gives

$$c_1 = \left(\frac{u_0 - u_1}{b - a} \right) ab \quad \text{and} \quad c_2 = \frac{u_1 b - u_0 a}{b - a}.$$

Thus

$$u(r) = \left(\frac{u_0 - u_1}{b - a} \right) \frac{ab}{r} + \frac{u_1 b - u_0 a}{b - a}.$$

28. The auxiliary equation is $m^2 = 0$ so that $u(r) = c_1 + c_2 \ln r$. The boundary conditions $u(a) = u_0$ and $u(b) = u_1$ yield the system $c_1 + c_2 \ln a = u_0$, $c_1 + c_2 \ln b = u_1$. Solving gives

$$c_1 = \frac{u_1 \ln a - u_0 \ln b}{\ln(a/b)} \quad \text{and} \quad c_2 = \frac{u_0 - u_1}{\ln(a/b)}.$$

Thus

$$u(r) = \frac{u_1 \ln a - u_0 \ln b}{\ln(a/b)} + \frac{u_0 - u_1}{\ln(a/b)} \ln r = \frac{u_0 \ln(r/b) - u_1 \ln(r/a)}{\ln(a/b)}.$$

29. The solution of the initial-value problem

$$x'' + \omega^2 x = 0, \quad x(0) = 0, \quad x'(0) = v_0, \quad \omega^2 = 10/m$$

is $x(t) = (v_0/\omega) \sin \omega t$. To satisfy the additional boundary condition $x(1) = 0$ we require that $\omega = n\pi$, $n = 1, 2, 3, \dots$. The eigenvalues $\lambda = \omega^2 = n^2\pi^2$ and eigenfunctions of the problem are then $x(t) = (v_0/n\pi) \sin n\pi t$. Using $\omega^2 = 10/m$ we find that the *only* masses that can pass through the equilibrium position at $t = 1$ are $m_n = 10/n^2\pi^2$. Note for $n = 1$, the heaviest mass $m_1 = 10/\pi^2$ will *not* pass through the

3.9 Linear Models: Boundary-Value Problems

equilibrium position on the interval $0 < t < 1$ (the period of $x(t) = (v_0/\pi) \sin \pi t$ is $T = 2$, so on $0 \leq t \leq 1$ its graph passes through $x = 0$ only at $t = 0$ and $t = 1$). Whereas for $n > 1$, masses of lighter weight will pass through the equilibrium position $n - 1$ times prior to passing through at $t = 1$. For example, if $n = 2$, the period of $x(t) = (v_0/2\pi) \sin 2\pi t$ is $2\pi/2\pi = 1$, the mass will pass through $x = 0$ only *once* ($t = \frac{1}{2}$) prior to $t = 1$; if $n = 3$, the period of $x(t) = (v_0/3\pi) \sin 3\pi t$ is $\frac{2}{3}$, the mass will pass through $x = 0$ *twice* ($t = \frac{1}{3}$ and $t = \frac{2}{3}$) prior to $t = 1$; and so on.

30. The initial-value problem is

$$x'' + \frac{2}{m}x' + \frac{k}{m}x = 0, \quad x(0) = 0, \quad x'(0) = v_0.$$

With $k = 10$, the auxiliary equation has roots $\gamma = -1/m \pm \sqrt{1 - 10m}/m$. Consider the three cases:

(i) $m = \frac{1}{10}$. The roots are $\gamma_1 = \gamma_2 = 10$ and the solution of the differential equation is $x(t) = c_1 e^{-10t} + c_2 t e^{-10t}$. The initial conditions imply $c_1 = 0$ and $c_2 = v_0$ and so $x(t) = v_0 t e^{-10t}$. The condition $x(1) = 0$ implies $v_0 e^{-10} = 0$ which is impossible because $v_0 \neq 0$.

(ii) $1 - 10m > 0$ or $0 < m < \frac{1}{10}$. The roots are

$$\gamma_1 = -\frac{1}{m} - \frac{1}{m}\sqrt{1 - 10m} \quad \text{and} \quad \gamma_2 = -\frac{1}{m} + \frac{1}{m}\sqrt{1 - 10m}$$

and the solution of the differential equation is $x(t) = c_1 e^{\gamma_1 t} + c_2 e^{\gamma_2 t}$. The initial conditions imply

$$c_1 + c_2 = 0$$

$$\gamma_1 c_1 + \gamma_2 c_2 = v_0$$

so $c_1 = v_0/(\gamma_1 - \gamma_2)$, $c_2 = -v_0/(\gamma_1 - \gamma_2)$, and

$$x(t) = \frac{v_0}{\gamma_1 - \gamma_2} (e^{\gamma_1 t} - e^{\gamma_2 t}).$$

Again, $x(1) = 0$ is impossible because $v_0 \neq 0$.

(iii) $1 - 10m < 0$ or $m > \frac{1}{10}$. The roots of the auxiliary equation are

$$\gamma_1 = -\frac{1}{m} - \frac{1}{m}\sqrt{10m - 1}i \quad \text{and} \quad \gamma_2 = -\frac{1}{m} + \frac{1}{m}\sqrt{10m - 1}i$$

and the solution of the differential equation is

$$x(t) = c_1 e^{-t/m} \cos \frac{1}{m} \sqrt{10m - 1} t + c_2 e^{-t/m} \sin \frac{1}{m} \sqrt{10m - 1} t.$$

The initial conditions imply $c_1 = 0$ and $c_2 = mv_0/\sqrt{10m - 1}$, so that

$$x(t) = \frac{mv_0}{\sqrt{10m - 1}} e^{-t/m} \sin \left(\frac{1}{m} \sqrt{10m - 1} t \right),$$

The condition $x(1) = 0$ implies

$$\begin{aligned} \frac{mv_0}{\sqrt{10m - 1}} e^{-1/m} \sin \frac{1}{m} \sqrt{10m - 1} &= 0 \\ \sin \frac{1}{m} \sqrt{10m - 1} &= 0 \\ \frac{1}{m} \sqrt{10m - 1} &= n\pi \\ \frac{10m - 1}{m^2} &= n^2 \pi^2, \quad n = 1, 2, 3, \dots \\ (n^2 \pi^2)m^2 - 10m + 1 &= 0 \end{aligned}$$

3.9 Linear Models: Boundary-Value Problems

$$m = \frac{10\sqrt{100 - 4n^2\pi^2}}{2n^2\pi^2} = \frac{5 \pm \sqrt{25 - n^2\pi^2}}{n^2\pi^2}.$$

Since m is real, $25 - n^2\pi^2 \geq 0$. If $25 - n^2\pi^2 = 0$, then $n^2 = 25/\pi^2$, and n is not an integer. Thus, $25 - n^2\pi^2 = (5 - n\pi)(5 + n\pi) > 0$ and since $n > 0$, $5 + n\pi > 0$, so $5 - n\pi > 0$ also. Then $n < 5/\pi$, and so $n = 1$. Therefore, the mass m will pass through the equilibrium position when $t = 1$ for

$$m_1 = \frac{5 + \sqrt{25 - \pi^2}}{\pi^2} \quad \text{and} \quad m_2 = \frac{5 - \sqrt{25 - \pi^2}}{\pi^2}.$$

- 31. (a)** The general solution of the differential equation is $y = c_1 \cos 4x + c_2 \sin 4x$. From $y_0 = y(0) = c_1$ we see that $y = y_0 \cos 4x + c_2 \sin 4x$. From $y_1 = y(\pi/2) = y_0$ we see that any solution must satisfy $y_0 = y_1$. We also see that when $y_0 = y_1$, $y = y_0 \cos 4x + c_2 \sin 4x$ is a solution of the boundary-value problem for any choice of c_2 . Thus, the boundary-value problem does not have a unique solution for any choice of y_0 and y_1 .
- (b)** Whenever $y_0 = y_1$ there are infinitely many solutions.
- (c)** When $y_0 \neq y_1$ there will be no solutions.
- (d)** The boundary-value problem will have the trivial solution when $y_0 = y_1 = 0$. This solution will not be unique.
- 32. (a)** The general solution of the differential equation is $y = c_1 \cos 4x + c_2 \sin 4x$. From $1 = y(0) = c_1$ we see that $y = \cos 4x + c_2 \sin 4x$. From $1 = y(L) = \cos 4L + c_2 \sin 4L$ we see that $c_2 = (1 - \cos 4L)/\sin 4L$. Thus,

$$y = \cos 4x + \left(\frac{1 - \cos 4L}{\sin 4L} \right) \sin 4x$$

will be a unique solution when $\sin 4L \neq 0$; that is, when $L \neq k\pi/4$ where $k = 1, 2, 3, \dots$.

- (b)** There will be infinitely many solutions when $\sin 4L = 0$ and $1 - \cos 4L = 0$; that is, when $L = k\pi/2$ where $k = 1, 2, 3, \dots$.
- (c)** There will be no solution when $\sin 4L \neq 0$ and $1 - \cos 4L \neq 0$; that is, when $L = k\pi/4$ where $k = 1, 3, 5, \dots$.
- (d)** There can be no trivial solution since it would fail to satisfy the boundary conditions.
- 33. (a)** A solution curve has the same y -coordinate at both ends of the interval $[-\pi, \pi]$ and the tangent lines at the endpoints of the interval are parallel.
- (b)** For $\lambda = 0$ the solution of $y'' = 0$ is $y = c_1x + c_2$. From the first boundary condition we have

$$y(-\pi) = -c_1\pi + c_2 = y(\pi) = c_1\pi + c_2$$

or $2c_1\pi = 0$. Thus, $c_1 = 0$ and $y = c_2$. This constant solution is seen to satisfy the boundary-value problem.

For $\lambda = -\alpha^2 < 0$ we have $y = c_1 \cosh \alpha x + c_2 \sinh \alpha x$. In this case the first boundary condition gives

$$\begin{aligned} y(-\pi) &= c_1 \cosh(-\alpha\pi) + c_2 \sinh(-\alpha\pi) \\ &= c_1 \cosh \alpha\pi - c_2 \sinh \alpha\pi \\ &= y(\pi) = c_1 \cosh \alpha\pi + c_2 \sinh \alpha\pi \end{aligned}$$

or $2c_2 \sinh \alpha\pi = 0$. Thus $c_2 = 0$ and $y = c_1 \cosh \alpha x$. The second boundary condition implies in a similar fashion that $c_1 = 0$. Thus, for $\lambda < 0$, the only solution of the boundary-value problem is $y = 0$.

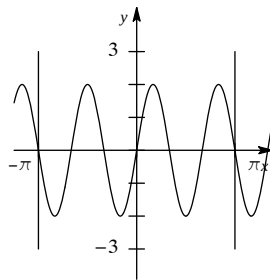
3.9 Linear Models: Boundary-Value Problems

For $\lambda = \alpha^2 > 0$ we have $y = c_1 \cos \alpha x + c_2 \sin \alpha x$. The first boundary condition implies

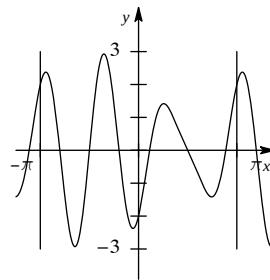
$$\begin{aligned} y(-\pi) &= c_1 \cos(-\alpha\pi) + c_2 \sin(-\alpha\pi) \\ &= c_1 \cos \alpha\pi - c_2 \sin \alpha\pi \\ &= y(\pi) = c_1 \cos \alpha\pi + c_2 \sin \alpha\pi \end{aligned}$$

or $2c_2 \sin \alpha\pi = 0$. Similarly, the second boundary condition implies $2c_1 \alpha \sin \alpha\pi = 0$. If $c_1 = c_2 = 0$ the solution is $y = 0$. However, if $c_1 \neq 0$ or $c_2 \neq 0$, then $\sin \alpha\pi = 0$, which implies that α must be an integer, n . Therefore, for c_1 and c_2 not both 0, $y = c_1 \cos nx + c_2 \sin nx$ is a nontrivial solution of the boundary-value problem. Since $\cos(-nx) = \cos nx$ and $\sin(-nx) = -\sin nx$, we may assume without loss of generality that the eigenvalues are $\lambda_n = \alpha^2 = n^2$, for n a positive integer. The corresponding eigenfunctions are $y_n = \cos nx$ and $y_n = \sin nx$.

(c)



$$y = 2 \sin 3x$$



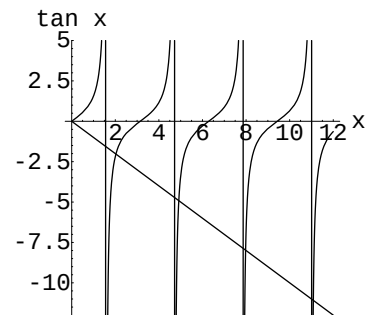
$$y = \sin 4x - 2 \cos 3x$$

34. For $\lambda = \alpha^2 > 0$ the general solution is $y = c_1 \cos \sqrt{\alpha} x + c_2 \sin \sqrt{\alpha} x$. Setting $y(0) = 0$ we find $c_1 = 0$, so that $y = c_2 \sin \sqrt{\alpha} x$. The boundary condition $y(1) + y'(1) = 0$ implies

$$c_2 \sin \sqrt{\alpha} + c_2 \sqrt{\alpha} \cos \sqrt{\alpha} = 0.$$

Taking $c_2 \neq 0$, this equation is equivalent to $\tan \sqrt{\alpha} = -\sqrt{\alpha}$. Thus, the eigenvalues are $\lambda_n = \alpha_n^2 = x_n^2$, $n = 1, 2, 3, \dots$, where the x_n are the consecutive positive roots of $\tan \sqrt{\alpha} = -\sqrt{\alpha}$.

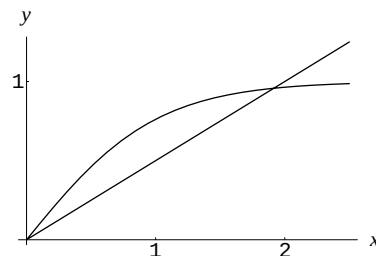
35. We see from the graph that $\tan x = -x$ has infinitely many roots. Since $\lambda_n = \alpha_n^2$, there are no new eigenvalues when $\alpha_n < 0$. For $\lambda = 0$, the differential equation $y'' = 0$ has general solution $y = c_1 x + c_2$. The boundary conditions imply $c_1 = c_2 = 0$, so $y = 0$.



36. Using a CAS we find that the first four nonnegative roots of $\tan x = -x$ are approximately 2.02876, 4.91318, 7.97867, and 11.0855. The corresponding eigenvalues are 4.11586, 24.1393, 63.6591, and 122.889, with eigenfunctions $\sin(2.02876x)$, $\sin(4.91318x)$, $\sin(7.97867x)$, and $\sin(11.0855x)$.

3.9 Linear Models: Boundary-Value Problems

37. In the case when $\lambda = -\alpha^2 < 0$, the solution of the differential equation is $y = c_1 \cosh \alpha x + c_2 \sinh \alpha x$. The condition $y(0) = 0$ gives $c_1 = 0$. The condition $y(1) - \frac{1}{2}y'(1) = 0$ applied to $y = c_2 \sinh \alpha x$ gives $c_2(\sinh \alpha - \frac{1}{2}\alpha \cosh \alpha) = 0$ or $\tanh \alpha = \frac{1}{2}\alpha$. As can be seen from the figure, the graphs of $y = \tanh x$ and $y = \frac{1}{2}x$ intersect at a single point with approximate x -coordinate $\alpha_1 = 1.915$. Thus, there is a single negative eigenvalue $\lambda_1 = -\alpha_1^2 \approx -3.667$ and the corresponding eigenfunction is $y_1 = \sinh 1.915x$.



For $\lambda = 0$ the only solution of the boundary-value problem is $y = 0$.

For $\lambda = \alpha^2 > 0$ the solution of the differential equation is $y = c_1 \cos \alpha x + c_2 \sin \alpha x$. The condition $y(0) = 0$ gives $c_1 = 0$, so $y = c_2 \sin \alpha x$. The condition $y(1) - \frac{1}{2}y'(1) = 0$ gives $c_2(\sin \alpha - \frac{1}{2}\alpha \cos \alpha) = 0$, so the eigenvalues are $\lambda_n = \alpha_n^2$ when α_n , $n = 2, 3, 4, \dots$, are the positive roots of $\tan \alpha = \frac{1}{2}\alpha$. Using a CAS we find that the first three values of α are $\alpha_2 = 4.27487$, $\alpha_3 = 7.59655$, and $\alpha_4 = 10.8127$. The first three eigenvalues are then $\lambda_2 = \alpha_2^2 = 18.2738$, $\lambda_3 = \alpha_3^2 = 57.7075$, and $\lambda_4 = \alpha_4^2 = 116.9139$ with corresponding eigenfunctions $y_2 = \sin 4.27487x$, $y_3 = \sin 7.59655x$, and $y_4 = \sin 10.8127x$.

38. For $\lambda = \alpha^4$, $\alpha > 0$, the solution of the differential equation is

$$y = c_1 \cos \alpha x + c_2 \sin \alpha x + c_3 \cosh \alpha x + c_4 \sinh \alpha x.$$

The boundary conditions $y(0) = 0$, $y'(0) = 0$, $y(1) = 0$, $y'(1) = 0$ give, in turn,

$$c_1 + c_3 = 0$$

$$\alpha c_2 + \alpha c_4 = 0,$$

$$c_1 \cos \alpha + c_2 \sin \alpha + c_3 \cosh \alpha + c_4 \sinh \alpha = 0$$

$$-c_1 \alpha \sin \alpha + c_2 \alpha \cos \alpha + c_3 \alpha \sinh \alpha + c_4 \alpha \cosh \alpha = 0.$$

The first two equations enable us to write

$$c_1(\cos \alpha - \cosh \alpha) + c_2(\sin \alpha - \sinh \alpha) = 0$$

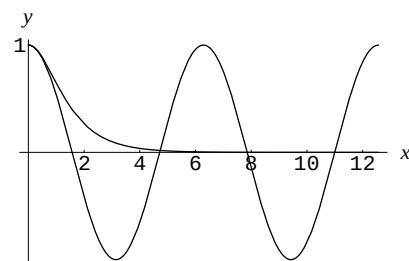
$$c_1(-\sin \alpha - \sinh \alpha) + c_2(\cos \alpha - \cosh \alpha) = 0.$$

The determinant

$$\begin{vmatrix} \cos \alpha - \cosh \alpha & \sin \alpha - \sinh \alpha \\ -\sin \alpha - \sinh \alpha & \cos \alpha - \cosh \alpha \end{vmatrix} = 0$$

simplifies to $\cos \alpha \cosh \alpha = 1$. From the figure showing the graphs of $1/\cosh x$ and $\cos x$, we see that this equation has an infinite number of positive roots. With the aid of a CAS the first four roots are found to be $\alpha_1 = 4.73004$, $\alpha_2 = 7.8532$, $\alpha_3 = 10.9956$, and $\alpha_4 = 14.1372$, and the corresponding eigenvalues are $\lambda_1 = 500.5636$, $\lambda_2 = 3803.5281$, $\lambda_3 = 14,617.5885$, and $\lambda_4 = 39,944.1890$. Using the third equation in the system to eliminate c_2 , we find that the eigenfunctions are

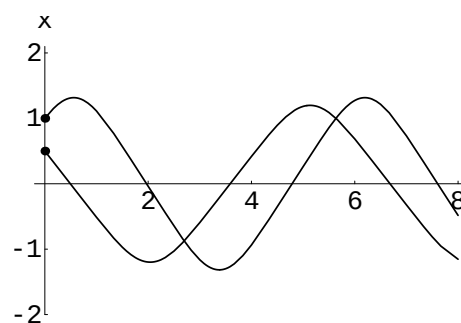
$$y_n = (-\sin \alpha_n + \sinh \alpha_n)(\cos \alpha_n x - \cosh \alpha_n x) + (\cos \alpha_n - \cosh \alpha_n)(\sin \alpha_n x - \sinh \alpha_n x).$$



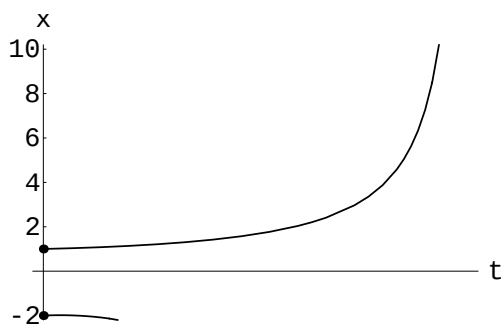
EXERCISES 3.10

Nonlinear Models

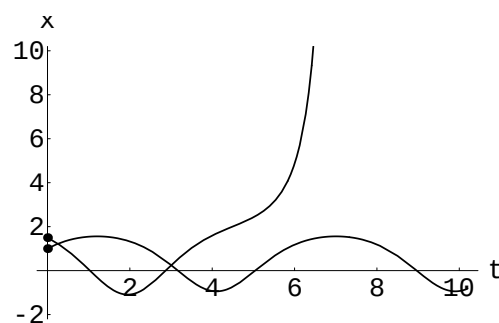
1. The period corresponding to $x(0) = 1, x'(0) = 1$ is approximately 5.6.
The period corresponding to $x(0) = 1/2, x'(0) = -1$ is approximately 6.2.



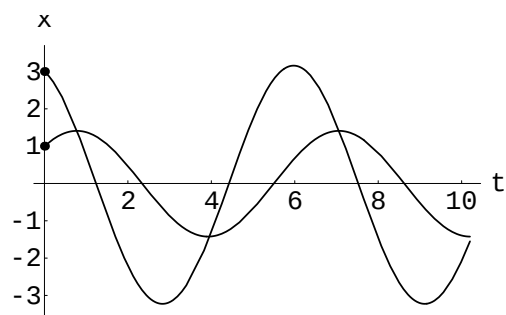
2. The solutions are not periodic.



3. The period corresponding to $x(0) = 1, x'(0) = 1$ is approximately 5.8. The second initial-value problem does not have a periodic solution.

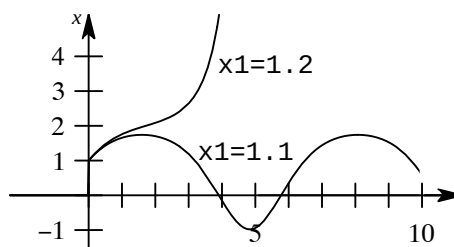


4. Both solutions have periods of approximately 6.3.

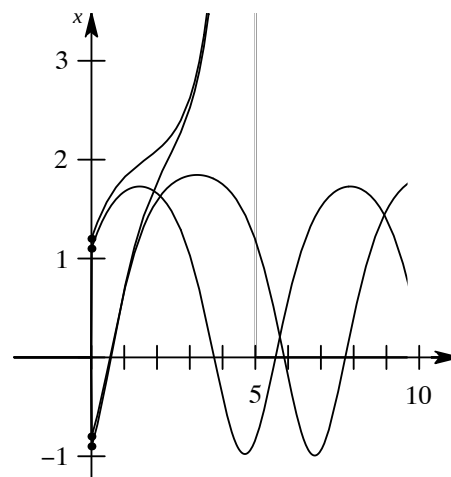


3.10 Nonlinear Models

5. From the graph we see that $|x_1| \approx 1.2$.



6. From the graphs we see that the interval is approximately $(-0.8, 1.1)$.

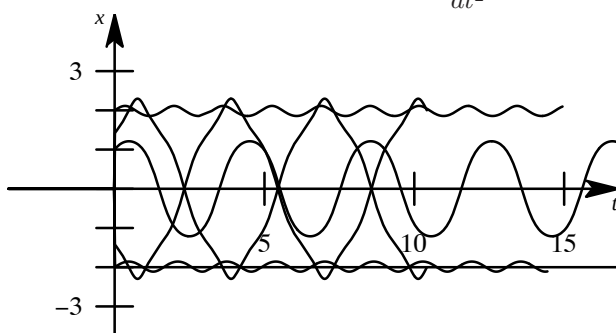


7. Since

$$xe^{0.01x} = x[1 + 0.01x + \frac{1}{2!}(0.01x)^2 + \dots] \approx x$$

for small values of x , a linearization is $\frac{d^2x}{dt^2} + x = 0$.

- 8.



For $x(0) = 1$ and $x'(0) = 1$ the oscillations are symmetric about the line $x = 0$ with amplitude slightly greater than 1.

For $x(0) = -2$ and $x'(0) = 0.5$ the oscillations are symmetric about the line $x = -2$ with small amplitude.

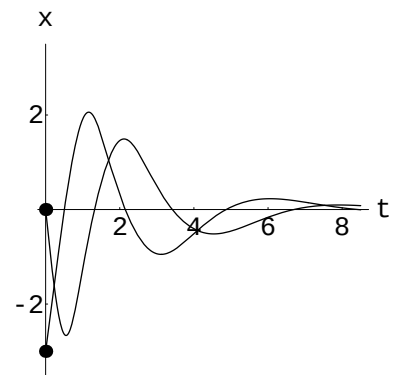
For $x(0) = \sqrt{2}$ and $x'(0) = 1$ the oscillations are symmetric about the line $x = 0$ with amplitude a little greater than 2.

For $x(0) = 2$ and $x'(0) = 0.5$ the oscillations are symmetric about the line $x = 2$ with small amplitude.

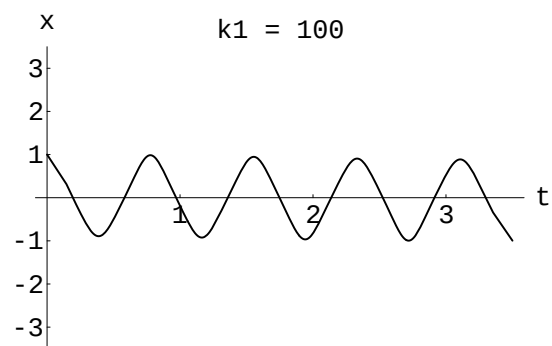
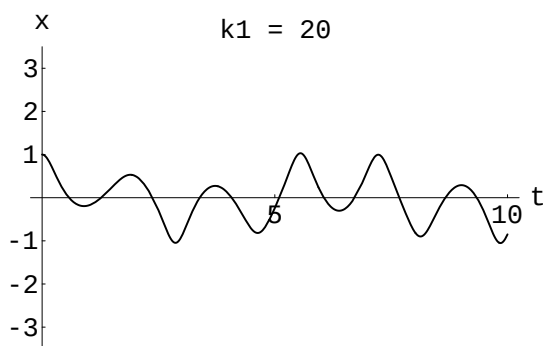
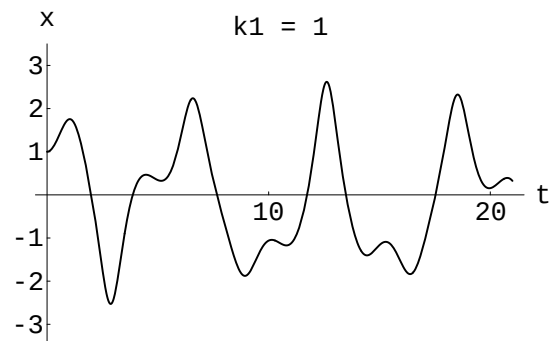
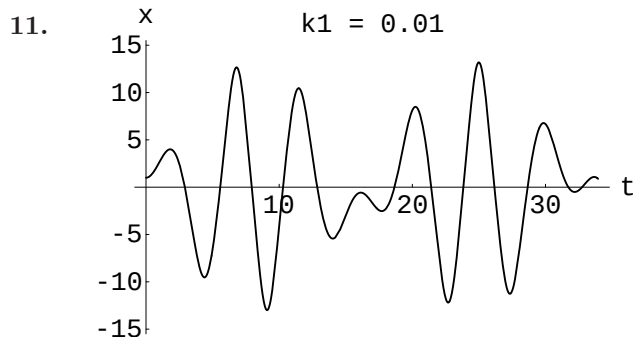
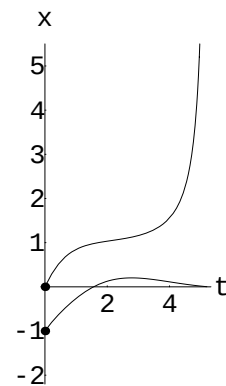
For $x(0) = -2$ and $x'(0) = 0$ there is no oscillation; the solution is constant.

For $x(0) = -\sqrt{2}$ and $x'(0) = -1$ the oscillations are symmetric about the line $x = 0$ with amplitude a little greater than 2.

9. This is a damped hard spring, so x will approach 0 as t approaches ∞ .



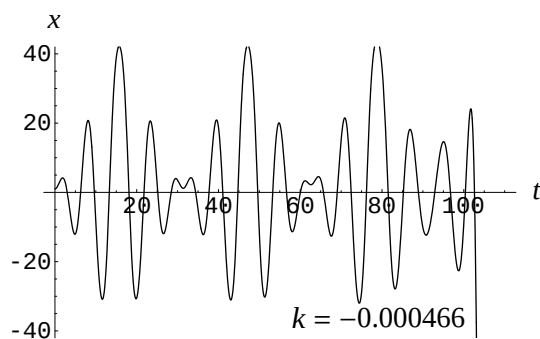
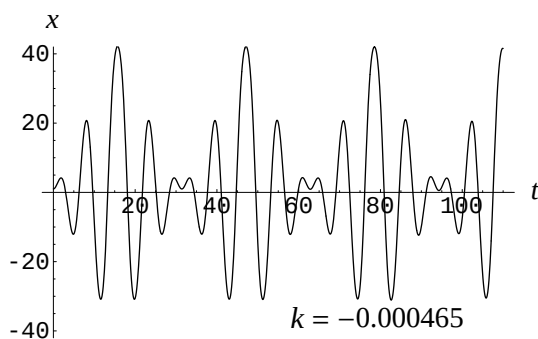
10. This is a damped soft spring, so we might expect no oscillatory solutions. However, if the initial conditions are sufficiently small the spring can oscillate.



When k_1 is very small the effect of the nonlinearity is greatly diminished, and the system is close to pure resonance.

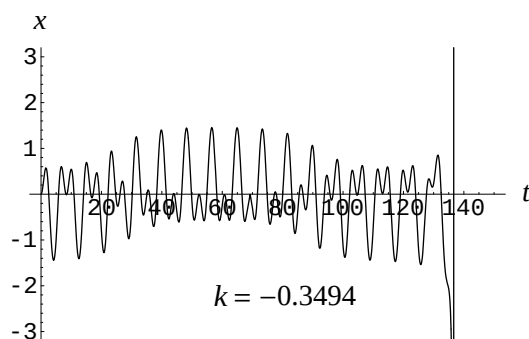
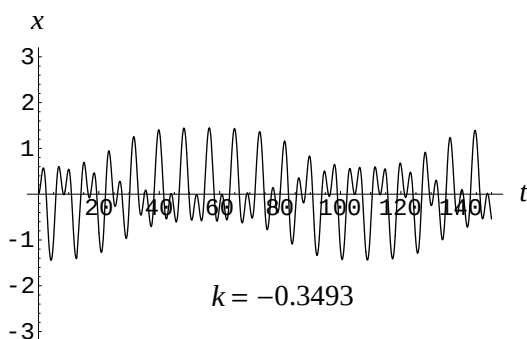
3.10 Nonlinear Models

12. (a)



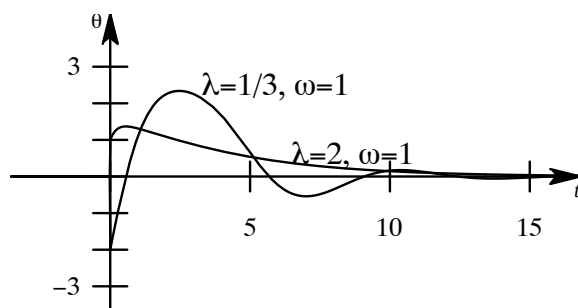
The system appears to be oscillatory for $-0.000465 \leq k_1 < 0$ and nonoscillatory for $k_1 \leq -0.000466$.

(b)



The system appears to be oscillatory for $-0.3493 \leq k_1 < 0$ and nonoscillatory for $k_1 \leq -0.3494$.

13. For $\lambda^2 - \omega^2 > 0$ we choose $\lambda = 2$ and $\omega = 1$ with $x(0) = 1$ and $x'(0) = 2$. For $\lambda^2 - \omega^2 < 0$ we choose $\lambda = 1/3$ and $\omega = 1$ with $x(0) = -2$ and $x'(0) = 4$. In both cases the motion corresponds to the overdamped and underdamped cases for spring/mass systems.



14. (a) Setting $dy/dt = v$, the differential equation in (13) becomes $dv/dt = -gR^2/y^2$. But, by the chain rule, $dv/dt = (dv/dy)(dy/dt) = v dv/dy$, so $v dv/dy = -gR^2/y^2$. Separating variables and integrating we obtain

$$v dv = -gR^2 \frac{dy}{y^2} \quad \text{and} \quad \frac{1}{2}v^2 = \frac{gR^2}{y} + c.$$

Setting $v = v_0$ and $y = R$ we find $c = -gR + \frac{1}{2}v_0^2$ and

$$v^2 = 2g \frac{R^2}{y} - 2gR + v_0^2.$$

(b) As $y \rightarrow \infty$ we assume that $v \rightarrow 0^+$. Then $v_0^2 = 2gR$ and $v_0 = \sqrt{2gR}$.

(c) Using $g = 32$ ft/s and $R = 4000(5280)$ ft we find

$$v_0 = \sqrt{2(32)(4000)(5280)} \approx 36765.2 \text{ ft/s} \approx 25067 \text{ mi/hr.}$$

(d) $v_0 = \sqrt{2(0.165)(32)(1080)} \approx 7760 \text{ ft/s} \approx 5291 \text{ mi/hr}$

15. (a) Intuitively, one might expect that only half of a 10-pound chain could be lifted by a 5-pound vertical force.
- (b) Since $x = 0$ when $t = 0$, and $v = dx/dt = \sqrt{160 - 64x/3}$, we have $v(0) = \sqrt{160} \approx 12.65$ ft/s.
- (c) Since x should always be positive, we solve $x(t) = 0$, getting $t = 0$ and $t = \frac{3}{2}\sqrt{5/2} \approx 2.3717$. Since the graph of $x(t)$ is a parabola, the maximum value occurs at $t_m = \frac{3}{4}\sqrt{5/2}$. (This can also be obtained by solving $x'(t) = 0$.) At this time the height of the chain is $x(t_m) \approx 7.5$ ft. This is higher than predicted because of the momentum generated by the force. When the chain is 5 feet high it still has a positive velocity of about 7.3 ft/s, which keeps it going higher for a while.
16. (a) Setting $dx/dt = v$, the differential equation becomes $(L - x)dv/dt - v^2 = Lg$. But, by the Chain Rule, $dv/dt = (dv/dx)(dx/dt) = v dv/dx$, so $(L - x)v dv/dx - v^2 = Lg$. Separating variables and integrating we obtain

$$\frac{v}{v^2 + Lg} dv = \frac{1}{L - x} dx \quad \text{and} \quad \frac{1}{2} \ln(v^2 + Lg) = -\ln(L - x) + \ln c,$$

so $\sqrt{v^2 + Lg} = c/(L - x)$. When $x = 0$, $v = 0$, and $c = L\sqrt{Lg}$. Solving for v and simplifying we get

$$\frac{dx}{dt} = v(x) = \frac{\sqrt{Lg(2Lx - x^2)}}{L - x}.$$

Again, separating variables and integrating we obtain

$$\frac{L - x}{\sqrt{Lg(2Lx - x^2)}} dx = dt \quad \text{and} \quad \frac{\sqrt{2Lx - x^2}}{\sqrt{Lg}} = t + c_1.$$

Since $x(0) = 0$, we have $c_1 = 0$ and $\sqrt{2Lx - x^2}/\sqrt{Lg} = t$. Solving for x we get

$$x(t) = L - \sqrt{L^2 - Lgt^2} \quad \text{and} \quad v(t) = \frac{dx}{dt} = \frac{\sqrt{Lg}t}{\sqrt{L - gt^2}}.$$

- (b) The chain will be completely on the ground when $x(t) = L$ or $t = \sqrt{L/g}$.
- (c) The predicted velocity of the upper end of the chain when it hits the ground is infinity.
17. (a) The weight of x feet of the chain is $2x$, so the corresponding mass is $m = 2x/32 = x/16$. The only force acting on the chain is the weight of the portion of the chain hanging over the edge of the platform. Thus, by Newton's second law,

$$\frac{d}{dt}(mv) = \frac{d}{dt}\left(\frac{x}{16}v\right) = \frac{1}{16}\left(x\frac{dv}{dt} + v\frac{dx}{dt}\right) = \frac{1}{16}\left(x\frac{dv}{dt} + v^2\right) = 2x$$

and $x dv/dt + v^2 = 32x$. Now, by the Chain Rule, $dv/dt = (dv/dx)(dx/dt) = v dv/dx$, so $xv dv/dx + v^2 = 32x$.

- (b) We separate variables and write the differential equation as $(v^2 - 32x) dx + xv dv = 0$. This is not an exact form, but $\mu(x) = x$ is an integrating factor. Multiplying by x we get $(xv^2 - 32x^2) dx + x^2v dv = 0$. This form is the total differential of $u = \frac{1}{2}x^2v^2 - \frac{32}{3}x^3$, so an implicit solution is $\frac{1}{2}x^2v^2 - \frac{32}{3}x^3 = c$. Letting $x = 3$ and $v = 0$ we find $c = -288$. Solving for v we get

$$\frac{dx}{dt} = v = \frac{8\sqrt{x^3 - 27}}{\sqrt{3}x}, \quad 3 \leq x \leq 8.$$

- (c) Separating variables and integrating we obtain

$$\frac{x}{\sqrt{x^3 - 27}} dx = \frac{8}{\sqrt{3}} dt \quad \text{and} \quad \int_3^x \frac{s}{\sqrt{s^3 - 27}} ds = \frac{8}{\sqrt{3}} t + c.$$

3.10 Nonlinear Models

Since $x = 3$ when $t = 0$, we see that $c = 0$ and

$$t = \frac{\sqrt{3}}{8} \int_3^x \frac{s}{\sqrt{s^3 - 27}} ds.$$

We want to find t when $x = 7$. Using a CAS we find $t(7) = 0.576$ seconds.

18. (a) There are two forces acting on the chain as it falls from the platform. One is the force due to gravity on the portion of the chain hanging over the edge of the platform. This is $F_1 = 2x$. The second is due to the motion of the portion of the chain stretched out on the platform. By Newton's second law this is

$$\begin{aligned} F_2 &= \frac{d}{dt}[mv] = \frac{d}{dt} \left[\frac{(8-x)2}{32} v \right] = \frac{d}{dt} \left[\frac{8-x}{16} v \right] \\ &= \frac{8-x}{16} \frac{dv}{dt} - \frac{1}{16} v \frac{dx}{dt} = \frac{1}{16} \left[(8-x) \frac{dv}{dt} - v^2 \right]. \end{aligned}$$

From $\frac{d}{dt}[mv] = F_1 - F_2$ we have

$$\begin{aligned} \frac{d}{dt} \left[\frac{2x}{32} v \right] &= 2x - \frac{1}{16} \left[(8-x) \frac{dv}{dt} - v^2 \right] \\ \frac{x}{16} \frac{dv}{dt} + \frac{1}{16} v \frac{dx}{dt} &= 2x - \frac{1}{16} \left[(8-x) \frac{dv}{dt} - v^2 \right] \\ x \frac{dv}{dt} + v^2 &= 32x - (8-x) \frac{dv}{dt} + v^2 \\ x \frac{dv}{dt} &= 32x - 8 \frac{dv}{dt} + x \frac{dv}{dt} \\ 8 \frac{dv}{dt} &= 32x. \end{aligned}$$

By the Chain Rule, $dv/dt = (dv/dx)(dx/dt) = v dv/dx$, so

$$8 \frac{dv}{dt} = 8v \frac{dv}{dx} = 32x \quad \text{and} \quad v \frac{dv}{dx} = 4x.$$

- (b) Integrating $v dv = 4x dx$ we get $\frac{1}{2}v^2 = 2x^2 + c$. Since $v = 0$ when $x = 3$, we have $c = -18$. Then $v^2 = 4x^2 - 36$ and $v = \sqrt{4x^2 - 36}$. Using $v = dx/dt$, separating variables, and integrating we obtain

$$\frac{dx}{\sqrt{x^2 - 9}} = 2 dt \quad \text{and} \quad \cosh^{-1} \frac{x}{3} = 2t + c_1.$$

Solving for x we get $x(t) = 3 \cosh(2t + c_1)$. Since $x = 3$ when $t = 0$, we have $\cosh c_1 = 1$ and $c_1 = 0$. Thus, $x(t) = 3 \cosh 2t$. Differentiating, we find $v(t) = dx/dt = 6 \sinh 2t$.

- (c) To find the time when the back end of the chain leaves the platform we solve $x(t) = 3 \cosh 2t = 8$. This gives $t_1 = \frac{1}{2} \cosh^{-1} \frac{8}{3} \approx 0.8184$ seconds. The velocity at this instant is

$$v(t_1) = 6 \sinh \left(\cosh^{-1} \frac{8}{3} \right) = 2\sqrt{55} \approx 14.83 \text{ ft/s}.$$

- (d) Replacing 8 with L and 32 with g in part (a) we have $L dv/dt = gx$. Then

$$L \frac{dv}{dt} = Lv \frac{dv}{dx} = gx \quad \text{and} \quad v \frac{dv}{dx} = \frac{g}{L} x.$$

Integrating we get $\frac{1}{2}v^2 = (g/2L)x^2 + c$. Setting $x = x_0$ and $v = 0$, we find $c = -(g/2L)x_0^2$. Solving for v we find

$$v(x) = \sqrt{\frac{g}{L}x^2 - \frac{g}{L}x_0^2}.$$

Then the velocity at which the end of the chain leaves the edge of the platform is

$$v(L) = \sqrt{\frac{g}{L}(L^2 - x_0^2)}.$$

19. Let (x, y) be the coordinates of S_2 on the curve C . The slope at (x, y) is then

$$dy/dx = (v_1 t - y)/(0 - x) = (y - v_1 t)/x \quad \text{or} \quad xy' - y = -v_1 t.$$

Differentiating with respect to x and using $r = v_1/v_2$ gives

$$\begin{aligned} xy'' + y' - y' &= -v_1 \frac{dt}{dx} \\ xy'' &= -v_1 \frac{dt}{ds} \frac{ds}{dx} \\ xy'' &= -v_1 \frac{1}{v_2} (-\sqrt{1 + (y')^2}) \\ xy'' &= r \sqrt{1 + (y')^2}. \end{aligned}$$

Letting $u = y'$ and separating variables, we obtain

$$\begin{aligned} x \frac{du}{dx} &= r \sqrt{1 + u^2} \\ \frac{du}{\sqrt{1 + u^2}} &= \frac{r}{x} dx \\ \sinh^{-1} u &= r \ln x + \ln c = \ln(cx^r) \\ u &= \sinh(\ln cx^r) \\ \frac{dy}{dx} &= \frac{1}{2} \left(cx^r - \frac{1}{cx^r} \right). \end{aligned}$$

At $t = 0$, $dy/dx = 0$ and $x = a$, so $0 = ca^r - 1/ca^r$. Thus $c = 1/a^r$ and

$$\frac{dy}{dx} = \frac{1}{2} \left[\left(\frac{x}{a} \right)^r - \left(\frac{a}{x} \right)^r \right] = \frac{1}{2} \left[\left(\frac{x}{a} \right)^r - \left(\frac{x}{a} \right)^{-r} \right].$$

If $r > 1$ or $r < 1$, integrating gives

$$y = \frac{a}{2} \left[\frac{1}{1+r} \left(\frac{x}{a} \right)^{1+r} - \frac{1}{1-r} \left(\frac{x}{a} \right)^{1-r} \right] + c_1.$$

When $t = 0$, $y = 0$ and $x = a$, so $0 = (a/2)[1/(1+r) - 1/(1-r)] + c_1$. Thus $c_1 = ar/(1-r^2)$ and

$$y = \frac{a}{2} \left[\frac{1}{1+r} \left(\frac{x}{a} \right)^{1+r} - \frac{1}{1-r} \left(\frac{x}{a} \right)^{1-r} \right] + \frac{ar}{1-r^2}.$$

To see if the paths ever intersect we first note that if $r > 1$, then $v_1 > v_2$ and $y \rightarrow \infty$ as $x \rightarrow 0^+$. In other words, S_2 always lags behind S_1 . Next, if $r < 1$, then $v_1 < v_2$ and $y = ar/(1-r^2)$ when $x = 0$. In other words, when the submarine's speed is greater than the ship's, their paths will intersect at the point $(0, ar/(1-r^2))$.

Finally, if $r = 1$, then integration gives

$$y = \frac{1}{2} \left[\frac{x^2}{2a} - \frac{1}{a} \ln x \right] + c_2.$$

When $t = 0$, $y = 0$ and $x = a$, so $0 = (1/2)[a/2 - (1/a) \ln a] + c_2$. Thus $c_2 = -(1/2)[a/2 - (1/a) \ln a]$ and

$$y = \frac{1}{2} \left[\frac{x^2}{2a} - \frac{1}{a} \ln x \right] - \frac{1}{2} \left[\frac{a}{2} - \frac{1}{a} \ln a \right] = \frac{1}{2} \left[\frac{1}{2a}(x^2 - a^2) + \frac{1}{a} \ln \frac{a}{x} \right].$$

Since $y \rightarrow \infty$ as $x \rightarrow 0^+$, S_2 will never catch up with S_1 .

3.10 Nonlinear Models

20. (a) Let (r, θ) denote the polar coordinates of the destroyer S_1 . When S_1 travels the 6 miles from $(9, 0)$ to $(3, 0)$ it stands to reason, since S_2 travels half as fast as S_1 , that the polar coordinates of S_2 are $(3, \theta_2)$, where θ_2 is unknown. In other words, the distances of the ships from $(0, 0)$ are the same and $r(t) = 15t$ then gives the radial distance of both ships. This is necessary if S_1 is to intercept S_2 .

- (b) The differential of arc length in polar coordinates is $(ds)^2 = (r d\theta)^2 + (dr)^2$, so that

$$\left(\frac{ds}{dt}\right)^2 = r^2 \left(\frac{d\theta}{dt}\right)^2 + \left(\frac{dr}{dt}\right)^2.$$

Using $ds/dt = 30$ and $dr/dt = 15$ then gives

$$900 = 225t^2 \left(\frac{d\theta}{dt}\right)^2 + 225$$

$$675 = 225t^2 \left(\frac{d\theta}{dt}\right)^2$$

$$\frac{d\theta}{dt} = \frac{\sqrt{3}}{t}$$

$$\theta(t) = \sqrt{3} \ln t + c = \sqrt{3} \ln \frac{r}{15} + c.$$

When $r = 3$, $\theta = 0$, so $c = -\sqrt{3} \ln \frac{1}{5}$ and

$$\theta(t) = \sqrt{3} \left(\ln \frac{r}{15} - \ln \frac{1}{5} \right) = \sqrt{3} \ln \frac{r}{3}.$$

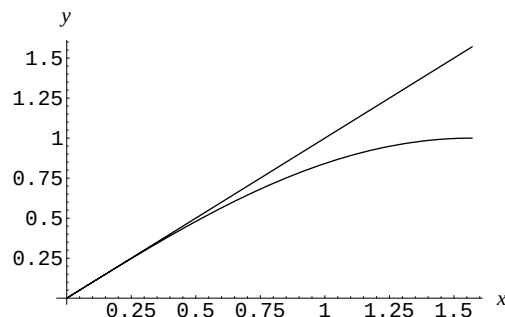
Thus $r = 3e^{\theta/\sqrt{3}}$, whose graph is a logarithmic spiral.

- (c) The time for S_1 to go from $(9, 0)$ to $(3, 0) = \frac{1}{5}$ hour. Now S_1 must intercept the path of S_2 for some angle β , where $0 < \beta < 2\pi$. At the time of interception t_2 we have $15t_2 = 3e^{\beta/\sqrt{3}}$ or $t = \frac{1}{5}e^{\beta/\sqrt{3}}$. The total time is then

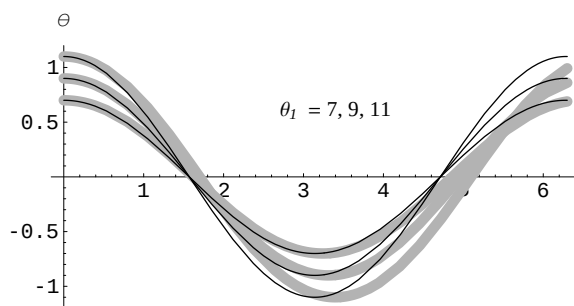
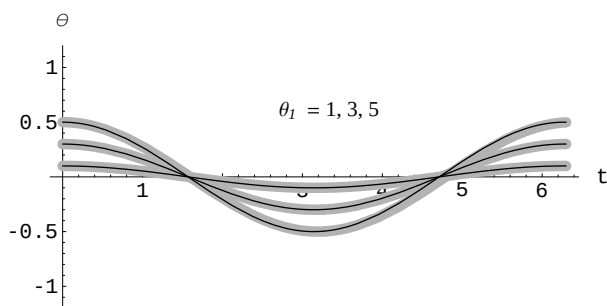
$$t = \frac{1}{5} + \frac{1}{5}e^{\beta/\sqrt{3}} < \frac{1}{5}(1 + e^{2\pi/\sqrt{3}}).$$

21. Since $(dx/dt)^2$ is always positive, it is necessary to use $|dx/dt|(dx/dt)$ in order to account for the fact that the motion is oscillatory and the velocity (or its square) should be negative when the spring is contracting.

22. (a) From the graph we see that the approximations appears to be quite good for $0 \leq x \leq 0.4$. Using an equation solver to solve $\sin x - x = 0.05$ and $\sin x - x = 0.005$, we find that the approximation is accurate to one decimal place for $\theta_1 = 0.67$ and to two decimal places for $\theta_1 = 0.31$.



(b)



23. (a) Write the differential equation as

$$\frac{d^2\theta}{dt^2} + \omega^2 \sin \theta = 0,$$

where $\omega^2 = g/l$. To test for differences between the earth and the moon we take $l = 3$, $\theta(0) = 1$, and $\theta'(0) = 2$.

Using $g = 32$ on the earth and $g = 5.5$ on the moon we obtain the graphs shown in the figure. Comparing the

apparent periods of the graphs, we see that the pendulum oscillates faster on the earth than on the moon.

(b) The amplitude is greater on the moon than on the earth.

(c) The linear model is

$$\frac{d^2\theta}{dt^2} + \omega^2 \theta = 0,$$

where $\omega^2 = g/l$. When $g = 32$, $l = 3$, $\theta(0) = 1$, and $\theta'(0) = 2$, the solution is

$$\theta(t) = \cos 3.266t + 0.612 \sin 3.266t.$$

When $g = 5.5$ the solution is

$$\theta(t) = \cos 1.354t + 1.477 \sin 1.354t.$$

As in the nonlinear case, the pendulum oscillates faster on the earth than on the moon and still has greater amplitude on the moon.

24. (a) The general solution of

$$\frac{d^2\theta}{dt^2} + \theta = 0$$

is $\theta(t) = c_1 \cos t + c_2 \sin t$. From $\theta(0) = \pi/12$ and $\theta'(0) = -1/3$ we find

$$\theta(t) = (\pi/12) \cos t - (1/3) \sin t.$$

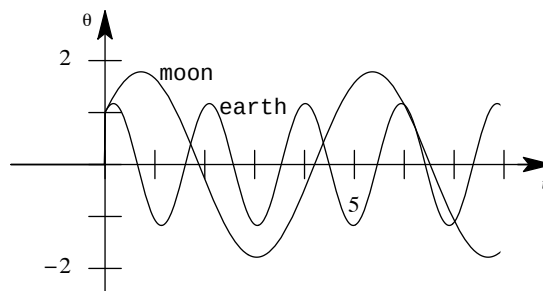
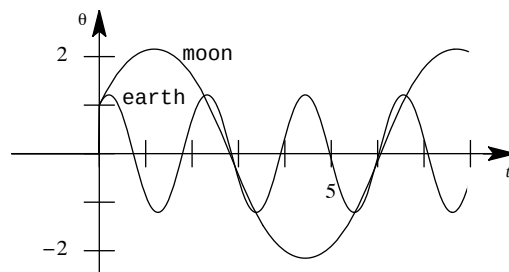
Setting $\theta(t) = 0$ we have $\tan t = \pi/4$ which implies $t_1 = \tan^{-1}(\pi/4) \approx 0.66577$.

(b) We set $\theta(t) = \theta(0) + \theta'(0)t + \frac{1}{2}\theta''(0)t^2 + \frac{1}{6}\theta'''(0)t^3 + \dots$ and use $\theta''(t) = -\sin \theta(t)$ together with $\theta(0) = \pi/12$ and $\theta'(0) = -1/3$. Then

$$\theta''(0) = -\sin(\pi/12) = -\sqrt{2}(\sqrt{3}-1)/4$$

and

$$\theta'''(0) = -\cos \theta(0) \cdot \theta'(0) = -\cos(\pi/12)(-1/3) = \sqrt{2}(\sqrt{3}+1)/12.$$



3.10 Nonlinear Models

Thus

$$\theta(t) = \frac{\pi}{12} - \frac{1}{3}t - \frac{\sqrt{2}(\sqrt{3}-1)}{8}t^2 + \frac{\sqrt{2}(\sqrt{3}+1)}{72}t^3 + \dots$$

(c) Setting $\pi/12 - t/3 = 0$ we obtain $t_1 = \pi/4 \approx 0.785398$.

(d) Setting

$$\frac{\pi}{12} - \frac{1}{3}t - \frac{\sqrt{2}(\sqrt{3}-1)}{8}t^2 = 0$$

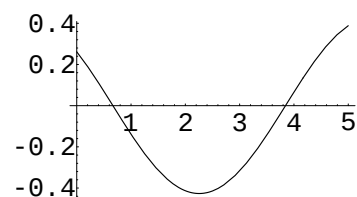
and using the positive root we obtain $t_1 \approx 0.63088$.

(e) Setting

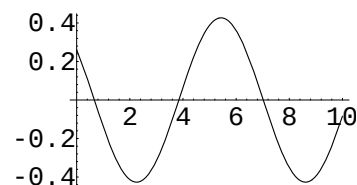
$$\frac{\pi}{12} - \frac{1}{3}t - \frac{\sqrt{2}(\sqrt{3}-1)}{8}t^2 + \frac{\sqrt{2}(\sqrt{3}+1)}{72}t^3 = 0$$

we find with the help of a CAS that $t_1 \approx 0.661973$ is the first positive root.

(f) From the output we see that $y(t)$ is an interpolating function on the interval $0 \leq t \leq 5$, whose graph is shown. The positive root of $y(t) = 0$ near $t = 1$ is $t_1 = 0.666404$.



(g) To find the next two positive roots we change the interval used in **NDSolve** and **Plot** from $\{t, 0, 5\}$ to $\{t, 0, 10\}$. We see from the graph that the second and third positive roots are near 4 and 7, respectively. Replacing $\{t, 1\}$ in **FindRoot** with $\{t, 4\}$ and then $\{t, 7\}$ we obtain $t_2 = 3.84411$ and $t_3 = 7.0218$.



25. From the table below we see that the pendulum first passes the vertical position between 1.7 and 1.8 seconds. To refine our estimate of t_1 we estimate the solution of the differential equation on $[1.7, 1.8]$ using a step size of $h = 0.01$. From the resulting table we see that t_1 is between 1.76 and 1.77 seconds. Repeating the process with $h = 0.001$ we conclude that $t_1 \approx 1.767$. Then the period of the pendulum is approximately $4t_1 = 7.068$. The error when using $t_1 = 2\pi$ is $7.068 - 6.283 = 0.785$ and the percentage relative error is $(0.785/7.068)100 = 11.1$.

h=0.1		h=0.01	
t_n	θ_n	t_n	θ_n
0.00	0.78540	1.70	0.07706
0.10	0.78523	1.71	0.06572
0.20	0.78407	1.72	0.05428
0.30	0.78092	1.73	0.04275
0.40	0.77482	1.74	0.03111
0.50	0.76482	1.75	0.01938
0.60	0.75004	1.76	0.00755
0.70	0.72962	1.77	-0.00438
0.80	0.70275	1.78	-0.01641
0.90	0.66872	1.79	-0.02854
1.00	0.62687	1.80	-0.04076
1.10	0.57660	h=0.001	
1.20	0.51744		
1.30	0.44895		
1.40	0.37085		
1.50	0.28289		
1.60	0.18497		
1.70	0.07706		
1.80	-0.04076		
1.90	-0.16831		
2.00	-0.30531		
		1.763	0.00398
		1.764	0.00279
		1.765	0.00160
		1.766	0.00040
		1.767	-0.00079
		1.768	-0.00199
		1.769	-0.00318
		1.770	-0.00438

EXERCISES 3.11

Solving Systems of Linear Equations

1. From $Dx = 2x - y$ and $Dy = x$ we obtain $y = 2x - Dx$, $Dy = 2Dx - D^2x$, and $(D^2 - 2D + 1)x = 0$. The solution is

$$\begin{aligned}x &= c_1 e^t + c_2 t e^t \\y &= (c_1 - c_2) e^t + c_2 t e^t.\end{aligned}$$

2. From $Dx = 4x + 7y$ and $Dy = x - 2y$ we obtain $y = \frac{1}{7}Dx - \frac{4}{7}x$, $Dy = \frac{1}{7}D^2x - \frac{4}{7}Dx$, and $(D^2 - 2D - 15)x = 0$. The solution is

$$\begin{aligned}x &= c_1 e^{5t} + c_2 e^{-3t} \\y &= \frac{1}{7} c_1 e^{5t} - c_2 e^{-3t}.\end{aligned}$$

3. From $Dx = -y + t$ and $Dy = x - t$ we obtain $y = t - Dx$, $Dy = 1 - D^2x$, and $(D^2 + 1)x = 1 + t$. The solution is

$$\begin{aligned}x &= c_1 \cos t + c_2 \sin t + 1 + t \\y &= c_1 \sin t - c_2 \cos t + t - 1.\end{aligned}$$

4. From $Dx - 4y = 1$ and $x + Dy = 2$ we obtain $y = \frac{1}{4}Dx - \frac{1}{4}$, $Dy = \frac{1}{4}D^2x$, and $(D^2 + 1)x = 2$. The solution is

$$\begin{aligned}x &= c_1 \cos t + c_2 \sin t + 2 \\y &= \frac{1}{4} c_2 \cos t - \frac{1}{4} c_1 \sin t - \frac{1}{4}.\end{aligned}$$

5. From $(D^2 + 5)x - 2y = 0$ and $-2x + (D^2 + 2)y = 0$ we obtain $y = \frac{1}{2}(D^2 + 5)x$, $D^2y = \frac{1}{2}(D^4 + 5D^2)x$, and $(D^2 + 1)(D^2 + 6)x = 0$. The solution is

$$\begin{aligned}x &= c_1 \cos t + c_2 \sin t + c_3 \cos \sqrt{6}t + c_4 \sin \sqrt{6}t \\y &= 2c_1 \cos t + 2c_2 \sin t - \frac{1}{2}c_3 \cos \sqrt{6}t - \frac{1}{2}c_4 \sin \sqrt{6}t.\end{aligned}$$

6. From $(D + 1)x + (D - 1)y = 2$ and $3x + (D + 2)y = -1$ we obtain $x = -\frac{1}{3} - \frac{1}{3}(D + 2)y$, $Dx = -\frac{1}{3}(D^2 + 2D)y$, and $(D^2 + 5)y = -7$. The solution is

$$\begin{aligned}y &= c_1 \cos \sqrt{5}t + c_2 \sin \sqrt{5}t - \frac{7}{5} \\x &= \left(-\frac{2}{3}c_1 - \frac{\sqrt{5}}{3}c_2\right) \cos \sqrt{5}t + \left(\frac{\sqrt{5}}{3}c_1 - \frac{2}{3}c_2\right) \sin \sqrt{5}t + \frac{3}{5}.\end{aligned}$$

7. From $D^2x = 4y + e^t$ and $D^2y = 4x - e^t$ we obtain $y = \frac{1}{4}D^2x - \frac{1}{4}e^t$, $D^2y = \frac{1}{4}D^4x - \frac{1}{4}e^t$, and $(D^2 + 4)(D - 2)(D + 2)x = -3e^t$. The solution is

$$\begin{aligned}x &= c_1 \cos 2t + c_2 \sin 2t + c_3 e^{2t} + c_4 e^{-2t} + \frac{1}{5}e^t \\y &= -c_1 \cos 2t - c_2 \sin 2t + c_3 e^{2t} + c_4 e^{-2t} - \frac{1}{5}e^t.\end{aligned}$$

3.11 Solving Systems of Linear Equations

8. From $(D^2 + 5)x + Dy = 0$ and $(D + 1)x + (D - 4)y = 0$ we obtain $(D - 5)(D^2 + 4)x = 0$ and $(D - 5)(D^2 + 4)y = 0$. The solution is

$$x = c_1 e^{5t} + c_2 \cos 2t + c_3 \sin 2t$$

$$y = c_4 e^{5t} + c_5 \cos 2t + c_6 \sin 2t.$$

Substituting into $(D + 1)x + (D - 4)y = 0$ gives

$$(6c_1 + c_4)e^{5t} + (c_2 + 2c_3 - 4c_5 + 2c_6) \cos 2t + (-2c_2 + c_3 - 2c_5 - 4c_6) \sin 2t = 0$$

so that $c_4 = -6c_1$, $c_5 = \frac{1}{2}c_3$, $c_6 = -\frac{1}{2}c_2$, and

$$y = -6c_1 e^{5t} + \frac{1}{2}c_3 \cos 2t - \frac{1}{2}c_2 \sin 2t.$$

9. From $Dx + D^2y = e^{3t}$ and $(D + 1)x + (D - 1)y = 4e^{3t}$ we obtain $D(D^2 + 1)x = 34e^{3t}$ and $D(D^2 + 1)y = -8e^{3t}$. The solution is

$$y = c_1 + c_2 \sin t + c_3 \cos t - \frac{4}{15}e^{3t}$$

$$x = c_4 + c_5 \sin t + c_6 \cos t + \frac{17}{15}e^{3t}.$$

Substituting into $(D + 1)x + (D - 1)y = 4e^{3t}$ gives

$$(c_4 - c_1) + (c_5 - c_6 - c_3 - c_2) \sin t + (c_6 + c_5 + c_2 - c_3) \cos t = 0$$

so that $c_4 = c_1$, $c_5 = c_3$, $c_6 = -c_2$, and

$$x = c_1 - c_2 \cos t + c_3 \sin t + \frac{17}{15}e^{3t}.$$

10. From $D^2x - Dy = t$ and $(D + 3)x + (D + 3)y = 2$ we obtain $D(D + 1)(D + 3)x = 1 + 3t$ and $D(D + 1)(D + 3)y = -1 - 3t$. The solution is

$$x = c_1 + c_2 e^{-t} + c_3 e^{-3t} - t + \frac{1}{2}t^2$$

$$y = c_4 + c_5 e^{-t} + c_6 e^{-3t} + t - \frac{1}{2}t^2.$$

Substituting into $(D + 3)x + (D + 3)y = 2$ and $D^2x - Dy = t$ gives

$$3(c_1 + c_4) + 2(c_2 + c_5)e^{-t} = 2$$

and

$$(c_2 + c_5)e^{-t} + 3(3c_3 + c_6)e^{-3t} = 0$$

so that $c_4 = -c_1$, $c_5 = -c_2$, $c_6 = -3c_3$, and

$$y = -c_1 - c_2 e^{-t} - 3c_3 e^{-3t} + t - \frac{1}{2}t^2.$$

11. From $(D^2 - 1)x - y = 0$ and $(D - 1)x + Dy = 0$ we obtain $y = (D^2 - 1)x$, $Dy = (D^3 - D)x$, and $(D - 1)(D^2 + D + 1)x = 0$. The solution is

$$x = c_1 e^t + e^{-t/2} \left[c_2 \cos \frac{\sqrt{3}}{2}t + c_3 \sin \frac{\sqrt{3}}{2}t \right]$$

$$y = \left(-\frac{3}{2}c_2 - \frac{\sqrt{3}}{2}c_3 \right) e^{-t/2} \cos \frac{\sqrt{3}}{2}t + \left(\frac{\sqrt{3}}{2}c_2 - \frac{3}{2}c_3 \right) e^{-t/2} \sin \frac{\sqrt{3}}{2}t.$$

3.11 Solving Systems of Linear Equations

12. From $(2D^2 - D - 1)x - (2D + 1)y = 1$ and $(D - 1)x + Dy = -1$ we obtain $(2D + 1)(D - 1)(D + 1)x = -1$ and $(2D + 1)(D + 1)y = -2$. The solution is

$$\begin{aligned}x &= c_1 e^{-t/2} + c_2 e^{-t} + c_3 e^t + 1 \\y &= c_4 e^{-t/2} + c_5 e^{-t} - 2.\end{aligned}$$

Substituting into $(D - 1)x + Dy = -1$ gives

$$\left(-\frac{3}{2}c_1 - \frac{1}{2}c_4\right)e^{-t/2} + (-2c_2 - c_5)e^{-t} = 0$$

so that $c_4 = -3c_1$, $c_5 = -2c_2$, and

$$y = -3c_1 e^{-t/2} - 2c_2 e^{-t} - 2.$$

13. From $(2D - 5)x + Dy = e^t$ and $(D - 1)x + Dy = 5e^t$ we obtain $Dy = (5 - 2D)x + e^t$ and $(4 - D)x = 4e^t$. Then

$$x = c_1 e^{4t} + \frac{4}{3}e^t$$

and $Dy = -3c_1 e^{4t} + 5e^t$ so that

$$y = -\frac{3}{4}c_1 e^{4t} + c_2 + 5e^t.$$

14. From $Dx + Dy = e^t$ and $(-D^2 + D + 1)x + y = 0$ we obtain $y = (D^2 - D - 1)x$, $Dy = (D^3 - D^2 - D)x$, and $D^2(D - 1)x = e^t$. The solution is

$$\begin{aligned}x &= c_1 + c_2 t + c_3 e^t + t e^t \\y &= -c_1 - c_2 - c_2 t - c_3 e^t - t e^t + e^t.\end{aligned}$$

15. Multiplying the first equation by $D + 1$ and the second equation by $D^2 + 1$ and subtracting we obtain $(D^4 - D^2)x = 1$. Then

$$x = c_1 + c_2 t + c_3 e^t + c_4 e^{-t} - \frac{1}{2}t^2.$$

Multiplying the first equation by $D + 1$ and subtracting we obtain $D^2(D + 1)y = 1$. Then

$$y = c_5 + c_6 t + c_7 e^{-t} - \frac{1}{2}t^2.$$

Substituting into $(D - 1)x + (D^2 + 1)y = 1$ gives

$$(-c_1 + c_2 + c_5 - 1) + (-2c_4 + 2c_7)e^{-t} + (-1 - c_2 + c_6)t = 1$$

so that $c_5 = c_1 - c_2 + 2$, $c_6 = c_2 + 1$, and $c_7 = c_4$. The solution of the system is

$$\begin{aligned}x &= c_1 + c_2 t + c_3 e^t + c_4 e^{-t} - \frac{1}{2}t^2 \\y &= (c_1 - c_2 + 2) + (c_2 + 1)t + c_4 e^{-t} - \frac{1}{2}t^2.\end{aligned}$$

16. From $D^2x - 2(D^2 + D)y = \sin t$ and $x + Dy = 0$ we obtain $x = -Dy$, $D^2x = -D^3y$, and $D(D^2 + 2D + 2)y = -\sin t$. The solution is

$$\begin{aligned}y &= c_1 + c_2 e^{-t} \cos t + c_3 e^{-t} \sin t + \frac{1}{5} \cos t + \frac{2}{5} \sin t \\x &= (c_2 + c_3)e^{-t} \sin t + (c_2 - c_3)e^{-t} \cos t + \frac{1}{5} \sin t - \frac{2}{5} \cos t.\end{aligned}$$

3.11 Solving Systems of Linear Equations

17. From $Dx = y$, $Dy = z$, and $Dz = x$ we obtain $x = D^2y = D^3x$ so that $(D - 1)(D^2 + D + 1)x = 0$,

$$x = c_1e^t + e^{-t/2} \left[c_2 \sin \frac{\sqrt{3}}{2}t + c_3 \cos \frac{\sqrt{3}}{2}t \right],$$

$$y = c_1e^t + \left(-\frac{1}{2}c_2 - \frac{\sqrt{3}}{2}c_3 \right) e^{-t/2} \sin \frac{\sqrt{3}}{2}t + \left(\frac{\sqrt{3}}{2}c_2 - \frac{1}{2}c_3 \right) e^{-t/2} \cos \frac{\sqrt{3}}{2}t,$$

and

$$z = c_1e^t + \left(-\frac{1}{2}c_2 + \frac{\sqrt{3}}{2}c_3 \right) e^{-t/2} \sin \frac{\sqrt{3}}{2}t + \left(-\frac{\sqrt{3}}{2}c_2 - \frac{1}{2}c_3 \right) e^{-t/2} \cos \frac{\sqrt{3}}{2}t.$$

18. From $Dx + z = e^t$, $(D - 1)x + Dy + Dz = 0$, and $x + 2y + Dz = e^t$ we obtain $z = -Dx + e^t$, $Dz = -D^2x + e^t$, and the system $(-D^2 + D - 1)x + Dy = -e^t$ and $(-D^2 + 1)x + 2y = 0$. Then $y = \frac{1}{2}(D^2 - 1)x$, $Dy = \frac{1}{2}D(D^2 - 1)x$, and $(D - 2)(D^2 + 1)x = -2e^t$ so that the solution is

$$x = c_1e^{2t} + c_2 \cos t + c_3 \sin t + e^t$$

$$y = \frac{3}{2}c_1e^{2t} - c_2 \cos t - c_3 \sin t$$

$$z = -2c_1e^{2t} - c_3 \cos t + c_2 \sin t.$$

19. Write the system in the form

$$Dx - 6y = 0$$

$$x - Dy + z = 0$$

$$x + y - Dz = 0.$$

Multiplying the second equation by D and adding to the third equation we obtain $(D + 1)x - (D^2 - 1)y = 0$.

Eliminating y between this equation and $Dx - 6y = 0$ we find

$$(D^3 - D - 6D - 6)x = (D + 1)(D + 2)(D - 3)x = 0.$$

Thus

$$x = c_1e^{-t} + c_2e^{-2t} + c_3e^{3t},$$

and, successively substituting into the first and second equations, we get

$$y = -\frac{1}{6}c_1e^{-t} - \frac{1}{3}c_2e^{-2t} + \frac{1}{2}c_3e^{3t}$$

$$z = -\frac{5}{6}c_1e^{-t} - \frac{1}{3}c_2e^{-2t} + \frac{1}{2}c_3e^{3t}.$$

20. Write the system in the form

$$(D + 1)x - z = 0$$

$$(D + 1)y - z = 0$$

$$x - y + Dz = 0.$$

Multiplying the third equation by $D + 1$ and adding to the second equation we obtain $(D + 1)x + (D^2 + D - 1)z = 0$. Eliminating z between this equation and $(D + 1)x - z = 0$ we find $D(D + 1)^2x = 0$.

Thus

$$x = c_1 + c_2e^{-t} + c_3te^{-t},$$

and, successively substituting into the first and third equations, we get

$$y = c_1 + (c_2 - c_3)e^{-t} + c_3te^{-t}$$

$$z = c_1 + c_3e^{-t}.$$

3.11 Solving Systems of Linear Equations

21. From $(D + 5)x + y = 0$ and $4x - (D + 1)y = 0$ we obtain $y = -(D + 5)x$ so that $Dy = -(D^2 + 5D)x$. Then $4x + (D^2 + 5D)x + (D + 5)x = 0$ and $(D + 3)^2x = 0$. Thus

$$\begin{aligned}x &= c_1 e^{-3t} + c_2 t e^{-3t} \\y &= -(2c_1 + c_2)e^{-3t} - 2c_2 t e^{-3t}.\end{aligned}$$

Using $x(1) = 0$ and $y(1) = 1$ we obtain

$$\begin{aligned}c_1 e^{-3} + c_2 e^{-3} &= 0 \\-(2c_1 + c_2)e^{-3} - 2c_2 e^{-3} &= 1\end{aligned}$$

or

$$\begin{aligned}c_1 + c_2 &= 0 \\2c_1 + 3c_2 &= -e^3.\end{aligned}$$

Thus $c_1 = e^3$ and $c_2 = -e^3$. The solution of the initial value problem is

$$\begin{aligned}x &= e^{-3t+3} - t e^{-3t+3} \\y &= -e^{-3t+3} + 2t e^{-3t+3}.\end{aligned}$$

22. From $Dx - y = -1$ and $3x + (D - 2)y = 0$ we obtain $x = -\frac{1}{3}(D - 2)y$ so that $Dx = -\frac{1}{3}(D^2 - 2D)y$. Then $-\frac{1}{3}(D^2 - 2D)y = y - 1$ and $(D^2 - 2D + 3)y = 3$. Thus

$$y = e^t (c_1 \cos \sqrt{2}t + c_2 \sin \sqrt{2}t) + 1$$

and

$$x = \frac{1}{3}e^t \left[(c_1 - \sqrt{2}c_2) \cos \sqrt{2}t + (\sqrt{2}c_1 + c_2) \sin \sqrt{2}t \right] + \frac{2}{3}.$$

Using $x(0) = y(0) = 0$ we obtain

$$\begin{aligned}c_1 + 1 &= 0 \\\frac{1}{3}(c_1 - \sqrt{2}c_2) + \frac{2}{3} &= 0.\end{aligned}$$

Thus $c_1 = -1$ and $c_2 = \sqrt{2}/2$. The solution of the initial value problem is

$$\begin{aligned}x &= e^t \left(-\frac{2}{3} \cos \sqrt{2}t - \frac{\sqrt{2}}{6} \sin \sqrt{2}t \right) + \frac{2}{3} \\y &= e^t \left(-\cos \sqrt{2}t + \frac{\sqrt{2}}{2} \sin \sqrt{2}t \right) + 1.\end{aligned}$$

23. Equating Newton's law with the net forces in the x - and y -directions gives $m d^2x/dt^2 = 0$ and $m d^2y/dt^2 = -mg$, respectively. From $mD^2x = 0$ we obtain $x(t) = c_1t + c_2$, and from $mD^2y = -mg$ or $D^2y = -g$ we obtain $y(t) = -\frac{1}{2}gt^2 + c_3t + c_4$.

24. From Newton's second law in the x -direction we have

$$m \frac{d^2x}{dt^2} = -k \cos \theta = -k \frac{1}{v} \frac{dx}{dt} = -|c| \frac{dx}{dt}.$$

In the y -direction we have

$$m \frac{d^2y}{dt^2} = -mg - k \sin \theta = -mg - k \frac{1}{v} \frac{dy}{dt} = -mg - |c| \frac{dy}{dt}.$$

From $mD^2x + |c|Dx = 0$ we have $D(mD + |c|)x = 0$ so that $(mD + |c|)x = c_1$ or $(D + |c|/m)x = c_2$. This is a linear first-order differential equation. An integrating factor is $e^{\int |c|dt/m} = e^{|c|t/m}$ so that

$$\frac{d}{dt}[e^{|c|t/m}x] = c_2 e^{|c|t/m}$$

3.11 Solving Systems of Linear Equations

and $e^{|c|t/m}x = (c_2m/|c|)e^{|c|t/m} + c_3$. The general solution of this equation is $x(t) = c_4 + c_3e^{-|c|t/m}$. From $(mD^2 + |c|D)y = -mg$ we have $D(mD + |c|)y = -mg$ so that $(mD + |c|)y = -mgt + c_1$ or $(D + |c|/m)y = -gt + c_2$. This is a linear first-order differential equation with integrating factor $e^{\int |c|dt/m} = e^{|c|t/m}$. Thus

$$\begin{aligned}\frac{d}{dt}[e^{|c|t/m}y] &= (-gt + c_2)e^{|c|t/m} \\ e^{|c|t/m}y &= -\frac{mg}{|c|}te^{|c|t/m} + \frac{m^2g}{c^2}e^{|c|t/m} + c_3e^{|c|t/m} + c_4\end{aligned}$$

and

$$y(t) = -\frac{mg}{|c|}t + \frac{m^2g}{c^2} + c_3 + c_4e^{-|c|t/m}.$$

25. The **FindRoot** application of *Mathematica* gives a solution of $x_1(t) = x_2(t)$ as approximately $t = 13.73$ minutes. So tank *B* contains more salt than tank *A* for $t > 13.73$ minutes.

26. (a) Separating variables in the first equation, we have $dx_1/x_1 = -dt/50$, so $x_1 = c_1e^{-t/50}$. From $x_1(0) = 15$ we get $c_1 = 15$. The second differential equation then becomes

$$\frac{dx_2}{dt} = \frac{15}{50}e^{-t/50} - \frac{2}{75}x_2 \quad \text{or} \quad \frac{dx_2}{dt} + \frac{2}{75}x_2 = \frac{3}{10}e^{-t/50}.$$

This differential equation is linear and has the integrating factor $e^{\int 2dt/75} = e^{2t/75}$. Then

$$\frac{d}{dt}[e^{2t/75}x_2] = \frac{3}{10}e^{-t/50+2t/75} = \frac{3}{10}e^{t/150}$$

so

$$e^{2t/75}x_2 = 45e^{t/150} + c_2$$

and

$$x_2 = 45e^{-t/50} + c_2e^{-2t/75}.$$

From $x_2(0) = 10$ we get $c_2 = -35$. The third differential equation then becomes

$$\frac{dx_3}{dt} = \frac{90}{75}e^{-t/50} - \frac{70}{75}e^{-2t/75} - \frac{1}{25}x_3$$

or

$$\frac{dx_3}{dt} + \frac{1}{25}x_3 = \frac{6}{5}e^{-t/50} - \frac{14}{15}e^{-2t/75}.$$

This differential equation is linear and has the integrating factor $e^{\int dt/25} = e^{t/25}$. Then

$$\frac{d}{dt}[e^{t/25}x_3] = \frac{6}{5}e^{-t/50+t/25} - \frac{14}{15}e^{-2t/75+t/25} = \frac{6}{5}e^{t/50} - \frac{14}{15}e^{t/75},$$

so

$$e^{t/25}x_3 = 60e^{t/50} - 70e^{t/75} + c_3$$

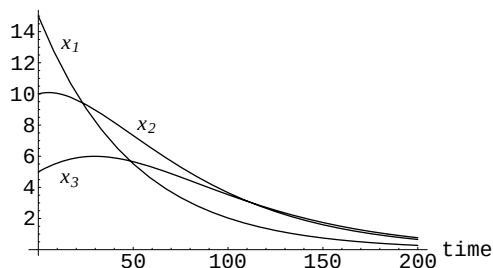
and

$$x_3 = 60e^{-t/50} - 70e^{-2t/75} + c_3e^{-t/25}.$$

From $x_3(0) = 5$ we get $c_3 = 15$. The solution of the initial-value problem is

$$\begin{aligned}x_1(t) &= 15e^{-t/50} \\ x_2(t) &= 45e^{-t/50} - 35e^{-2t/75} \\ x_3(t) &= 60e^{-t/50} - 70e^{-2t/75} + 15e^{-t/25}.\end{aligned}$$

(b) pounds salt



(c) Solving $x_1(t) = \frac{1}{2}$, $x_2(t) = \frac{1}{2}$, and $x_3(t) = \frac{1}{2}$, **FindRoot** gives, respectively, $t_1 = 170.06$ min, $t_2 = 214.7$ min, and $t_3 = 224.4$ min. Thus, all three tanks will contain less than or equal to 0.5 pounds of salt after 224.4 minutes.

27. (a) Write the system as

$$\begin{aligned} (D^2 + 3)x_1 - x_2 &= 0 \\ -2x_1 + (D^2 + 2)x_2 &= 0. \end{aligned}$$

Then

$$(D^2 + 2)(D^2 + 3)x_1 - 2x_1 = (D^2 + 1)(D^2 + 4)x_1 = 0,$$

and

$$x_1(t) = c_1 \cos t + c_2 \sin t + c_3 \cos 2t + c_4 \sin 2t.$$

Since $x_2 = (D^2 + 3)x_1$, we have

$$x_2(t) = 2c_1 \cos t + 2c_2 \sin t - c_3 \cos 2t - c_4 \sin 2t.$$

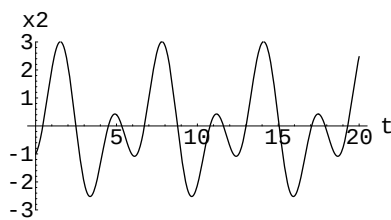
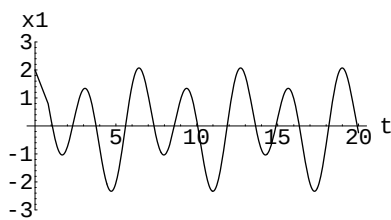
The initial conditions imply

$$\begin{aligned} x_1(0) &= c_1 + c_3 = 2 \\ x_1'(0) &= c_2 + 2c_4 = 1 \\ x_2(0) &= 2c_1 - c_3 = -1 \\ x_2'(0) &= 2c_2 - 2c_4 = 1, \end{aligned}$$

so $c_1 = \frac{1}{3}$, $c_2 = \frac{2}{3}$, $c_3 = \frac{5}{3}$, and $c_4 = \frac{1}{6}$. Thus

$$\begin{aligned} x_1(t) &= \frac{1}{3} \cos t + \frac{2}{3} \sin t + \frac{5}{3} \cos 2t + \frac{1}{6} \sin 2t \\ x_2(t) &= \frac{2}{3} \cos t + \frac{4}{3} \sin t - \frac{5}{3} \cos 2t - \frac{1}{6} \sin 2t. \end{aligned}$$

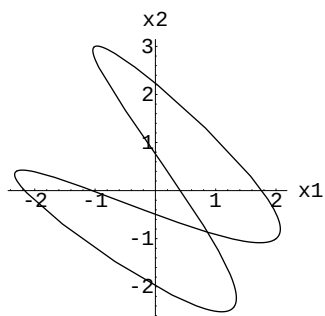
(b)



In this problem the motion appears to be periodic with period 2π . In Figure 3.59 of Example 4 in the text the motion does not appear to be periodic.

3.11 Solving Systems of Linear Equations

(c)



CHAPTER 3 REVIEW EXERCISES

1. $y = 0$
2. Since $y_c = c_1 e^x + c_2 e^{-x}$, a particular solution for $y'' - y = 1 + e^x$ is $y_p = A + Bxe^x$.
3. It is not true unless the differential equation is homogeneous. For example, $y_1 = x$ is a solution of $y'' + y = x$, but $y_2 = 5x$ is not.
4. True
5. 8 ft, since $k = 4$
6. $2\pi/5$, since $\frac{1}{4}x'' + 6.25x = 0$
7. $5/4$ m, since $x = -\cos 4t + \frac{3}{4}\sin 4t$
8. From $x(0) = (\sqrt{2}/2)\sin\phi = -1/2$ we see that $\sin\phi = -1/\sqrt{2}$, so ϕ is an angle in the third or fourth quadrant. Since $x'(t) = \sqrt{2}\cos(2t + \phi)$, $x'(0) = \sqrt{2}\cos\phi = 1$ and $\cos\phi > 0$. Thus ϕ is in the fourth quadrant and $\phi = -\pi/4$.
9. The set is linearly independent over $(-\infty, \infty)$ and linearly dependent over $(0, \infty)$.
10. (a) Since $f_2(x) = 2\ln x = 2f_1(x)$, the set of functions is linearly dependent.
 (b) Since x^{n+1} is not a constant multiple of x^n , the set of functions is linearly independent.
 (c) Since $x + 1$ is not a constant multiple of x , the set of functions is linearly independent.
 (d) Since $f_1(x) = \cos x \cos(\pi/2) - \sin x \sin(\pi/2) = -\sin x = -f_2(x)$, the set of functions is linearly dependent.
 (e) Since $f_1(x) = 0 \cdot f_2(x)$, the set of functions is linearly dependent.
 (f) Since $2x$ is not a constant multiple of x , the set of functions is linearly independent.
 (g) Since $3(x^2) + 2(1 - x^2) - (2 + x^2) = 0$, the set of functions is linearly dependent.
 (h) Since $xe^{x+1} + 0(4x - 5)e^x - exe^x = 0$, the set of functions is linearly dependent.

11. (a) The auxiliary equation is $(m-3)(m+5)(m-1) = m^3 + m^2 - 17m + 15 = 0$, so the differential equation is $y''' + y'' - 17y' + 15y = 0$.

(b) The form of the auxiliary equation is

$$m(m-1)(m-2) + bm(m-1) + cm + d = m^3 + (b-3)m^2 + (c-b+2)m + d = 0.$$

Since $(m-3)(m+5)(m-1) = m^3 + m^2 - 17m + 15 = 0$, we have $b-3 = 1$, $c-b+2 = -17$, and $d = 15$. Thus, $b = 4$ and $c = -15$, so the differential equation is $y''' + 4y'' - 15y' + 15y = 0$.

12. (a) The auxiliary equation is $am(m-1) + bm + c = am^2 + (b-a)m + c = 0$. If the roots are 3 and -1 , then we want $(m-3)(m+1) = m^2 - 2m - 3 = 0$. Thus, let $a = 1$, $b = -1$, and $c = -3$, so that the differential equation is $x^2y'' - xy' - 3y = 0$.

(b) In this case we want the auxiliary equation to be $m^2 + 1 = 0$, so let $a = 1$, $b = 1$, and $c = 1$. Then the differential equation is $x^2y'' + xy' + y = 0$.

13. From $m^2 - 2m - 2 = 0$ we obtain $m = 1 \pm \sqrt{3}$ so that

$$y = c_1 e^{(1+\sqrt{3})x} + c_2 e^{(1-\sqrt{3})x}.$$

14. From $2m^2 + 2m + 3 = 0$ we obtain $m = -1/2 \pm (\sqrt{5}/2)i$ so that

$$y = e^{-x/2} \left(c_1 \cos \frac{\sqrt{5}}{2}x + c_2 \sin \frac{\sqrt{5}}{2}x \right).$$

15. From $m^3 + 10m^2 + 25m = 0$ we obtain $m = 0$, $m = -5$, and $m = -5$ so that

$$y = c_1 + c_2 e^{-5x} + c_3 x e^{-5x}.$$

16. From $2m^3 + 9m^2 + 12m + 5 = 0$ we obtain $m = -1$, $m = -1$, and $m = -5/2$ so that

$$y = c_1 e^{-5x/2} + c_2 e^{-x} + c_3 x e^{-x}.$$

17. From $3m^3 + 10m^2 + 15m + 4 = 0$ we obtain $m = -1/3$ and $m = -3/2 \pm (\sqrt{7}/2)i$ so that

$$y = c_1 e^{-x/3} + e^{-3x/2} \left(c_2 \cos \frac{\sqrt{7}}{2}x + c_3 \sin \frac{\sqrt{7}}{2}x \right).$$

18. From $2m^4 + 3m^3 + 2m^2 + 6m - 4 = 0$ we obtain $m = 1/2$, $m = -2$, and $m = \pm\sqrt{2}i$ so that

$$y = c_1 e^{x/2} + c_2 e^{-2x} + c_3 \cos \sqrt{2}x + c_4 \sin \sqrt{2}x.$$

19. Applying D^4 to the differential equation we obtain $D^4(D^2 - 3D + 5) = 0$. Then

$$y = \underbrace{e^{3x/2} \left(c_1 \cos \frac{\sqrt{11}}{2}x + c_2 \sin \frac{\sqrt{11}}{2}x \right)}_{y_c} + c_3 + c_4 x + c_5 x^2 + c_6 x^3$$

and $y_p = A + Bx + Cx^2 + Dx^3$. Substituting y_p into the differential equation yields

$$(5A - 3B + 2C) + (5B - 6C + 6D)x + (5C - 9D)x^2 + 5Dx^3 = -2x + 4x^3.$$

Equating coefficients gives $A = -222/625$, $B = 46/125$, $C = 36/25$, and $D = 4/5$. The general solution is

$$y = e^{3x/2} \left(c_1 \cos \frac{\sqrt{11}}{2}x + c_2 \sin \frac{\sqrt{11}}{2}x \right) - \frac{222}{625} + \frac{46}{125}x + \frac{36}{25}x^2 + \frac{4}{5}x^3.$$

CHAPTER 3 REVIEW EXERCISES

20. Applying $(D - 1)^3$ to the differential equation we obtain $(D - 1)^3(D - 2D + 1) = (D - 1)^5 = 0$. Then

$$y = \underbrace{c_1 e^x + c_2 x e^x}_{y_c} + c_3 x^2 e^x + c_4 x^3 e^x + c_5 x^4 e^x$$

and $y_p = Ax^2 e^x + Bx^3 e^x + Cx^4 e^x$. Substituting y_p into the differential equation yields

$$12Cx^2 e^x + 6Bx e^x + 2Ae^x = x^2 e^x.$$

Equating coefficients gives $A = 0$, $B = 0$, and $C = 1/12$. The general solution is

$$y = c_1 e^x + c_2 x e^x + \frac{1}{12} x^4 e^x.$$

21. Applying $D(D^2 + 1)$ to the differential equation we obtain

$$D(D^2 + 1)(D^3 - 5D^2 + 6D) = D^2(D^2 + 1)(D - 2)(D - 3) = 0.$$

Then

$$y = \underbrace{c_1 + c_2 e^{2x} + c_3 e^{3x}}_{y_c} + c_4 x + c_5 \cos x + c_6 \sin x$$

and $y_p = Ax + B \cos x + C \sin x$. Substituting y_p into the differential equation yields

$$6A + (5B + 5C) \cos x + (-5B + 5C) \sin x = 8 + 2 \sin x.$$

Equating coefficients gives $A = 4/3$, $B = -1/5$, and $C = 1/5$. The general solution is

$$y = c_1 + c_2 e^{2x} + c_3 e^{3x} + \frac{4}{3}x - \frac{1}{5} \cos x + \frac{1}{5} \sin x.$$

22. Applying D to the differential equation we obtain $D(D^3 - D^2) = D^3(D - 1) = 0$. Then

$$y = \underbrace{c_1 + c_2 x + c_3 e^x}_{y_c} + c_4 x^2$$

and $y_p = Ax^2$. Substituting y_p into the differential equation yields $-2A = 6$. Equating coefficients gives $A = -3$. The general solution is

$$y = c_1 + c_2 x + c_3 e^x - 3x^2.$$

23. The auxiliary equation is $m^2 - 2m + 2 = [m - (1 + i)][m - (1 - i)] = 0$, so $y_c = c_1 e^x \sin x + c_2 e^x \cos x$ and

$$W = \begin{vmatrix} e^x \sin x & e^x \cos x \\ e^x \cos x + e^x \sin x & -e^x \sin x + e^x \cos x \end{vmatrix} = -e^{2x}.$$

Identifying $f(x) = e^x \tan x$ we obtain

$$u'_1 = -\frac{(e^x \cos x)(e^x \tan x)}{-e^{2x}} = \sin x$$

$$u'_2 = \frac{(e^x \sin x)(e^x \tan x)}{-e^{2x}} = -\frac{\sin^2 x}{\cos x} = \cos x - \sec x.$$

Then $u_1 = -\cos x$, $u_2 = \sin x - \ln |\sec x + \tan x|$, and

$$y = c_1 e^x \sin x + c_2 e^x \cos x - e^x \sin x \cos x + e^x \sin x \cos x - e^x \cos x \ln |\sec x + \tan x|$$

$$= c_1 e^x \sin x + c_2 e^x \cos x - e^x \cos x \ln |\sec x + \tan x|.$$

24. The auxiliary equation is $m^2 - 1 = 0$, so $y_c = c_1 e^x + c_2 e^{-x}$ and

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2.$$

Identifying $f(x) = 2e^x/(e^x + e^{-x})$ we obtain

$$u'_1 = \frac{1}{e^x + e^{-x}} = \frac{e^x}{1 + e^{2x}}$$

$$u'_2 = -\frac{e^{2x}}{e^x + e^{-x}} = -\frac{e^{3x}}{1 + e^{2x}} = -e^x + \frac{e^x}{1 + e^{2x}}.$$

Then $u_1 = \tan^{-1} e^x$, $u_2 = -e^x + \tan^{-1} e^x$, and

$$y = c_1 e^x + c_2 e^{-x} + e^x \tan^{-1} e^x - 1 + e^{-x} \tan^{-1} e^x.$$

25. The auxiliary equation is $6m^2 - m - 1 = 0$ so that

$$y = c_1 x^{1/2} + c_2 x^{-1/3}.$$

26. The auxiliary equation is $2m^3 + 13m^2 + 24m + 9 = (m + 3)^2(m + 1/2) = 0$ so that

$$y = c_1 x^{-3} + c_2 x^{-3} \ln x + c_3 x^{-1/2}.$$

27. The auxiliary equation is $m^2 - 5m + 6 = (m - 2)(m - 3) = 0$ and a particular solution is $y_p = x^4 - x^2 \ln x$ so that

$$y = c_1 x^2 + c_2 x^3 + x^4 - x^2 \ln x.$$

28. The auxiliary equation is $m^2 - 2m + 1 = (m - 1)^2 = 0$ and a particular solution is $y_p = \frac{1}{4}x^3$ so that

$$y = c_1 x + c_2 x \ln x + \frac{1}{4}x^3.$$

29. (a) The auxiliary equation is $m^2 + \omega^2 = 0$, so $y_c = c_1 \cos \omega t + c_2 \sin \omega t$. When $\omega \neq \alpha$, $y_p = A \cos \alpha t + B \sin \alpha t$ and

$$y = c_1 \cos \omega t + c_2 \sin \omega t + A \cos \alpha t + B \sin \alpha t.$$

When $\omega = \alpha$, $y_p = At \cos \omega t + Bt \sin \omega t$ and

$$y = c_1 \cos \omega t + c_2 \sin \omega t + At \cos \omega t + Bt \sin \omega t.$$

- (b) The auxiliary equation is $m^2 - \omega^2 = 0$, so $y_c = c_1 e^{\omega t} + c_2 e^{-\omega t}$. When $\omega \neq \alpha$, $y_p = Ae^{\alpha t}$ and

$$y = c_1 e^{\omega t} + c_2 e^{-\omega t} + Ae^{\alpha t}.$$

When $\omega = \alpha$, $y_p = Ate^{\omega t}$ and

$$y = c_1 e^{\omega t} + c_2 e^{-\omega t} + Ate^{\omega t}.$$

30. (a) If $y = \sin x$ is a solution then so is $y = \cos x$ and $m^2 + 1$ is a factor of the auxiliary equation $m^4 + 2m^3 + 11m^2 + 2m + 10 = 0$. Dividing by $m^2 + 1$ we get $m^2 + 2m + 10$, which has roots $-1 \pm 3i$. The general solution of the differential equation is

$$y = c_1 \cos x + c_2 \sin x + e^{-x}(c_3 \cos 3x + c_4 \sin 3x).$$

- (b) The auxiliary equation is $m(m + 1) = m^2 + m = 0$, so the associated homogeneous differential equation is $y'' + y' = 0$. Letting $y = c_1 + c_2 e^{-x} + \frac{1}{2}x^2 - x$ and computing $y'' + y'$ we get x . Thus, the differential equation is $y'' + y' = x$.

31. (a) The auxiliary equation is $m^4 - 2m^2 + 1 = (m^2 - 1)^2 = 0$, so the general solution of the differential equation is

$$y = c_1 \sinh x + c_2 \cosh x + c_3 x \sinh x + c_4 x \cosh x.$$

CHAPTER 3 REVIEW EXERCISES

(b) Since both $\sinh x$ and $x \sinh x$ are solutions of the associated homogeneous differential equation, a particular solution of $y^{(4)} - 2y'' + y = \sinh x$ has the form $y_p = Ax^2 \sinh x + Bx^2 \cosh x$.

32. Since $y'_1 = 1$ and $y''_1 = 0$, $x^2 y''_1 - (x^2 + 2x)y'_1 + (x+2)y_1 = -x^2 - 2x + x^2 + 2x = 0$, and $y_1 = x$ is a solution of the associated homogeneous equation. Using the method of reduction of order, we let $y = ux$. Then $y' = xu' + u$ and $y'' = xu'' + 2u'$, so

$$\begin{aligned} x^2 y'' - (x^2 + 2x)y' + (x+2)y &= x^3 u'' + 2x^2 u' - x^3 u' - 2x^2 u' - x^2 u - 2xu + x^2 u + 2xu \\ &= x^3 u'' - x^3 u' = x^3(u'' - u'). \end{aligned}$$

To find a second solution of the homogeneous equation we note that $u = e^x$ is a solution of $u'' - u' = 0$. Thus, $y_c = c_1 x + c_2 x e^x$. To find a particular solution we set $x^3(u'' - u') = x^3$ so that $u'' - u' = 1$. This differential equation has a particular solution of the form Ax . Substituting, we find $A = -1$, so a particular solution of the original differential equation is $y_p = -x^2$ and the general solution is $y = c_1 x + c_2 x e^x - x^2$.

33. The auxiliary equation is $m^2 - 2m + 2 = 0$ so that $m = 1 \pm i$ and $y = e^x(c_1 \cos x + c_2 \sin x)$. Setting $y(\pi/2) = 0$ and $y(\pi) = -1$ we obtain $c_1 = e^{-\pi}$ and $c_2 = 0$. Thus, $y = e^{x-\pi} \cos x$.
34. The auxiliary equation is $m^2 + 2m + 1 = (m+1)^2 = 0$, so that $y = c_1 e^{-x} + c_2 x e^{-x}$. Setting $y(-1) = 0$ and $y'(0) = 0$ we get $c_1 e - c_2 e = 0$ and $-c_1 + c_2 = 0$. Thus $c_1 = c_2$ and $y = c_1(e^{-x} + x e^{-x})$ is a solution of the boundary-value problem for any real number c_1 .
35. The auxiliary equation is $m^2 - 1 = (m-1)(m+1) = 0$ so that $m = \pm 1$ and $y = c_1 e^x + c_2 e^{-x}$. Assuming $y_p = Ax + B + C \sin x$ and substituting into the differential equation we find $A = -1$, $B = 0$, and $C = -\frac{1}{2}$. Thus $y_p = -x - \frac{1}{2} \sin x$ and

$$y = c_1 e^x + c_2 e^{-x} - x - \frac{1}{2} \sin x.$$

Setting $y(0) = 2$ and $y'(0) = 3$ we obtain

$$\begin{aligned} c_1 + c_2 &= 2 \\ c_1 - c_2 - \frac{3}{2} &= 3. \end{aligned}$$

Solving this system we find $c_1 = \frac{13}{4}$ and $c_2 = -\frac{5}{4}$. The solution of the initial-value problem is

$$y = \frac{13}{4} e^x - \frac{5}{4} e^{-x} - x - \frac{1}{2} \sin x.$$

36. The auxiliary equation is $m^2 + 1 = 0$, so $y_c = c_1 \cos x + c_2 \sin x$ and

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Identifying $f(x) = \sec^3 x$ we obtain

$$u'_1 = -\sin x \sec^3 x = -\frac{\sin x}{\cos^3 x}$$

$$u'_2 = \cos x \sec^3 x = \sec^2 x.$$

Then

$$u_1 = -\frac{1}{2} \frac{1}{\cos^2 x} = -\frac{1}{2} \sec^2 x$$

$$u_2 = \tan x.$$

Thus

$$y = c_1 \cos x + c_2 \sin x - \frac{1}{2} \cos x \sec^2 x + \sin x \tan x$$

$$\begin{aligned}
 &= c_1 \cos x + c_2 \sin x - \frac{1}{2} \sec x + \frac{1 - \cos^2 x}{\cos x} \\
 &= c_3 \cos x + c_2 \sin x + \frac{1}{2} \sec x.
 \end{aligned}$$

and

$$y' = -c_3 \sin x + c_2 \cos x + \frac{1}{2} \sec x \tan x.$$

The initial conditions imply

$$\begin{aligned}
 c_3 + \frac{1}{2} &= 1 \\
 c_2 &= \frac{1}{2}.
 \end{aligned}$$

Thus $c_3 = c_2 = 1/2$ and

$$y = \frac{1}{2} \cos x + \frac{1}{2} \sin x + \frac{1}{2} \sec x.$$

- 37.** Let $u = y'$ so that $u' = y''$. The equation becomes $u du/dx = 4x$. Separating variables we obtain

$$u du = 4x dx \implies \frac{1}{2} u^2 = 2x^2 + c_1 \implies u^2 = 4x^2 + c_2.$$

When $x = 1$, $y' = u = 2$, so $4 = 4 + c_2$ and $c_2 = 0$. Then

$$\begin{aligned}
 u^2 = 4x^2 &\implies \frac{dy}{dx} = 2x \quad \text{or} \quad \frac{dy}{dx} = -2x \\
 &\implies y = x^2 + c_3 \quad \text{or} \quad y = -x^2 + c_4.
 \end{aligned}$$

When $x = 1$, $y = 5$, so $5 = 1 + c_3$ and $5 = -1 + c_4$. Thus $c_3 = 4$ and $c_4 = 6$. We have $y = x^2 + 4$ and $y = -x^2 + 6$. Note however that when $y = -x^2 + 6$, $y' = -2x$ and $y'(1) = -2 \neq 2$. Thus, the solution of the initial-value problem is $y = x^2 + 4$.

- 38.** Let $u = y'$ so that $y'' = u du/dy$. The equation becomes $2u du/dy = 3y^2$. Separating variables we obtain

$$2u du = 3y^2 dy \implies u^2 = y^3 + c_1.$$

When $x = 0$, $y = 1$ and $y' = u = 1$ so $1 = 1 + c_1$ and $c_1 = 0$. Then

$$\begin{aligned}
 u^2 = y^3 &\implies \left(\frac{dy}{dx} \right)^2 = y^3 \implies \frac{dy}{dx} = y^{3/2} \implies y^{-3/2} dy = dx \\
 &\implies -2y^{-1/2} = x + c_2 \implies y = \frac{4}{(x + c_2)^2}.
 \end{aligned}$$

When $x = 0$, $y = 1$, so $1 = 4/c_2^2$ and $c_2 = \pm 2$. Thus, $y = 4/(x + 2)^2$ and $y = 4/(x - 2)^2$. Note, however, that when $y = 4/(x + 2)^2$, $y' = -8/(x + 2)^3$ and $y'(0) = -1 \neq 1$. Thus, the solution of the initial-value problem is $y = 4/(x - 2)^2$.

- 39. (a)** The auxiliary equation is $12m^4 + 64m^3 + 59m^2 - 23m - 12 = 0$ and has roots -4 , $-\frac{3}{2}$, $-\frac{1}{3}$, and $\frac{1}{2}$. The general solution is

$$y = c_1 e^{-4x} + c_2 e^{-3x/2} + c_3 e^{-x/3} + c_4 e^{x/2}.$$

- (b)** The system of equations is

$$\begin{aligned}
 c_1 + c_2 + c_3 + c_4 &= -1 \\
 -4c_1 - \frac{3}{2}c_2 - \frac{1}{3}c_3 + \frac{1}{2}c_4 &= 2 \\
 16c_1 + \frac{9}{4}c_2 + \frac{1}{9}c_3 + \frac{1}{4}c_4 &= 5 \\
 -64c_1 - \frac{27}{8}c_2 - \frac{1}{27}c_3 + \frac{1}{8}c_4 &= 0.
 \end{aligned}$$

CHAPTER 3 REVIEW EXERCISES

Using a CAS we find $c_1 = -\frac{73}{495}$, $c_2 = \frac{109}{35}$, $c_3 = -\frac{3726}{385}$, and $c_4 = \frac{257}{45}$. The solution of the initial-value problem is

$$y = -\frac{73}{495}e^{-4x} + \frac{109}{35}e^{-3x/2} - \frac{3726}{385}e^{-x/3} + \frac{257}{45}e^{x/2}.$$

40. Consider $xy'' + y' = 0$ and look for a solution of the form $y = x^m$. Substituting into the differential equation we have

$$xy'' + y' = m(m-1)x^{m-1} + mx^{m-1} = m^2x^{m-1}.$$

Thus, the general solution of $xy'' + y' = 0$ is $y_c = c_1 + c_2 \ln x$. To find a particular solution of $xy'' + y' = -\sqrt{x}$ we use variation of parameters. The Wronskian is

$$W = \begin{vmatrix} 1 & \ln x \\ 0 & 1/x \end{vmatrix} = \frac{1}{x}.$$

Identifying $f(x) = -x^{-1/2}$ we obtain

$$u'_1 = \frac{x^{-1/2} \ln x}{1/x} = \sqrt{x} \ln x \quad \text{and} \quad u'_2 = \frac{-x^{-1/2}}{1/x} = -\sqrt{x},$$

so that

$$u_1 = x^{3/2} \left(\frac{2}{3} \ln x - \frac{4}{9} \right) \quad \text{and} \quad u_2 = -\frac{2}{3}x^{3/2}.$$

Then

$$y_p = x^{3/2} \left(\frac{2}{3} \ln x - \frac{4}{9} \right) - \frac{2}{3}x^{3/2} \ln x = -\frac{4}{9}x^{3/2}$$

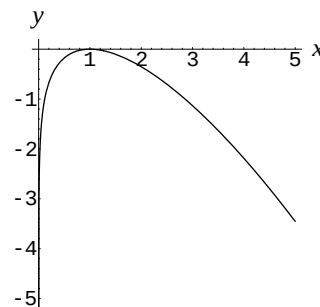
and the general solution of the differential equation is

$$y = c_1 + c_2 \ln x - \frac{4}{9}x^{3/2}.$$

The initial conditions are $y(1) = 0$ and $y'(1) = 0$. These imply that $c_1 = \frac{4}{9}$ and $c_2 = \frac{2}{3}$. The solution of the initial-value problem is

$$y = \frac{4}{9} + \frac{2}{3} \ln x - \frac{4}{9}x^{3/2}.$$

The graph is shown above.



41. From $(D-2)x + (D-2)y = 1$ and $Dx + (2D-1)y = 3$ we obtain $(D-1)(D-2)y = -6$ and $Dx = 3 - (2D-1)y$. Then

$$y = c_1 e^{2t} + c_2 e^t - 3 \quad \text{and} \quad x = -c_2 e^t - \frac{3}{2}c_1 e^{2t} + c_3.$$

Substituting into $(D-2)x + (D-2)y = 1$ gives $c_3 = \frac{5}{2}$ so that

$$x = -c_2 e^t - \frac{3}{2}c_1 e^{2t} + \frac{5}{2}.$$

42. From $(D-2)x - y = t - 2$ and $-3x + (D-4)y = -4t$ we obtain $(D-1)(D-5)x = 9 - 8t$. Then

$$x = c_1 e^t + c_2 e^{5t} - \frac{8}{5}t - \frac{3}{25}$$

and

$$y = (D-2)x - t + 2 = -c_1 e^t + 3c_2 e^{5t} + \frac{16}{25} + \frac{11}{25}t.$$

43. From $(D-2)x - y = -e^t$ and $-3x + (D-4)y = -7e^t$ we obtain $(D-1)(D-5)x = -4e^t$ so that

$$x = c_1 e^t + c_2 e^{5t} + t e^t.$$

Then

$$y = (D - 2)x + e^t = -c_1 e^t + 3c_2 e^{5t} - t e^t + 2e^t.$$

44. From $(D + 2)x + (D + 1)y = \sin 2t$ and $5x + (D + 3)y = \cos 2t$ we obtain $(D^2 + 5)y = 2 \cos 2t - 7 \sin 2t$. Then

$$y = c_1 \cos t + c_2 \sin t - \frac{2}{3} \cos 2t + \frac{7}{3} \sin 2t$$

and

$$\begin{aligned} x &= -\frac{1}{5}(D + 3)y + \frac{1}{5} \cos 2t \\ &= \left(\frac{1}{5}c_1 - \frac{3}{5}c_2\right) \sin t + \left(-\frac{1}{5}c_2 - \frac{3}{5}c_1\right) \cos t - \frac{5}{3} \sin 2t - \frac{1}{3} \cos 2t. \end{aligned}$$

45. The period of a spring/mass system is given by $T = 2\pi/\omega$ where $\omega^2 = k/m = kg/W$, where k is the spring constant, W is the weight of the mass attached to the spring, and g is the acceleration due to gravity. Thus, the period of oscillation is $T = (2\pi/\sqrt{kg})\sqrt{W}$. If the weight of the original mass is W , then $(2\pi/\sqrt{kg})\sqrt{W} = 3$ and $(2\pi/\sqrt{kg})\sqrt{W-8} = 2$. Dividing, we get $\sqrt{W}/\sqrt{W-8} = 3/2$ or $W = \frac{9}{4}(W-8)$. Solving for W we find that the weight of the original mass was 14.4 pounds.

46. (a) Solving $\frac{3}{8}x'' + 6x = 0$ subject to $x(0) = 1$ and $x'(0) = -4$ we obtain

$$x = \cos 4t - \sin 4t = \sqrt{2} \sin(4t + 3\pi/4).$$

(b) The amplitude is $\sqrt{2}$, period is $\pi/2$, and frequency is $2/\pi$.

(c) If $x = 1$ then $t = n\pi/2$ and $t = -\pi/8 + n\pi/2$ for $n = 1, 2, 3, \dots$

(d) If $x = 0$ then $t = \pi/16 + n\pi/4$ for $n = 0, 1, 2, \dots$. The motion is upward for n even and downward for n odd.

(e) $x'(3\pi/16) = 0$

(f) If $x' = 0$ then $4t + 3\pi/4 = \pi/2 + n\pi$ or $t = 3\pi/16 + n\pi$.

47. From $mx'' + 4x' + 2x = 0$ we see that nonoscillatory motion results if $16 - 8m \geq 0$ or $0 < m \leq 2$.

48. From $x'' + \beta x' + 64x = 0$ we see that oscillatory motion results if $\beta^2 - 256 < 0$ or $0 \leq \beta < 16$.

49. From $q'' + 10^4 q = 100 \sin 50t$, $q(0) = 0$, and $q'(0) = 0$ we obtain $q_c = c_1 \cos 100t + c_2 \sin 100t$, $q_p = \frac{1}{75} \sin 50t$, and

(a) $q = -\frac{1}{150} \sin 100t + \frac{1}{75} \sin 50t$,

(b) $i = -\frac{2}{3} \cos 100t + \frac{2}{3} \cos 50t$, and

(c) $q = 0$ when $\sin 50t(1 - \cos 50t) = 0$ or $t = n\pi/50$ for $n = 0, 1, 2, \dots$.

50. By Kirchhoff's second law,

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t).$$

Using $q'(t) = i(t)$ we can write the differential equation in the form

$$L \frac{di}{dt} + Ri + \frac{1}{C} q = E(t).$$

Then differentiating we obtain

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = E'(t).$$

51. For $\lambda = \alpha^2 > 0$ the general solution is $y = c_1 \cos \alpha x + c_2 \sin \alpha x$. Now

$$y(0) = c_1 \quad \text{and} \quad y(2\pi) = c_1 \cos 2\pi\alpha + c_2 \sin 2\pi\alpha,$$

CHAPTER 3 REVIEW EXERCISES

so the condition $y(0) = y(2\pi)$ implies

$$c_1 = c_1 \cos 2\pi\alpha + c_2 \sin 2\pi\alpha$$

which is true when $\alpha = \sqrt{\lambda} = n$ or $\lambda = n^2$ for $n = 1, 2, 3, \dots$. Since

$$y' = -\alpha c_1 \sin \alpha x + \alpha c_2 \cos \alpha x = -nc_1 \sin nx + nc_2 \cos nx,$$

we see that $y'(0) = nc_2 = y'(2\pi)$ for $n = 1, 2, 3, \dots$. Thus, the eigenvalues are n^2 for $n = 1, 2, 3, \dots$, with corresponding eigenfunctions $\cos nx$ and $\sin nx$. When $\lambda = 0$, the general solution is $y = c_1x + c_2$ and the corresponding eigenfunction is $y = 1$.

For $\lambda = -\alpha^2 < 0$ the general solution is $y = c_1 \cosh \alpha x + c_2 \sinh \alpha x$. In this case $y(0) = c_1$ and $y(2\pi) = c_1 \cosh 2\pi\alpha + c_2 \sinh 2\pi\alpha$, so $y(0) = y(2\pi)$ can only be valid for $\alpha = 0$. Thus, there are no eigenvalues corresponding to $\lambda < 0$.

52. (a) The differential equation is $d^2r/dt^2 - \omega^2 r = -g \sin \omega t$. The auxiliary equation is $m^2 - \omega^2 = 0$, so $r_c = c_1 e^{\omega t} + c_2 e^{-\omega t}$. A particular solution has the form $r_p = A \sin \omega t + B \cos \omega t$. Substituting into the differential equation we find $-2A\omega^2 \sin \omega t - 2B\omega^2 \cos \omega t = -g \sin \omega t$. Thus, $B = 0$, $A = g/2\omega^2$, and $r_p = (g/2\omega^2) \sin \omega t$. The general solution of the differential equation is $r(t) = c_1 e^{\omega t} + c_2 e^{-\omega t} + (g/2\omega^2) \sin \omega t$. The initial conditions imply $c_1 + c_2 = r_0$ and $g/2\omega - \omega c_1 + \omega c_2 = v_0$. Solving for c_1 and c_2 we get

$$c_1 = (2\omega^2 r_0 + 2\omega v_0 - g)/4\omega^2 \quad \text{and} \quad c_2 = (2\omega^2 r_0 - 2\omega v_0 + g)/4\omega^2,$$

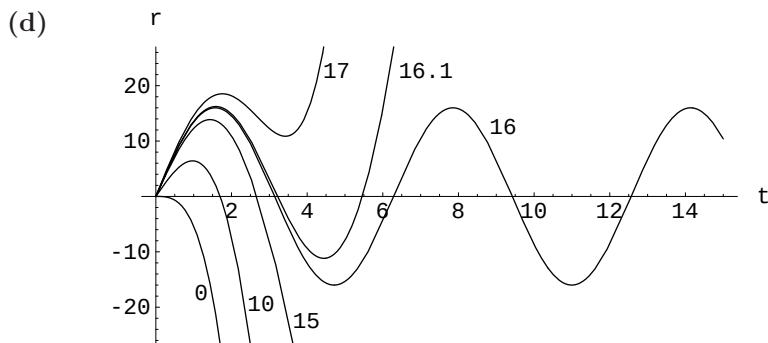
so that

$$r(t) = \frac{2\omega^2 r_0 + 2\omega v_0 - g}{4\omega^2} e^{\omega t} + \frac{2\omega^2 r_0 - 2\omega v_0 + g}{4\omega^2} e^{-\omega t} + \frac{g}{2\omega^2} \sin \omega t.$$

- (b) The bead will exhibit simple harmonic motion when the exponential terms are missing. Solving $c_1 = 0$, $c_2 = 0$ for r_0 and v_0 we find $r_0 = 0$ and $v_0 = g/2\omega$.

To find the minimum length of rod that will accommodate simple harmonic motion we determine the amplitude of $r(t)$ and double it. Thus $L = g/\omega^2$.

- (c) As t increases, $e^{\omega t}$ approaches infinity and $e^{-\omega t}$ approaches 0. Since $\sin \omega t$ is bounded, the distance, $r(t)$, of the bead from the pivot point increases without bound and the distance of the bead from P will eventually exceed $L/2$.



- (e) For each v_0 we want to find the smallest value of t for which $r(t) = \pm 20$. Whether we look for $r(t) = -20$ or $r(t) = 20$ is determined by looking at the graphs in part (d). The total times that the bead stays on the rod is shown in the table below.

v_0	0	10	15	16.1	17
r	-20	-20	-20	20	20
t	1.55007	2.35494	3.43088	6.11627	4.22339

When $v_0 = 16$ the bead never leaves the rod.

53. Unlike the derivation given in Section 3.8 in the text, the weight mg of the mass m does not appear in the net force since the spring is not stretched by the weight of the mass when it is in the equilibrium position (i.e. there is no $mg - ks$ term in the net force). The only force acting on the mass when it is in motion is the restoring force of the spring. By Newton's second law,

$$m \frac{d^2x}{dt^2} = -kx \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{k}{m}x = 0.$$

54. The force of kinetic friction opposing the motion of the mass is μN , where μ is the coefficient of sliding friction and N is the normal component of the weight. Since friction is a force opposite to the direction of motion and since N is pointed directly downward (it is simply the weight of the mass), Newton's second law gives, for motion to the right ($x' > 0$),

$$m \frac{d^2x}{dt^2} = -kx - \mu mg,$$

and for motion to the left ($x' < 0$),

$$m \frac{d^2x}{dt^2} = -kx + \mu mg.$$

Traditionally, these two equations are written as one expression

$$m \frac{d^2x}{dt^2} + f_x \operatorname{sgn}(x') + kx = 0,$$

where $f_k = \mu mg$ and

$$\operatorname{sgn}(x') = \begin{cases} 1, & x' > 0 \\ -1, & x' < 0. \end{cases}$$